

Australian/New Zealand Standard™

## Cold-formed steel structures



## **AS/NZS 4600:2005**

This Joint Australian/New Zealand Standard was prepared by Joint Technical Committee BD-082, Cold-formed Steel Structures. It was approved on behalf of the Council of Standards Australia on 28 September 2005 and on behalf of the Council of Standards New Zealand on 23 September 2005.  
This Standard was published on 30 December 2005.

---

The following are represented on Committee BD-082:

Association of Consulting Engineers Australia  
Australian Building Codes Board  
Australian Chamber of Commerce and Industry  
Australian Steel Institute  
Bureau of Steel Manufacturers of Australia  
Engineers Australia  
NZ Structural Engineering Society  
NZ Heavy Engineering Research Association  
NZ Metal Roofing and Cladding Manufacturers Association Inc.  
Queensland University of Technology  
University of Sydney  
University of Tasmania  
Welding Technology Institute of Australia

---

### **Keeping Standards up-to-date**

Standards are living documents which reflect progress in science, technology and systems. To maintain their currency, all Standards are periodically reviewed, and new editions are published. Between editions, amendments may be issued. Standards may also be withdrawn. It is important that readers assure themselves they are using a current Standard, which should include any amendments which may have been published since the Standard was purchased.

Detailed information about joint Australian/New Zealand Standards can be found by visiting the Standards Web Shop at [www.saiglobal.com.au](http://www.saiglobal.com.au) or Standards New Zealand web site at [www.standards.co.nz](http://www.standards.co.nz) and looking up the relevant Standard in the on-line catalogue.

For more frequent listings or notification of revisions, amendments and withdrawals, Standards Australia and Standards New Zealand offer a number of update options. For information about these services, users should contact their respective national Standards organization.

We also welcome suggestions for improvement in our Standards, and especially encourage readers to notify us immediately of any apparent inaccuracies or ambiguities. Please address your comments to the Chief Executive of either Standards Australia or Standards New Zealand at the address shown on the back cover.

---

*This Standard was issued in draft form for comment as DR 03518.*

---

# Australian/New Zealand Standard™

## Cold-formed steel structures

First published in Australia as AS 1538—1974.  
Second edition 1988.  
AS 1538—1988 jointly revised and redesignated AS/NZS 4600:1996.  
Second edition 2005.  
Reissued incorporating Amendment No. 1 (August 2010).

### COPYRIGHT

© Standards Australia Limited/Standards New Zealand

All rights are reserved. No part of this work may be reproduced or copied in any form or by any means, electronic or mechanical, including photocopying, without the written permission of the publisher, unless otherwise permitted under the Copyright Act 1968 (Australia) or the Copyright Act 1994 (New Zealand).

Jointly published by SAI Global Limited under licence from Standards Australia Limited, GPO Box 476, Sydney, NSW 2001 and by Standards New Zealand, Private Bag 2439, Wellington 6140

ISBN 0 7337 7073 8

## PREFACE

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee BD-082, Cold-formed Steel Structures, to supersede AS/NZS 4600:1996.

*This Standard incorporates Amendment No. 1 (August 2010). The changes required by the Amendment are indicated in the text by a marginal bar and amendment number against the clause, note, table, figure or part thereof affected.*

The objective of this Standard is to provide designers of cold-formed steel structures with specifications for cold-formed steel structural members used for load-carrying purposes in buildings and other structures.

This edition incorporates the following major changes to the previous edition:

- (a) Alignment of terminology with AS/NZS 1170 series for structural design actions.
- (b) The acceptance of welding of G450 steel to AS 1397 using existing rules with a minor change in capacity factors. This circumvents the confusion for welding of G450 steel.
- (c) Increase in the design stress of G550 steel to AS 1397, less than 0.9 mm thick and greater than or equal to 0.6 mm thick, from 75% to 90%, and 75% for thickness less than 0.6 mm of the specified values of yield stress and tensile strength.
- (d) The addition of web with holes to allow for holes in webs in shear and bearing.
- (e) A new set of design rules for unstiffened elements and edge stiffeners under stress gradient.
- (f) Minor modifications to the rules for uniformly compressed elements with edge and intermediate stiffeners to remove a discontinuity in the equations which formerly existed.
- (g) A new approach for edge-stiffened elements with intermediate stiffeners.
- (h) A new approach for multiple intermediate stiffeners in compression flanges where the stiffeners no longer need to be fully effective.
- (i) The significant liberalization of the lateral buckling rules for beams to allow the AISI design curve to be used with a rational buckling analysis. This will significantly increase the capacity of purlins throughout Australia and New Zealand.
- (j) The introduction of a whole new set of equations for web crippling (bearing) of webs without holes and removal of unconservatism in the previous edition which was discovered by Australian research.
- (k) Bearing of nested Z-section.
- (l) The removal of  $l/1000$  for angle sections in compression which are fully effective.
- (m) Additional design rules for fillet welds, flare welds and resistance welds.
- (n) Modification of the bearing coefficient for bolts to be a function of  $d/t$  for high values of  $d/t$  and a separate bearing capacity given for bolts where bolt hole deformation is considered.
- (o) Significant reduction in the edge distance provision from  $3.0d$  to  $1.5d$  for screw fasteners and blind rivets.
- (p) The addition of a new section on fatigue of cold-formed members.
- (q) Inclusion of new direct strength method as an alternative to the effective width method of design.

(r) Alignment of testing provisions with AS/NZS 1170.0.

This Standard will be referenced in the Building Code of Australia 2006, thereby superseding AS 4600—1996, which will be withdrawn 12 months from the date of publication of this Standard.

Notes to the text contain information and guidance. They are not an integral part of the Standard.

A statement expressed in mandatory terms in a note to a table is deemed to be a requirement of this Standard.

The terms ‘normative’ and ‘informative’ have been used in this Standard to define the application of the appendix to which they apply. A ‘normative’ appendix is an integral part of a Standard, whereas an ‘informative’ appendix is only for information and guidance.

## CONTENTS

|                                                                                                               | <i>Page</i> |
|---------------------------------------------------------------------------------------------------------------|-------------|
| SECTION 1 SCOPE AND GENERAL                                                                                   |             |
| 1.1 SCOPE .....                                                                                               | 6           |
| 1.2 NORMATIVE REFERENCES .....                                                                                | 6           |
| 1.3 DEFINITIONS .....                                                                                         | 6           |
| 1.4 NOTATION .....                                                                                            | 13          |
| 1.5 MATERIALS .....                                                                                           | 24          |
| 1.6 DESIGN REQUIREMENTS .....                                                                                 | 28          |
| SECTION 2 ELEMENTS                                                                                            |             |
| 2.1 SECTION PROPERTIES .....                                                                                  | 34          |
| 2.2 EFFECTIVE WIDTHS OF STIFFENED ELEMENTS .....                                                              | 36          |
| 2.3 EFFECTIVE WIDTHS OF UNSTIFFENED ELEMENTS .....                                                            | 41          |
| 2.4 EFFECTIVE WIDTHS OF UNIFORMLY COMPRESSED ELEMENTS<br>WITH AN EDGE STIFFENER .....                         | 44          |
| 2.5 EFFECTIVE WIDTHS OF UNIFORMLY COMPRESSED STIFFENED<br>ELEMENTS WITH ONE INTERMEDIATE STIFFENER .....      | 47          |
| 2.6 EFFECTIVE WIDTHS OF UNIFORMLY COMPRESSED STIFFENED<br>ELEMENTS WITH MULTIPLE INTERMEDIATE STIFFENER ..... | 48          |
| 2.7 EFFECTIVE WIDTHS OF UNIFORMLY COMPRESSED EDGE-STIFFENED<br>ELEMENTS WITH INTERMEDIATE STIFFENERS .....    | 51          |
| 2.8 ARCHED COMPRESSION ELEMENTS .....                                                                         | 52          |
| SECTION 3 MEMBERS                                                                                             |             |
| 3.1 GENERAL .....                                                                                             | 53          |
| 3.2 MEMBERS SUBJECT TO AXIAL TENSION .....                                                                    | 53          |
| 3.3 MEMBERS SUBJECT TO BENDING .....                                                                          | 54          |
| 3.4 CONCENTRICALLY LOADED COMPRESSION MEMBERS .....                                                           | 74          |
| 3.5 COMBINED AXIAL COMPRESSION OR TENSION, AND BENDING .....                                                  | 77          |
| 3.6 CYLINDRICAL TUBULAR MEMBERS .....                                                                         | 79          |
| SECTION 4 STRUCTURAL ASSEMBLIES                                                                               |             |
| 4.1 BUILT-UP SECTIONS .....                                                                                   | 81          |
| 4.2 MIXED SYSTEMS .....                                                                                       | 82          |
| 4.3 LATERAL RESTRAINTS .....                                                                                  | 82          |
| 4.4 WALL STUDS AND WALL STUD ASSEMBLIES .....                                                                 | 87          |
| SECTION 5 CONNECTIONS                                                                                         |             |
| 5.1 GENERAL .....                                                                                             | 88          |
| 5.2 WELDED CONNECTIONS .....                                                                                  | 88          |
| 5.3 BOLTED CONNECTIONS .....                                                                                  | 99          |
| 5.4 SCREWED CONNECTIONS .....                                                                                 | 104         |
| 5.5 BLIND RIVETED CONNECTIONS .....                                                                           | 107         |
| 5.6 RUPTURE .....                                                                                             | 109         |
| 5.7 OTHER CONNECTIONS USING ANY TYPE OF FASTENERS .....                                                       | 110         |

*Page*

## SECTION 6 FATIGUE

|     |                                                        |     |
|-----|--------------------------------------------------------|-----|
| 6.1 | GENERAL .....                                          | 111 |
| 6.2 | CALCULATION OF MAXIMUM STRESSES AND STRESS RANGE ..... | 114 |
| 6.3 | DETAIL CATEGORIES FOR CLASSIFIED DETAILS.....          | 114 |
| 6.4 | FATIGUE ASSESSMENT .....                               | 117 |

## SECTION 7 DIRECT STRENGTH METHOD

|     |               |     |
|-----|---------------|-----|
| 7.1 | GENERAL ..... | 119 |
| 7.2 | MEMBERS.....  | 120 |

## SECTION 8 TESTING

|     |                                                   |     |
|-----|---------------------------------------------------|-----|
| 8.1 | TESTING FOR DETERMINING MATERIAL PROPERTIES ..... | 125 |
| 8.2 | TESTING FOR ASSESSMENT OR VERIFICATION.....       | 126 |

## APPENDICES

|   |                                                                                                                           |     |
|---|---------------------------------------------------------------------------------------------------------------------------|-----|
| A | NORMATIVE REFERENCES .....                                                                                                | 128 |
| B | FLEXURAL MEMBERS SUBJECTED TO POSITIVE<br>AND NEGATIVE BENDING .....                                                      | 130 |
| C | PROTECTION .....                                                                                                          | 131 |
| D | DISTORTIONAL BUCKLING STRESSES OF<br>GENERAL CHANNELS, LIPPED CHANNELS AND Z-SECTIONS<br>IN COMPRESSION AND BENDING ..... | 133 |
| E | SECTION PROPERTIES .....                                                                                                  | 137 |
| F | STANDARD TESTS FOR SINGLE-POINT FASTENER CONNECTIONS .....                                                                | 141 |
| G | BIBLIOGRAPHY .....                                                                                                        | 146 |

## STANDARDS AUSTRALIA/STANDARDS NEW ZEALAND

---

**Australian/New Zealand Standard**  
**Cold-formed steel structures**

---

## SECTION 1 SCOPE AND GENERAL

**1.1 SCOPE**

This Standard sets out minimum requirements for the design of structural members cold-formed to shape from carbon or low-alloy steel sheet, strip, plate or bar not more than 25 mm in thickness and used for load-carrying purposes in buildings. It is also applicable for structures other than buildings provided appropriate allowances are made for dynamic effects.

This Standard does not apply to the design of structures subject to fire and brittle fracture.

**1.2 NORMATIVE REFERENCES**

Documents referred to in this Standard are listed in Appendix A and are indispensable for the application of this document.

**1.3 DEFINITIONS**

For the purpose of this Standard, the definitions below apply. Definitions peculiar to a particular clause or section are also given in that clause or section.

**1.3.1 Action**

Set of concentrated or distributed forces acting on a structure (direct action), or deformation imposed on a structure or constrained within it (indirect action).

**1.3.2 Action effect (internal effects of actions, load effects)**

Internal forces and bending moments due to actions (stress resultants).

**1.3.3 Arched compression element**

A circular or parabolic arch-shaped compression element having an inside radius-to-thickness ratio greater than 8, stiffened at both ends by edge stiffeners. (See Figure 1.3(d).)

**1.3.4 Assemblage of elements**

A system of interconnected cold-formed steel elements that act together to resist earthquake action in such a way that the strength and deformation capacity of the system is not adversely affected by the buckling or crippling of any one element of the assemblage.

**1.3.5 Bend**

Portion adjacent to flat elements and having a maximum inside radius-to-thickness ratio ( $r_i/t$ ) of 8. (See Figure 1.1.)

**1.3.6 Braced member**

Member for which the transverse displacement of one end of the member relative to the other is effectively prevented.



### 1.3.7 Can

Implies a capability or possibility and refers to the ability of the user of the Standard, or to a possibility that is available or that might occur.

### 1.3.8 Capacity design principles

Appropriate material standard design and detailing provisions which enable zones where post-elastic response is acceptable to be identified and detailed in a manner that ensures these zones are capable of accepting the inelastic demands placed upon them.

NOTE: All other zones are to be designed to ensure that all other undesirable inelastic response mechanisms are suppressed and detailed in a manner that the ultimate limit state horizontal deformations that they are expected to be subjected to, can be sustained without significant (e.g., greater than 20%) loss of load-carrying capacity after four complete cycles of loading.

### 1.3.9 Capacity reduction factor

A factor used to multiply the nominal capacity to obtain the design capacity.

### 1.3.10 Clinching

Structural fastening of two or more flat elements by single-point embossing or piercing without using additional material.

### 1.3.11 Cold-formed steel structural members

Shapes that are manufactured by press-braking blanks sheared from sheets, cut lengths of coils or plates, or by roll forming cold- or hot-rolled coils or sheets; both forming operations being performed at ambient room temperature, that is, without manifest addition of heat as required for hot-forming.

### 1.3.12 Direct strength method

An alternative design method that provides predictions of member resistance without the use of effective widths.

### 1.3.13 Design action effect

The action effect computed from the design values of the actions or design loads.

### 1.3.14 Design capacity

The product of the capacity reduction factor and the nominal capacity.

### 1.3.15 Distortional buckling

A mode of buckling involving change in cross-sectional shape, excluding local buckling.

### 1.3.16 Doubly-symmetric section

A section symmetric about two orthogonal axes through its centroid. (See Figure 1.5(a).)

### 1.3.17 Effective design width

Where the flat width of an element is reduced for design purposes, the reduced design width is termed the effective width or effective design width.

### 1.3.18 Elements

Simple shapes into which a cold-formed structural member is considered divided and may consist of the following shapes:

- (a) *Flat elements* Appearing in cross-section as rectangles. (See Figure 1.2.)
- (b) *Bends* Appearing in cross-section as sectors of circular rings, having the inside radius-to-thickness ratio less than or equal to eight ( $r_i/t \leq 8$ ). (See Figure 1.2.)
- (c) *Arched elements* Circular or parabolic elements having the inside radius-to-thickness ratio greater than eight ( $r_i/t > 8$ ). (See Figure 1.2.)

**1.3.19 Feed width ( $w_f$ )**

Width of coiled or flat steel used in the production of a cold-formed product.

**1.3.20 Flexural-torsional buckling**

A mode of buckling in which compression members can bend and twist simultaneously without change of cross-sectional shape.

**1.3.21 Length (of a compression member)**

The actual length ( $l$ ) of an axially loaded compression member, taken as the length centre-to-centre of intersections with supporting members, or the cantilevered length in the case of a freestanding member.

**1.3.22 Limit states**

States beyond which the structure no longer satisfies the design criteria.

NOTE: Limit states separate desired states (compliance) from undesired states (non-compliance).

**1.3.23 Limit states, serviceability**

States that correspond to conditions beyond which specified service criteria for a structure or structural element are no longer met.

**1.3.24 Limit states, stability**

States that correspond to the loss of static equilibrium of a structure considered as a rigid body.

**1.3.25 Limit states, ultimate**

States associated with collapse, or with other similar forms of structural failure.

NOTE: This generally corresponds to the maximum load-carrying resistance of a structure or structural element, but, in some cases, to the maximum applicable strain or deformation.

**1.3.26 Load**

The value of a force appropriate for an action.

**1.3.27 Local buckling**

A mode of buckling involving plate flexure alone without transverse deformation of the line or lines of intersection of adjoining plates.

**1.3.28 May**

Indicates the existence of an option.

**1.3.29 Multiple-stiffened element**

An element that is stiffened between webs, or between a web and a stiffened edge, by means of intermediate stiffeners that are parallel to the direction of stress. (See Figure 1.3(c).)

**1.3.30 Nominal action effect or nominal load**

An unfactored action effect or load determined in accordance with the relevant loading Standard.

**1.3.31 Nominal capacity**

The capacity of a member or connection, calculated using the parameters specified in this Standard.

**1.3.32 Nominal dimension**

A specified manufactured dimension.

### **1.3.33 Point-symmetric section**

A section symmetrical about a point (centroid) such as a Z-section having equal flanges. (See Figure 1.5(b).)

### **1.3.34 Primary structure**

The structural system provided to carry the earthquake forces generated in the structure to the ground.

### **1.3.35 Proof testing**

Application of test loads to a structure, sub-structure, member or connection, to ascertain the structural characteristics of only that one item under test.

### **1.3.36 Prototype testing**

Application of test loads to one or more structures, sub-structures, members or connections, to ascertain the structural characteristics of that class of structures, sub-structures, members or connections which are nominally identical to the units tested.

### **1.3.37 Pull-over (pull-through)**

Failure of a single-point connection by the sheet being pulled over the head of the fastener or the head of the fastener being pulled through the sheet.

### **1.3.38 Pull-out**

Failure of a single-point connection by the embedded part of the fastener being pulled out of the member.

### **1.3.39 Segment (in a member subjected to bending)**

The length between adjacent cross-sections, which are fully or partially restrained, or the length between an unrestrained end and the adjacent cross-section, which is fully or partially restrained.

### **1.3.40 Shall**

Indicates that a statement is mandatory.

### **1.3.41 Should**

Indicates a recommendation (non-mandatory).

### **1.3.42 Single-point fastener**

A mechanical connection at a single discrete point such as a screw or rivet.

### **1.3.43 Singly-symmetric (monosymmetric) section**

A section symmetric about only one axis through its centroid. (See Figure 1.5(c).)

### **1.3.44 Special study**

A procedure for the analysis or design, or both, of the structure, agreed between the authority having statutory powers to control the design and erection of a structure, and the design engineer.

### **1.3.45 Stiffened or partially stiffened compression element**

A flat compression element (i.e., a plane compression flange of a flexural member or a plane web or flange of a compression member) of which both edges parallel to the direction of stress are stiffened by a web, flange, edge stiffener, intermediate stiffener, or the like. (See Figure 1.3(a).)

### 1.3.46 Stiffeners

#### 1.3.46.1 Edge stiffener

Formed element at the edge of a flat compression element. (See Figure 1.4(a).)

#### 1.3.46.2 Intermediate stiffeners

Formed elements, employed in multiple-stiffened segments, and located between edges of stiffened elements. (See Figure 1.4(b).)

### 1.3.47 Structural ductility factor

A numerical assessment of the ability of a structure to sustain cyclic inelastic displacements.

### 1.3.48 Structural performance factor

A numerical assessment of the ability of a building to survive cyclic displacements.

### 1.3.49 Structural response factor

The level of force reduction available for a given system compared with an elastic structural system.

### 1.3.50 Sub-element

The portion between adjacent stiffeners, or between web and intermediate stiffener, or between edge and stiffener.

### 1.3.51 Tensile strength

The minimum ultimate strength in tension specified for the grade of steel in the appropriate Standard.

### 1.3.52 Thickness

The base steel thickness ( $t$ ), exclusive of coatings.

### 1.3.53 Unformed steel

Steel as received from the steel producer or warehouse before being cold-worked as a result of fabricating operations.

### 1.3.54 Unformed steel properties

Mechanical properties of unformed steel, such as yield stress, tensile strength and ductility.

### 1.3.55 Unstiffened compression element

A flat compression element which is stiffened at only one edge parallel to the direction of stress. (See Figure 1.3(b).)

### 1.3.56 Yield stress

The minimum yield stress in tension specified for the grade of steel in the appropriate Standard.

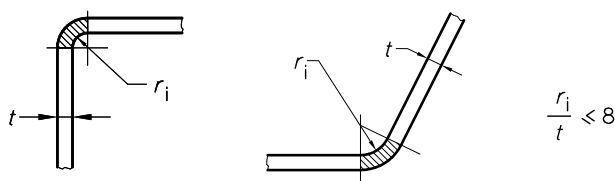
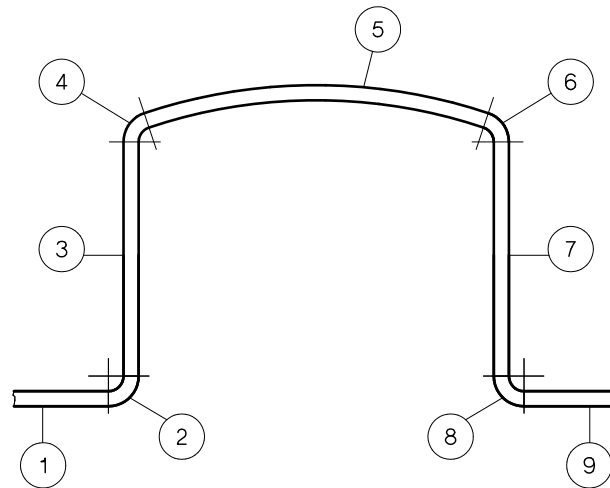


FIGURE 1.1 BENDS



NOTE: The member illustrated consists of the following nine elements:

- (a) Elements 1, 3, 7, 9 are flat elements (flats).
- (b) Elements 2, 4, 6, 8 are bends ( $r_i/t \leq 8$ ).
- (c) Element 5 is an arched element ( $r_i/t > 8$ ).

FIGURE 1.2 ELEMENTS

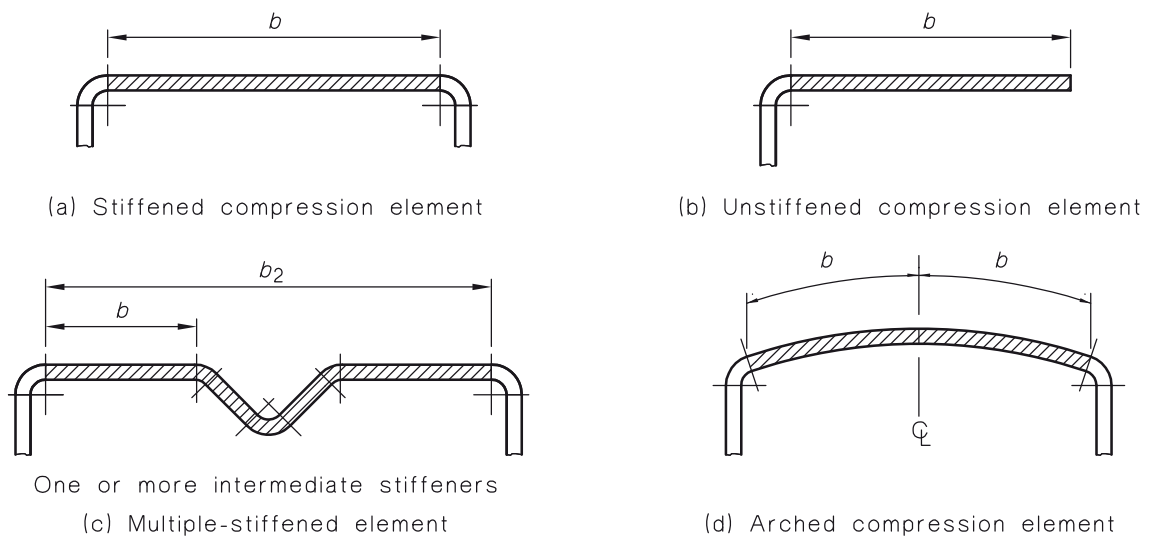


FIGURE 1.3 STIFFENING MODES

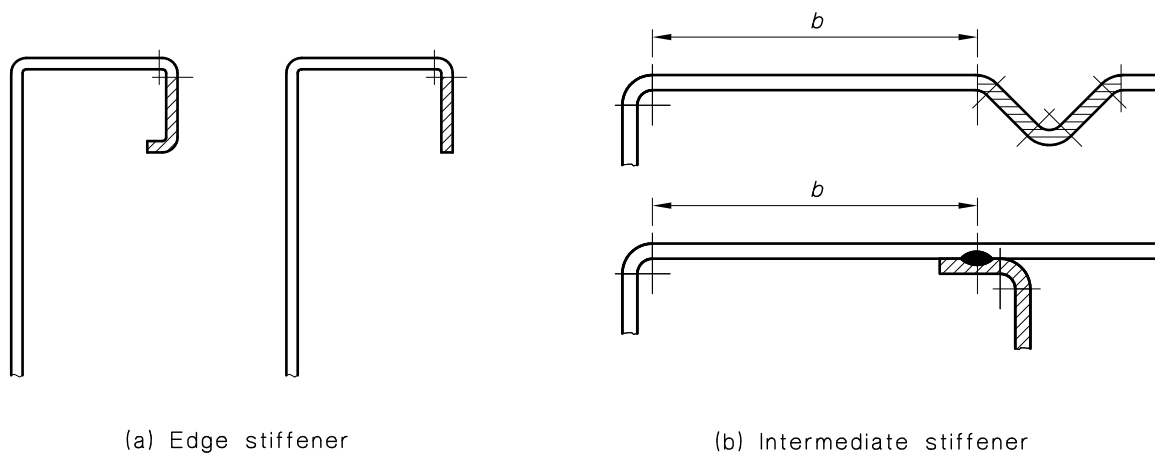
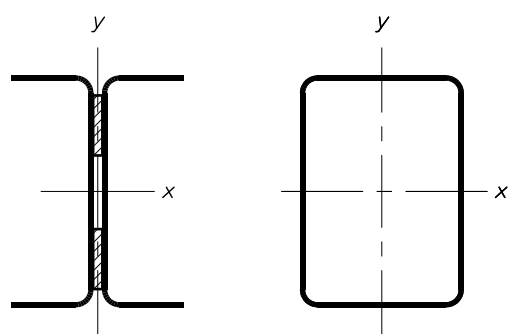
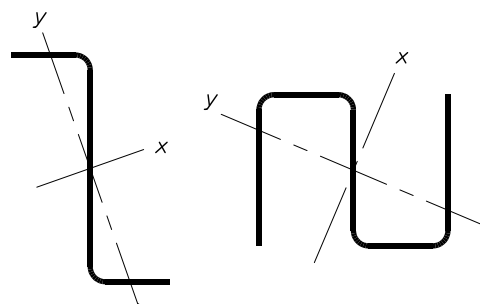


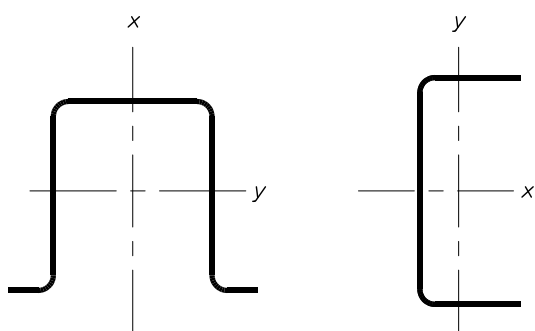
FIGURE 1.4 STIFFENERS



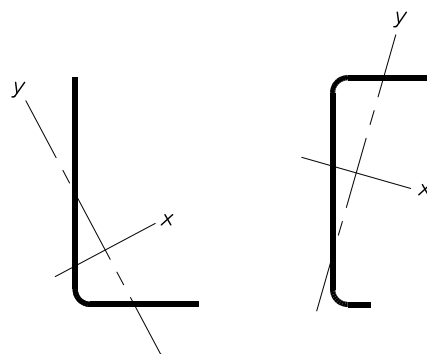
(a) Doubly-symmetric sections



(b) Point-symmetric sections



(c) Singly-symmetric (monosymmetric) sections



(d) Non-symmetric (asymmetric) sections

FIGURE 1.5 EXAMPLES OF SECTION SYMMETRY

## 1.4 NOTATION

The symbols used in this Standard are listed in Table 1.4.

Where non-dimensional ratios are involved, both the numerator and denominator are expressed in identical units.

The dimensional units for length and stress in all expressions or equations are to be taken as millimetres (mm) and megapascals (MPa) respectively, unless specifically noted otherwise.

An asterisk placed after a symbol denotes a design action effect due to the design load for the ultimate limit state.

**TABLE 1.4**  
**NOTATION**

| Symbol           | Description                                                                                                                                                                                                                                     | Clause reference                      |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| $A_c$            | minor diameter area of a bolt                                                                                                                                                                                                                   | 5.3.5.1                               |
| $A_e$            | effective area of the bearing stiffener subjected to uniform compressive stress; <i>or</i><br>effective area at the yield stress ( $f_y$ ) to calculate $N_s$ ; <i>or</i><br>effective area at the critical stress ( $f_n$ ) to calculate $N_c$ | 3.3.8.2, 3.4.1, 3.6.3                 |
| $A_g$            | gross area of the element including stiffeners; <i>or</i><br>gross area of the cross-section                                                                                                                                                    | 2.6.1, 3.2.2                          |
| $A_{gt}$         | gross area subject to tension in block shear rupture                                                                                                                                                                                            | 5.6.3                                 |
| $A_{gv}$         | gross area subject to shear in block shear rupture                                                                                                                                                                                              | 5.6.3                                 |
| $A_n$            | net area of the cross-section; <i>or</i><br>net area of the connected part                                                                                                                                                                      | 3.2.2, 5.3.3, 5.4.2.2, 5.5.2.2        |
| $A_{nt}$         | net area subject to tension in block shear rupture                                                                                                                                                                                              | 5.6.3                                 |
| $A_{nv}$         | net area subject to shear in block shear rupture                                                                                                                                                                                                | 5.6.3                                 |
| $A_o$            | reduced area due to local buckling; <i>or</i><br>plain shank area of a bolt                                                                                                                                                                     | 3.6.3, 5.3.5.1                        |
| $A_s$            | reduced area of a stiffener; <i>or</i><br>gross area of the stiffener; <i>or</i><br>cross-sectional area of a transverse stiffener; <i>or</i><br>tensile stress area of a bolt                                                                  | 2.5.2, 2.6.2.1, 3.3.8.1, 5.3.5.2      |
| $A_{se}$         | effective area of a stiffener                                                                                                                                                                                                                   | 2.4.2, 2.5.2                          |
| $A_{st}$         | gross area of a shear stiffener                                                                                                                                                                                                                 | 3.3.8.3                               |
| $A_{s1}, A_{s2}$ | area of a member in compression consisting of the transverse stiffeners and a portion of the web                                                                                                                                                | 3.3.8.1                               |
| $A_{wn}$         | net area of the web                                                                                                                                                                                                                             | 5.6.1                                 |
| $a$              | bracing interval; <i>or</i><br>shear panel length for unstiffened web elements; <i>or</i><br>distance between transverse stiffeners for stiffened web elements; <i>or</i><br>distance between centre-lines of braces                            | 3.3.3.2.1, 3.3.4.1, 3.3.8.3, 4.3.3.4  |
| $B_c$            | constant                                                                                                                                                                                                                                        | 1.5.1.2                               |
| $b$              | flat width of element excluding radii; <i>or</i><br>length of the web hole; <i>or</i><br>flat width of element excluding corners or bends; <i>or</i><br>half the length of the arched compression element                                       | 2.2.1.2, 2.2.4.1, 2.5.2, 3.3.5, 4.1.2 |

(continued)

TABLE 1.4 (continued)

| Symbol           | Description                                                                                                                                                                                                                                                                                                                                               | Clause reference                                                                                        |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| $b_e$            | effective width of uniformly compressed stiffened and unstiffened elements used for determining the capacity                                                                                                                                                                                                                                              | 2.2.1.2, 2.2.2.2, 2.2.3.2, 2.3.1.2, 2.3.1.3, 2.3.2.2, 2.3.2.3, 2.4.2, 2.4.3, 2.5.3, 2.6.1, 2.6.2.2, 2.7 |
| $b_{ed}$         | effective width of uniformly compressed stiffened and unstiffened elements used for determining the deflection                                                                                                                                                                                                                                            | 2.2.1.3, 2.2.2.3                                                                                        |
| $b_{e1}, b_{e2}$ | effective width of stiffened element with stress gradient                                                                                                                                                                                                                                                                                                 | 2.2.3.2, 2.2.3.3                                                                                        |
| $b_f$            | flange width of a channel- or Z-section                                                                                                                                                                                                                                                                                                                   | 3.4.7, 4.3.3.3                                                                                          |
| $b_o$            | total flat width of the stiffened element                                                                                                                                                                                                                                                                                                                 | 2.6.1, 2.6.2.1                                                                                          |
| $b_p$            | greatest sub-element flat width                                                                                                                                                                                                                                                                                                                           | 2.6.3.1                                                                                                 |
| $b_1$            | width of the flange projecting beyond the web for I-beams and similar sections; <i>or</i><br>half the distance between webs for box- or U-type sections; <i>or</i><br>sum of the flange projection beyond the web and the depth of the lip for I-beams and similar sections; <i>or</i><br>width of stiffened element                                      | 2.1.3.2, 2.1.3.3, 2.3.2.2                                                                               |
| $b_2$            | width of unstiffened element; <i>or</i><br>flat width of element with intermediate stiffener excluding radii; <i>or</i><br>total flat width of the edge-stiffened element                                                                                                                                                                                 | 2.3.2.2, 2.5.2, 2.7                                                                                     |
| $C$              | for compression members, ratio of the total bend cross-sectional area to the total cross-sectional area of the full section; <i>and</i><br>for flexural members, ratio of the total bend cross-sectional area of the controlling flange to the full cross-sectional area of the controlling flange; <i>or</i><br>coefficient; <i>or</i><br>bearing factor | 1.5.1.2, 3.3.6.2, 5.3.4.2, 5.4.2.3                                                                      |
| $C_b$            | coefficient depending on moment distribution in the laterally unbraced segment                                                                                                                                                                                                                                                                            | 3.3.3.2.1                                                                                               |
| $C_i$            | horizontal distance from the edge of the element to the centre-line of the stiffener                                                                                                                                                                                                                                                                      | 2.6.3.1                                                                                                 |
| $C_{TF}$         | coefficient for unequal end moment                                                                                                                                                                                                                                                                                                                        | 3.3.3.2.1, 3.5.1                                                                                        |
| $C_l$            | coefficient of bearing length                                                                                                                                                                                                                                                                                                                             | 3.3.6.2                                                                                                 |
| $C_{ms}$         | coefficient used to determine $N_{ib}^*$ for multiple-span system with midspan restraints                                                                                                                                                                                                                                                                 | 4.3.3.3                                                                                                 |
| $C_{mx}, C_{my}$ | coefficient for unequal end moment                                                                                                                                                                                                                                                                                                                        | 3.5.1                                                                                                   |
| $C_r$            | coefficient of inside bent radius                                                                                                                                                                                                                                                                                                                         | 3.3.6.2                                                                                                 |
| $C_s$            | coefficient for moment causing compression or tension on the shear centre side of the centroid                                                                                                                                                                                                                                                            | 3.3.3.2.1                                                                                               |
| $C_{th}$         | coefficient used to determine $N_{ib}^*$ for multiple-span system with third-point restraints                                                                                                                                                                                                                                                             | 4.3.3.3                                                                                                 |
| $C_{tr}$         | coefficient used to determine $N_{ib}^*$ for multiple-span system with restraints at the support                                                                                                                                                                                                                                                          | 4.3.3.3                                                                                                 |
| $C_w$            | coefficient of web slenderness                                                                                                                                                                                                                                                                                                                            | 3.3.6.2                                                                                                 |
| $C_y$            | compression strain factor                                                                                                                                                                                                                                                                                                                                 | 3.3.2.3                                                                                                 |
| $c_f$            | amount of curling                                                                                                                                                                                                                                                                                                                                         | 2.1.3.2                                                                                                 |

(continued)



TABLE 1.4 (continued)

| Symbol   | Description                                                                                                                                                                                                                                                                                                                                                                                                     | Clause reference                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| $d$      | depth of a section; <i>or</i><br>actual stiffener dimension                                                                                                                                                                                                                                                                                                                                                     | 2.1.3.2, Figure 2.4.2(a),<br>3.3.6.3, 3.4.7                                                              |
| $d_a$    | average diameter of an arc spot weld at mid-thickness of $t_c$ ; <i>or</i><br>average width of an arc seam weld                                                                                                                                                                                                                                                                                                 | 5.2.4.2, 5.2.5.2                                                                                         |
| $d_e$    | effective diameter of a fused area of an arc spot weld; <i>or</i><br>effective width of an arc seam weld at fused surfaces                                                                                                                                                                                                                                                                                      | 5.2.4.1, 5.2.4.2, 5.2.5.2                                                                                |
| $d_f$    | nominal diameter of a bolt, screw, blind rivet                                                                                                                                                                                                                                                                                                                                                                  | Table 5.3.1, 5.3.2, 5.3.4.2,<br>5.4.1, 5.4.2.1, 5.4.2.2,<br>5.4.2.3, 5.5.1, 5.5.2.1,<br>5.5.2.2, 5.5.2.3 |
| $d_h$    | diameter of a hole                                                                                                                                                                                                                                                                                                                                                                                              | 2.2.2.2, Table 5.3.1, 5.3.2,<br>5.6.1                                                                    |
| $d_l$    | actual stiffener dimension; <i>or</i><br>overall depth of lip                                                                                                                                                                                                                                                                                                                                                   | Figure 2.4.2(a)                                                                                          |
| $d_o$    | outside diameter of a tubular member                                                                                                                                                                                                                                                                                                                                                                            | 3.6.1, 3.6.2                                                                                             |
| $d_s$    | reduced effective width of a stiffener; <i>or</i><br>effective stiffener dimension                                                                                                                                                                                                                                                                                                                              | Figure 2.4.2(b)                                                                                          |
| $d_{se}$ | effective width of a stiffener; <i>or</i><br>effective stiffener dimension                                                                                                                                                                                                                                                                                                                                      | Figure 2.4.2(b)                                                                                          |
| $d_{sh}$ | nominal shank diameter                                                                                                                                                                                                                                                                                                                                                                                          | Figure F1, Appendix F                                                                                    |
| $d_w$    | depth of the compressed portion of the web; <i>or</i><br>visible diameter of the outer surface of an arc spot weld; <i>or</i><br>width of an arc seam weld; <i>or</i><br>screw head or washer diameter                                                                                                                                                                                                          | 3.3.2.3, 5.2.4.2, 5.2.5.2,<br>5.4.3.2                                                                    |
| $d_{wc}$ | coped depth of a web                                                                                                                                                                                                                                                                                                                                                                                            | 5.6.1                                                                                                    |
| $d_{wh}$ | depth of the web hole                                                                                                                                                                                                                                                                                                                                                                                           | 2.2.4.1, 3.3.4.2                                                                                         |
| $d_1$    | depth of the flat portion of a web measured along the plane<br>of the web; <i>or</i><br>width of elements adjoining the stiffened element                                                                                                                                                                                                                                                                       | 2.1.3.4, 2.2.4.1, 2.6.1,<br>3.3.4.1, 3.3.4.2, 3.3.6.2                                                    |
| $E$      | Young's modulus of elasticity ( $200 \times 10^3$ MPa)                                                                                                                                                                                                                                                                                                                                                          | 2.2.1.2, 3.3.2.3, 5.2.4.2                                                                                |
| $e$      | edge distance measured in the line of the force from centre-<br>line of an arc spot weld, arc seam weld or from centre of a<br>bolt hole to the nearest edge of an adjacent weld or bolt<br>hole, or to the end of the connected part toward which the<br>force is directed; <i>or</i><br>distance measured in the line of force from the centre of a<br>standard hole to the nearest end of the connected part | 5.2.4.3, 5.2.5.3, 5.3.2,<br>5.4.2.4, 5.5.2.4                                                             |
| $e_y$    | yield strain                                                                                                                                                                                                                                                                                                                                                                                                    | 3.3.2.3                                                                                                  |
| $F_p^*$  | vertical design load supported by all purlin lines being<br>restrained                                                                                                                                                                                                                                                                                                                                          | 4.3.3.3                                                                                                  |
| $f_c$    | stress at service load in the cover plate or sheet; <i>or</i><br>fatigue strength corrected for thickness of material                                                                                                                                                                                                                                                                                           | 4.1.2, 6.1.3                                                                                             |
| $f_{cr}$ | plate elastic buckling stress                                                                                                                                                                                                                                                                                                                                                                                   | 2.2.1.2, 3.4.2                                                                                           |
| $f_f$    | uncorrected fatigue strength                                                                                                                                                                                                                                                                                                                                                                                    | 6.1.3                                                                                                    |
| $f_n$    | critical stress                                                                                                                                                                                                                                                                                                                                                                                                 | 3.3.8.1, 3.4.1, 3.6.3                                                                                    |
| $f_{oc}$ | elastic flexural, torsional and flexural-torsional buckling<br>stress                                                                                                                                                                                                                                                                                                                                           | 3.4.1, 3.4.2, 3.4.3, 3.6.3                                                                               |

(continued)

TABLE 1.4 (continued)

|    | Symbol    | Description                                                                                                                                                                                                                               | Clause reference                                                                 |
|----|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| A1 | $f_{od}$  | elastic distortional buckling stress of the cross-section                                                                                                                                                                                 | 3.3.3.3, 7.2.1.4, 7.2.2.4, Paragraphs D1, D2, D3, Appendix D                     |
| A1 | $f_{ol}$  | elastic local buckling stress                                                                                                                                                                                                             | 7.2.1.3, 7.2.2.3                                                                 |
|    | $f_{ox}$  | elastic buckling stress in an axially loaded compression member for flexural buckling about the $x$ -axis                                                                                                                                 | 3.3.3.2.1                                                                        |
|    | $f_{oy}$  | elastic buckling stress in an axially loaded compression member for flexural buckling about the $y$ -axis                                                                                                                                 | 3.3.3.2.1                                                                        |
|    | $f_{oz}$  | elastic buckling stress in an axially loaded compression member for torsional buckling                                                                                                                                                    | 3.3.3.2.1, 3.4.3                                                                 |
|    | $f_{rn}$  | detail category reference fatigue strength at $n_r$ -normal stress                                                                                                                                                                        | 6.1.3                                                                            |
|    | $f_{rnc}$ | corrected detail category reference fatigue strength for normal stress                                                                                                                                                                    | 6.1.3                                                                            |
|    | $f_{rs}$  | detail category reference fatigue strength at $n_r$ -shear stress                                                                                                                                                                         | 6.1.3                                                                            |
|    | $f_{rsc}$ | corrected detail category reference fatigue strength for shear stress                                                                                                                                                                     | 6.1.3                                                                            |
|    | $f_u$     | tensile strength used in design; <i>or</i><br>tensile strength of sheet                                                                                                                                                                   | 1.5.1.1, 1.5.1.4, 1.5.1.6, 1.5.2, 3.2.2, 5.3.4.2                                 |
|    | $f_{uf}$  | minimum tensile strength of a bolt                                                                                                                                                                                                        | 5.3.5.1                                                                          |
|    | $f_{uv}$  | tensile strength of unformed steel                                                                                                                                                                                                        | 1.5.1.2                                                                          |
|    | $f_{uw}$  | nominal tensile strength of a weld metal                                                                                                                                                                                                  | 5.2.2.2, 5.2.3.4                                                                 |
|    | $f_{u1}$  | tensile strength used in the design of the connected plate of the thickness $t_1$ ; <i>or</i><br>tensile strength of the sheet in contact with the screw head or with the rivet head                                                      | 5.2.3.3, 5.4.2.3, 5.5.2.3                                                        |
|    | $f_{u2}$  | tensile strength used in the design of the connected plate of the thickness $t_2$ ; <i>or</i><br>tensile strength of the sheet not in contact with the screw head or with the rivet head                                                  | 5.2.3.3, 5.4.2.3, 5.5.2.3                                                        |
|    | $f_y$     | yield stress used in design; <i>or</i><br>yield stress of web steel; <i>or</i><br>yield stress of stiffener; <i>or</i><br>yield stress used in design for the lower strength base steel; <i>or</i><br>tensile or compressive yield stress | 1.5.1.1, 1.5.1.4, 1.5.1.6, 1.5.2, 3.2.2, 3.3.2.3, 3.3.8.2, 5.2.2.1, 6.1.3, 8.1.3 |
|    | $f_{wy}$  | lower yield stress value of the beam web ( $f_y$ ) or of the stiffener section ( $f_{ys}$ )                                                                                                                                               | 3.3.8.1                                                                          |
|    | $f_{ya}$  | average design yield stress of a full section                                                                                                                                                                                             | 1.5.1.2                                                                          |
|    | $f_{yc}$  | tensile yield stress of bends                                                                                                                                                                                                             | 1.5.1.2                                                                          |
|    | $f_{yf}$  | yield stress of flat portions; <i>or</i><br>yield stress of unformed steel if tests are not made; <i>or</i><br>yield stress of flat coupons of formed members                                                                             | 1.5.1.2, 8.1.4.1                                                                 |
|    | $f_{ys}$  | yield stress of stiffener steel                                                                                                                                                                                                           | 3.3.8.1                                                                          |
|    | $f_{yv}$  | tensile yield stress of unformed steel                                                                                                                                                                                                    | 1.5.1.2                                                                          |
|    | $f_3$     | detail category fatigue strength at constant amplitude fatigue limit ( $5 \times 10^6$ cycles)                                                                                                                                            | 6.1.3                                                                            |

(continued)

TABLE 1.4 (continued)

| Symbol         | Description                                                                                                                                                       | Clause reference                    |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| $f_{3c}$       | corrected detail category fatigue strength at constant amplitude fatigue limit                                                                                    | 6.1.3                               |
| $f_5$          | detail category fatigue strength at cut off limit (108 cycles)                                                                                                    | 6.1.3                               |
| $f_{5c}$       | corrected detail category fatigue strength at cut off limit                                                                                                       | 6.1.3                               |
| $f^*$          | design stress in the compression element calculated on the basis of the effective design width; <i>or</i> design stress range                                     | 2.2.1.2, 2.4.2, 6.1.3               |
| $f_{av}^*$     | average design stress in the full, unreduced flange width                                                                                                         | 2.1.3.2                             |
| $f_d^*$        | design compressive stress in the element being considered, based on the effective section at the load for which deflections are determined                        | 2.2.1.3, 2.2.2.3, 2.6.2.2, 2.6.3.2  |
| $f_{d1}^*$     | calculated stress $f_1^*$                                                                                                                                         | 2.2.3.3                             |
| $f_{d2}^*$     | calculated stress $f_2^*$                                                                                                                                         | 2.2.3.3                             |
| $f_i^*$        | design stress range for loading event $i$                                                                                                                         | 6.1.3                               |
| $f_1^*, f_2^*$ | web stresses calculated on the basis of the effective section specified in Clause 2.2.3.2 or the full section specified in Appendix F                             | 2.2.3.2, 2.3.2.2                    |
| $G$            | shear modulus of elasticity ( $80 \times 10^3$ MPa)                                                                                                               | 3.3.3.2.1                           |
| $I_a$          | adequate second moment of area of a stiffener, so that each component element behaves as a stiffened element                                                      | 2.4.2, 2.5.2                        |
| $I_b$          | second moment of area of the full, unreduced cross-section about the bending axis                                                                                 | 3.5.1                               |
| $I_{eff}$      | effective second moment of area for deflection                                                                                                                    | 7.1.4                               |
| $I_g$          | gross second moment of area                                                                                                                                       | 7.1.4                               |
| $I_{min.}$     | minimum second moment of area                                                                                                                                     | 2.8                                 |
| $I_s$          | second moment of area of a full stiffener about its own centroidal axis parallel to the element to be stiffened                                                   | 2.4.2, 2.5.2                        |
| $I_{sp}$       | second moment of area of a stiffener about the centre-line of the flat portion of the element                                                                     | 2.6.2.1                             |
| $I_w$          | warping constant for a cross-section                                                                                                                              | 3.3.3.2.1, Paragraph E1, Appendix E |
| $I_x, I_y$     | second moment of area of the cross-section about the principal x- and y-axes                                                                                      | 3.3.3.2.1, 4.3.3.4                  |
| $I_{x'}$       | second moment of area of the cross-section about its centroidal axis perpendicular to the web                                                                     | 4.3.3.4                             |
| $I_{x'y'}$     | product of second moment of area of the full section about its major and minor principal axes parallel and perpendicular to the web                               | 4.3.3.4                             |
| $I_{yc}$       | second moment of area of the compression portion of a section about the centroidal axis of the full section parallel to the web, using the full unreduced section | 3.3.3.2.1                           |
| $i$            | index for stiffener ' $i$ '                                                                                                                                       | 2.6.3.1                             |
| $J$            | torsion constant for a cross-section                                                                                                                              | 3.3.3.2.1, Paragraph E1, Appendix E |

(continued)

TABLE 1.4 (continued)

| Symbol                   | Description                                                                                                                                                                                                                                                                                                                               | Clause reference                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $k$                      | plate buckling coefficient; <i>or</i><br>non-dimensional yield stress                                                                                                                                                                                                                                                                     | 2.2.1.2, 2.3.2.2, Table 2.4.2,<br>2.5.2, 2.6.1                                                                           |
| $k_d$                    | plate buckling coefficient for distortional buckling                                                                                                                                                                                                                                                                                      | 2.6.1                                                                                                                    |
| $k_f$                    | total population variation due to fabrication                                                                                                                                                                                                                                                                                             | 8.2.2                                                                                                                    |
| $k_{loc}$                | plate buckling coefficient for local sub-element buckling                                                                                                                                                                                                                                                                                 | 2.6.1                                                                                                                    |
| $k_m$                    | total population of variation due to material                                                                                                                                                                                                                                                                                             | 8.2.2                                                                                                                    |
| $k_s$                    | shear stiffener coefficient                                                                                                                                                                                                                                                                                                               | 3.3.8.3                                                                                                                  |
| $k_{st}$                 | stiffener type coefficient                                                                                                                                                                                                                                                                                                                | 3.3.8.2                                                                                                                  |
| $k_t$                    | correction factor for distribution of forces; <i>or</i><br>factor to allow for variability of structural units                                                                                                                                                                                                                            | 3.2.2, Tables 3.2 and 8.2.3                                                                                              |
| $k_v$                    | shear buckling coefficient                                                                                                                                                                                                                                                                                                                | 3.3.4.1, 3.3.8.3                                                                                                         |
| $k'$                     | coefficient used to determine $N_{ib}^*$ where neither flange is<br>connected to the sheeting or connected to the sheeting with<br>concealed fasteners                                                                                                                                                                                    | 4.3.3.4                                                                                                                  |
| $l$                      | actual length of a compression member; <i>or</i><br>full span for simple beams; <i>or</i><br>distance between inflection points for continuous beams;<br><i>or</i><br>twice the length of cantilever beams; <i>or</i><br>unbraced length of a member; <i>or</i><br>laterally unbraced length of a member; <i>or</i><br>length of a member | 1.3.21, 2.1.3.3, 3.3.3.2.1,<br>3.3.3.2.2, 4.1.1, 4.3.3.3,<br>6.1.3                                                       |
| $l_a$                    | lap length                                                                                                                                                                                                                                                                                                                                | Figure F1, Appendix F                                                                                                    |
| $l_b$                    | actual length of bearing                                                                                                                                                                                                                                                                                                                  | 3.3.6.2, 3.3.6.3, 4.3.3.4                                                                                                |
| $l_{br}$                 | unsupported length of bracing or other restraint that<br>restricts distortional buckling of the element                                                                                                                                                                                                                                   | 2.6.2.1                                                                                                                  |
| $l_c$                    | unclamped length of the specimen                                                                                                                                                                                                                                                                                                          | Figure F1, Appendix F                                                                                                    |
| $l_e$                    | effective length of the member                                                                                                                                                                                                                                                                                                            | 3.4.2                                                                                                                    |
| $l_{ex}, l_{ey}, l_{ez}$ | effective buckling for bending about the $x$ - and $y$ -axes, and<br>for twisting, respectively                                                                                                                                                                                                                                           | 3.3.3.2.1                                                                                                                |
| $l_{eb}$                 | effective length in the plane of bending                                                                                                                                                                                                                                                                                                  | 3.5.1                                                                                                                    |
| $l_g$                    | gauge length for measuring the joint displacement                                                                                                                                                                                                                                                                                         | Figure F1, Appendix F                                                                                                    |
| $l_{st}$                 | length of transverse stiffener                                                                                                                                                                                                                                                                                                            | 3.3.8.1                                                                                                                  |
| $l_{sb}$                 | length of bearing stiffener                                                                                                                                                                                                                                                                                                               | 3.3.8.1                                                                                                                  |
| $l_u$                    | limit of unbraced length by which lateral-torsional<br>buckling is not considered                                                                                                                                                                                                                                                         | 3.3.3.2.2                                                                                                                |
| $l_w$                    | length of the full size of the weld; <i>or</i><br>length of fillet weld                                                                                                                                                                                                                                                                   | 5.2.2.1, 5.2.3.3, 5.2.3.4,<br>5.2.5.2                                                                                    |
| $l_{w1}, l_{w2}$         | leg lengths of fillet weld                                                                                                                                                                                                                                                                                                                | 5.2.3.4                                                                                                                  |
| $M$                      | moment due to nominal loads on member to be considered                                                                                                                                                                                                                                                                                    | 7.1.4                                                                                                                    |
| $M_b$                    | nominal member moment capacity                                                                                                                                                                                                                                                                                                            | 2.2.1.2, 3.3.1, 3.3.3.1,<br>3.3.3.2.1, 3.3.3.2.2, 3.3.3.3,<br>3.3.3.4, 3.3.5, 3.6.2, 7.2.2.1<br>Paragraph B2, Appendix B |

(continued)

TABLE 1.4 (continued)

| Symbol             | Description                                                                                                                                                                                                 | Clause reference                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| $M_{bx}, M_{by}$   | nominal member moment capacities about the $x$ - and $y$ -axes, respectively                                                                                                                                | 3.5.1, 3.5.2                                                           |
| $M_c$              | critical moment                                                                                                                                                                                             | 3.3.3.2.1, 3.3.3.3                                                     |
| $M_{bd}$           | nominal member capacity for distortional buckling                                                                                                                                                           | 7.2.2.1, 7.2.2.4                                                       |
| $M_{be}$           | nominal member capacity for lateral-torsional buckling                                                                                                                                                      | 7.2.2.1, 7.2.2.2                                                       |
| $M_{bl}$           | nominal member capacity for local buckling                                                                                                                                                                  | 7.2.2.1, 7.2.2.3                                                       |
| $M_{max.}$         | absolute value of the maximum moment in the unbraced segment                                                                                                                                                | 3.3.3.2.1                                                              |
| $M_n$              | nominal flexural capacity                                                                                                                                                                                   | 7.1.4                                                                  |
| $M_o$              | elastic buckling moment; <i>or</i><br>elastic lateral-torsional buckling moment                                                                                                                             | 3.3.3.2.1, 7.2.2.2                                                     |
| $M_{od}$           | elastic buckling moment in the distortional mode                                                                                                                                                            | 3.3.3.3, 7.2.2.4                                                       |
| $M_{ol}$           | elastic local buckling moment                                                                                                                                                                               | 7.2.2.3                                                                |
| $M_s$              | nominal section moment capacity                                                                                                                                                                             | 2.2.1.2, 3.3.1, 3.3.2.1,<br>3.3.2.2, 3.3.2.3, 3.3.3.5,<br>3.3.5, 3.3.7 |
| $M_{sxf}, M_{syf}$ | nominal section yield moment capacity of the full section<br>about the $x$ - and $y$ -axes, respectively                                                                                                    | 3.5.2                                                                  |
| $M_y$              | moment causing initial yield at the extreme compression<br>fibre of a full section                                                                                                                          | 2.2.1.2, 3.3.3.2.1, 3.3.3.3                                            |
| $M_1$              | smaller bending moment at the ends of the unbraced length                                                                                                                                                   | 3.3.3.2                                                                |
| $M_2$              | larger bending moment at the ends of the unbraced length                                                                                                                                                    | 3.3.3.2                                                                |
| $M_3$              | absolute value of the moment at quarter point of the<br>unbraced segment                                                                                                                                    | 3.3.3.2.1                                                              |
| $M_4$              | absolute value of the moment at mid-point of the unbraced<br>segment                                                                                                                                        | 3.3.3.2.1                                                              |
| $M_5$              | absolute value of the moment at three-quarter point of the<br>unbraced segment                                                                                                                              | 3.3.3.2.1                                                              |
| $M^*$              | design bending moment                                                                                                                                                                                       | 3.3.1, 3.3.5, 3.3.7, 3.6.2,<br>Paragraph B2, Appendix B                |
| $M_x^*, M_y^*$     | design bending moment about the $x$ - and $y$ -axes,<br>respectively                                                                                                                                        | 3.5.1, 3.5.2                                                           |
| $m$                | constant; <i>or</i><br>non-dimensional thickness; <i>or</i><br>distance from the shear centre of one channel to the mid-<br>plane of its web; <i>or</i><br>distance from the concentrated load to the brace | 1.5.1.2, 4.1.1, 4.3.3.4,<br>Paragraph E1, Appendix E                   |
| $N_c$              | nominal member capacity of a member in compression                                                                                                                                                          | 2.2.1.3, 3.3.8.1, 3.4.1, 3.4.7,<br>3.5.1, 7.2.1.1                      |
| $N_{cd}$           | nominal member capacity for distortional buckling                                                                                                                                                           | 7.2.1.1, 7.2.1.4                                                       |
| $N_{ce}$           | nominal member capacity for flexural, torsional or<br>flexural-torsional buckling                                                                                                                           | 7.2.1.1, 7.2.1.2                                                       |
| $N_{cl}$           | nominal member capacity for local buckling                                                                                                                                                                  | 7.2.1.1, 7.2.1.3                                                       |
| $N_e$              | elastic buckling load                                                                                                                                                                                       | 3.5.1                                                                  |
| $N_f$              | nominal tensile capacity of the section of the connected<br>part                                                                                                                                            | 5.3.3                                                                  |

(continued)

TABLE 1.4 (continued)

| Symbol     | Description                                                                                                                                                           | Clause reference                                           |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| $N_{ft}$   | nominal tensile capacity of a bolt                                                                                                                                    | 5.3.5.2                                                    |
| $N_{oc}$   | least of the elastic column buckling load in flexural, torsional and flexural-torsional buckling                                                                      | 7.2.1.2                                                    |
| $N_{od}$   | elastic distortional compression member buckling load                                                                                                                 | 7.2.1.4                                                    |
| $N_{ol}$   | elastic local buckling load                                                                                                                                           | 7.2.1.3                                                    |
| $N_{ou}$   | nominal pull-out capacity of a screw                                                                                                                                  | 5.4.3.2                                                    |
| $N_{ov}$   | nominal pull-over (pull-through) capacity of a screw                                                                                                                  | 5.4.3.2                                                    |
| $N_s$      | nominal section capacity of a member in compression                                                                                                                   | 2.2.1.2, 3.3.8.1, 3.4.1, 3.5.1                             |
| $N_{sl}$   | nominal axial capacity for local buckling                                                                                                                             | 7.2.1.1                                                    |
| $N_t$      | nominal section capacity of a member in tension; <i>or</i> nominal capacity of the connection in tension; <i>or</i> capacity of the net section of the connected part | 3.2.1, 3.5.2, 5.4.2.2, 5.4.3.2, 5.5.2.2                    |
| $N_w$      | nominal tensile or compressive capacity of a butt weld or an arc spot weld                                                                                            | 5.2.2.1, 5.2.4.4                                           |
| $N_y$      | nominal yield capacity of a member in compression                                                                                                                     | 7.2.1.2                                                    |
| $N^*$      | design axial force, tensile or compressive; <i>or</i> design concentrated load or reaction                                                                            | 1.5.1.4, 3.2.1, 3.3.8.1, 3.4.1, 3.5.1, 3.5.2, 3.6.3, 4.1.1 |
| $N_f^*$    | design tensile force on the net section of the connected part                                                                                                         | 5.3.3                                                      |
| $N_{ft}^*$ | design tensile force on a bolt                                                                                                                                        | 5.3.5.2, 5.3.5.3                                           |
| $N_{ib}^*$ | design force to be resisted by intermediate beam brace                                                                                                                | 4.3.3.3, 4.3.3.4                                           |
| $N_t^*$    | design tensile force on the net section of a connected part using screws or blind rivets                                                                              | 5.4.2.2, 5.4.3.2, 5.5.2.2                                  |
| $N_w^*$    | design tensile or compressive force normal to the area of a butt weld or an arc spot weld                                                                             | 5.2.2.1, 5.2.4.4                                           |
| $n$        | exponent                                                                                                                                                              | 2.5.2                                                      |
| $n_c$      | number of compression flange stiffeners                                                                                                                               | Table 7.1.2                                                |
| $n_h$      | number of holes in the critical plane                                                                                                                                 | 5.6.1                                                      |
| $n_i$      | number of cycles of nominal loading event $i$ , producing $f_i^*$                                                                                                     | 6.1.3                                                      |
| $n_n$      | number of the shear planes with threads intercepting the shear plane                                                                                                  | 5.3.5.1                                                    |
| $n_p$      | number of parallel purlin lines                                                                                                                                       | 4.3.3.3                                                    |
| $n_r$      | reference number of stress cycles ( $2 \times 10^6$ cycles)                                                                                                           | 6.1.3                                                      |
| $n_{sc}$   | number of stress cycles                                                                                                                                               | 6.1.3                                                      |
| $n_t$      | number of tension flange stiffeners                                                                                                                                   | Table 7.1.2                                                |
| $n_w$      | number of web stiffeners/folds                                                                                                                                        | Table 7.1.2                                                |
| $n_x$      | number of shear planes without threads intercepting the shear plane                                                                                                   | 5.3.5.1                                                    |
| $q$        | intensity of the design load on a beam                                                                                                                                | 4.1.1                                                      |
| $R$        | modification factor for the distortional plate buckling coefficient; <i>or</i> reduction factor; <i>or</i> radius of outside bend surface                             | 2.6.1, 3.3.3.4, 3.3.3.5, 3.6.3, 5.2.6.2                    |

(continued)

**TABLE 1.4** (continued)

| Symbol     | Description                                                                                                                                                                                                                                                                                                                                                                                                                   | Clause reference                |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| $R_b$      | nominal capacity for concentrated load or reaction for one solid web connecting top and bottom flanges                                                                                                                                                                                                                                                                                                                        | 3.3.6.1, 3.3.6.2, 3.3.7         |
| $R_d$      | design capacity                                                                                                                                                                                                                                                                                                                                                                                                               | 1.6.3, 8.2.3                    |
| $R_f$      | structural response factor                                                                                                                                                                                                                                                                                                                                                                                                    | 1.6.4.1                         |
| $R_{min.}$ | minimum value of the test results                                                                                                                                                                                                                                                                                                                                                                                             | 8.2.3                           |
| $R_n$      | nominal capacity for block shear rupture of the beam-end or tension member connection                                                                                                                                                                                                                                                                                                                                         | 5.6.3                           |
| $R_u$      | nominal capacity                                                                                                                                                                                                                                                                                                                                                                                                              | 1.5.1.4, 1.6.3                  |
| $R_{wc}$   | web crippling capacity for channel-section flexural member                                                                                                                                                                                                                                                                                                                                                                    | 3.3.8.2                         |
| $R^*$      | design concentrated load or reaction in the presence of bending moment                                                                                                                                                                                                                                                                                                                                                        | 3.3.6.1, 3.3.7                  |
| $R_b^*$    | design concentrated load or reaction                                                                                                                                                                                                                                                                                                                                                                                          | 4.1.1                           |
| $r$        | radius of gyration of the full, unreduced cross-section; <i>or</i> centre-line radius                                                                                                                                                                                                                                                                                                                                         | 3.4.2, Table 7.1.1              |
| $r_{cy}$   | radius of gyration of one channel about its centroidal axis parallel to the web                                                                                                                                                                                                                                                                                                                                               | 4.1.1                           |
| $r_f$      | ratio of the force transmitted by the bolts or screws, or rivets at the section considered, divided by the tensile force in the member at that section                                                                                                                                                                                                                                                                        | 5.4.2.2, 5.5.2.2                |
| $r_i$      | inside bend radius                                                                                                                                                                                                                                                                                                                                                                                                            | 1.5.1.2, 3.3.6.2                |
| $r_{o1}$   | polar radius of gyration of the cross-section about the shear centre                                                                                                                                                                                                                                                                                                                                                          | 3.3.3.2.1, 3.4.3                |
| $r_x, r_y$ | radii of gyration of the cross-section about the $x$ - and $y$ -axes, respectively                                                                                                                                                                                                                                                                                                                                            | 3.3.3.2                         |
| $r_1$      | radius of gyration of I-section about the axis perpendicular to the direction in which buckling occurs for the given conditions of end support and intermediate bracing                                                                                                                                                                                                                                                       | 4.1.1                           |
| $S$        | slenderness factor; <i>or</i> fastener distance from the centre-line of the web divided by the flange width for Z-section; <i>or</i> flange width minus the fastener distance from the centre-line of the web divided by the flange width for channel-sections; <i>or</i> spacing in line of the stress of welds, bolts, rivets connecting a cover plate, sheet or a non-integral stiffener in compression to another element | 2.4.2, 2.5.2, 2.7, 3.4.7, 4.1.2 |
| $S_e$      | elastic section modulus of the effective section calculated with extreme compression or tension fibre at $f_y$                                                                                                                                                                                                                                                                                                                | 3.3.3.5                         |
| $S_p$      | structural performance factor                                                                                                                                                                                                                                                                                                                                                                                                 | 1.6.4.2.4                       |
| $S^*$      | design action effects [design actions]                                                                                                                                                                                                                                                                                                                                                                                        | 5.6.3                           |
| $s$        | fastener distance from the centre-line of the web divided by the flange width for Z-sections                                                                                                                                                                                                                                                                                                                                  | 3.4.7, 4.1.2                    |
| $s_f$      | spacing of bolts, screws or rivets perpendicular to the line of the force; <i>or</i> width of sheet, in the case of a single bolt, screw or rivet                                                                                                                                                                                                                                                                             | 5.3.3, 5.4.2.2, 5.5.2.2         |

(continued)

TABLE 1.4 (continued)

| Symbol     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Clause reference                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $s_g$      | vertical distance between two rows of connections nearest to the top and bottom flanges; <i>or</i><br>gauge, the distance measured at right angles to the direction of the design action in the member, centre-to-centre of holes in consecutive lines                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4.1.1, 5.3.1                                                                                                                             |
| $s_{max.}$ | maximum longitudinal spacing of welds or other connectors joining two channels to form an I-section                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4.1.1                                                                                                                                    |
| $s_p$      | staggered pitch distance measured parallel to the direction of the design action in the member, centre-to-centre of holes in consecutive lines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.3.1                                                                                                                                    |
| $s_w$      | weld spacing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4.1.1                                                                                                                                    |
| $t$        | nominal base steel thickness of any element or section exclusive of coatings; <i>or</i><br>thickness of the uniformly compressed stiffened elements; <i>or</i><br>base thickness of beam web; <i>or</i><br>thickness of a channel- or Z-section; <i>or</i><br>thickness of the cover plate or sheet; <i>or</i><br>thickness of the thinnest connected part; <i>or</i><br>thickness of element; <i>or</i><br>thickness of thinnest outside sheet; <i>or</i><br>thickness of the connected part; <i>or</i><br>thickness of the holed material; <i>or</i><br>base metal thickness; <i>or</i><br>thickness of the part in which the end distance is measured; <i>or</i><br>thickness of coped web | 1.3.52, 2.1.3.1, 2.2.1.2, 2.6.1, 3.3.8.1, 3.4.7, 4.1.2, 4.3.3.3, 5.2.4.3, 5.2.5.2, 5.2.7, 5.3.1, 5.3.2, 5.3.4.2, 5.4.2.4, 5.5.2.4, 5.6.1 |
| $t_c$      | total combined base steel thickness (exclusive of coatings) of sheets involved in shear transfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5.2.4.2                                                                                                                                  |
| $t_f$      | thickness of the flange                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2.1.3.2                                                                                                                                  |
| $t_p$      | plate thickness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6.1.3                                                                                                                                    |
| $t_s$      | thickness of the stiffener                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3.3.8.1                                                                                                                                  |
| $t_t$      | design throat thickness of a butt weld                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5.2.2.1, 5.2.3.4, 5.2.6.2                                                                                                                |
| $t_w$      | thickness of a web                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2.1.3.4, 3.3.4.1, 3.3.6.2, 3.3.7                                                                                                         |
| $t_1$      | thickness of the connecting plate of the tensile strength $f_{u1}$ ; <i>or</i><br>thickness of the sheet in contact with the screw head or rivet head                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5.2.3.3, 5.4.2.3, 5.5.2.3                                                                                                                |
| $t_2$      | thickness of the connecting plate of the tensile strength $f_{u2}$ ; <i>or</i><br>thickness of the sheet not in contact with the screw head or rivet head                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5.2.3.3, 5.4.2.3, 5.5.2.3                                                                                                                |
| $V_b$      | nominal bearing capacity of the connected part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 5.3.4.2, 5.3.4.3, 5.4.2.3, 5.5.2.3                                                                                                       |
| $V_f$      | nominal shear capacity of the connected part along two parallel lines in the direction of the applied force                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5.3.2                                                                                                                                    |
| $V_{fv}$   | nominal shear capacity of a bolt or screw                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5.3.5.1, 5.4.2.1, 5.5.2.1                                                                                                                |
| $V_n$      | nominal shear capacity of an arc seam weld or of a beam-end connection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5.2.5.2, 5.6.1                                                                                                                           |

(continued)



TABLE 1.4 (continued)

| Symbol                          | Description                                                                                                                                                | Clause reference                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| $V_{sc}$                        | coefficient of variation of structural characteristic                                                                                                      | 8.2.2, Table 8.2.3                                    |
| $V_v$                           | nominal shear capacity of the web                                                                                                                          | 3.3.4.1, 3.3.4.2, 3.3.5                               |
| $V_w$                           | nominal shear capacity of a butt, fillet, arc spot, flare or resistance weld; <i>or</i><br>nominal shear force transmitted by the weld                     | 5.2.2.2, 5.2.3.1, 5.2.4.2,<br>5.2.4.3, 5.2.6.2, 5.2.7 |
| $V^*$                           | design shear force                                                                                                                                         | 3.3.2.3, 3.3.4.1, 3.3.5                               |
| $V_b^*$                         | design bearing force on a screw or on a rivet; <i>or</i><br>design bearing force on the connected part                                                     | 5.4.2.3, 5.5.2.3                                      |
| $V_f^*$                         | design shear force of the connected part                                                                                                                   | 5.3.2                                                 |
| $V_{fv}^*$                      | design shear force on a bolt, screw or rivet                                                                                                               | 5.3.5.1, 5.3.5.3, 5.4.2.4                             |
| $V_n^*$                         | design shear force on an arc seam weld or a beam-end connection                                                                                            | 5.2.5.2, 5.6.1                                        |
| $V_w^*$                         | design shear force on a butt, fillet, arc spot, flare or resistance weld                                                                                   | 5.2.2.2, 5.2.3.1, 5.2.4.2,<br>5.2.4.3, 5.2.6.2, 5.2.7 |
| $w$                             | width of the specimen                                                                                                                                      | Figure F1, Appendix F                                 |
| $w_f$                           | feed width of the coiled or flat sheet                                                                                                                     | 1.3.19, Note 2 to Figure E1,<br>Appendix E            |
| $x, y$                          | principal axes of the cross-section                                                                                                                        | 3.3.3.2.1, 3.3.6.3                                    |
| $x_o, y_o$                      | coordinates of the shear centre of the cross-section                                                                                                       | 3.3.3.2                                               |
| $Z_c$                           | effective section modulus calculated at a stress $f_c$ in the extreme compression fibre                                                                    | 3.3.3.3                                               |
| $Z_e$                           | effective section modulus calculated with the extreme compression or tension fibre at $f_y$                                                                | 3.3.2.2, 3.3.3.2.1                                    |
| $Z_f$                           | full unreduced section modulus for the extreme compression fibre                                                                                           | 3.3.3.2.1, 3.3.3.3                                    |
| $Z_{ft}$                        | section modulus of the full unreduced section for the extreme tension fibre about the appropriate axis                                                     | 3.5.2                                                 |
| $\alpha$                        | coefficient; <i>or</i><br>modification factor for type of bearing connection                                                                               | 4.3.3.3, 5.3.4.2                                      |
| $\alpha_{nx}, \alpha_{ny}$      | moment amplification factors                                                                                                                               | 3.5.1                                                 |
| $\alpha_s$                      | inverse of the slope of the S-N curve                                                                                                                      | 6.1.3                                                 |
| $\beta$                         | coefficient                                                                                                                                                | 2.6.2.1                                               |
| $\beta_x, \beta_y$              | monosymmetry section constant about the $x$ - and $y$ -axes, respectively                                                                                  | 3.3.3.2.1, Paragraph E2,<br>Appendix E                |
| $\beta_{tf}$                    | thickness correction factor                                                                                                                                | 6.1.3                                                 |
| $\gamma$                        | importance factor                                                                                                                                          | 2.6.2.1                                               |
| $\delta$                        | coefficient                                                                                                                                                | 2.6.2.1                                               |
| $\theta$                        | angle between the plane of the web and the plane of the bearing surface; <i>or</i><br>angle between the vertical and the plane of the web of the Z-section | 3.3.6.2, 4.3.3.3                                      |
| $\lambda, \lambda_1, \lambda_2$ | slenderness ratio                                                                                                                                          | 2.2.1.2, 3.3.2.3, 3.3.7                               |
| $\lambda_b$                     | non-dimensional slenderness used to determine $M_c$ for members subject to lateral buckling                                                                | 3.3.3.2.1                                             |

(continued)

TABLE 1.4 (continued)

| Symbol      | Description                                                                                                                                                                               | Clause reference                                                                                                                                                                |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\lambda_c$ | non-dimensional slenderness used to determine $f_n$ ; <i>or</i> non-dimensional slenderness used to determine $N_{ce}$ ; <i>or</i> non-dimensional slenderness used to determine $M_{cd}$ | 3.4.1, 3.6.3, 7.2.1.2                                                                                                                                                           |
| $\lambda_d$ | non-dimensional slenderness used to determine $M_c$ for members subject to distortional buckling; <i>or</i> non-dimensional slenderness used to determine $N_{cd}$ and $M_{bd}$           | 3.3.3.3, 7.2.1.4, 7.2.2.4                                                                                                                                                       |
| $\lambda_l$ | non-dimensional slenderness used to determine $N_{cl}$ ; <i>or</i> non-dimensional slenderness used to determine $M_{bl}$                                                                 | 7.2.1.3, 7.2.2.3                                                                                                                                                                |
| $\mu$       | structural ductility factor                                                                                                                                                               | 1.6.4.2.2                                                                                                                                                                       |
| $\nu$       | Poisson's ratio = 0.3                                                                                                                                                                     | 2.2.1.2                                                                                                                                                                         |
| $\phi$      | capacity reduction factor                                                                                                                                                                 | 1.5.1.4, 5.2.2.1, 5.2.2.2, 5.2.3.1, 5.2.4.2, 5.2.4.3, 5.2.5.2, 5.2.6.2, 5.2.7, 5.3.2, 5.3.3, 5.3.5.1, 5.3.5.2, 5.4.2.2, 5.4.2.3, 5.5.2.2, 5.5.2.3, 5.5.2.4, 5.6.1, 5.6.3, 6.1.3 |
| $\phi_b$    | capacity reduction factor for bending                                                                                                                                                     | 3.3.1, 3.5.1                                                                                                                                                                    |
| $\phi_c$    | capacity reduction factor for compression                                                                                                                                                 | 3.3.8.1, 3.4.1, 3.5.1                                                                                                                                                           |
| $\phi_t$    | capacity reduction factor for tension                                                                                                                                                     | 3.2.1                                                                                                                                                                           |
| $\phi_v$    | capacity reduction factor for shear                                                                                                                                                       | 3.3.4.1                                                                                                                                                                         |
| $\phi_w$    | capacity reduction factor for bearing                                                                                                                                                     | 3.3.6.1, 3.3.8.2                                                                                                                                                                |
| $\rho$      | quantity for load capacity; <i>or</i> effective width factor                                                                                                                              | 1.5.1.2, 2.2.1.2, 2.3.2.2, 2.6.1                                                                                                                                                |
| $\omega_i$  | coefficient                                                                                                                                                                               | 2.6.3.1                                                                                                                                                                         |
| $\psi$      | stress ratio $f_2^* / f_1^*$                                                                                                                                                              | 2.2.3.2, 2.3.2.2, 3.3.8.3                                                                                                                                                       |

## 1.5 MATERIALS

### 1.5.1 Structural steel

#### 1.5.1.1 Applicable steels

Structural members or steel used in manufacturing shall comply with—

- AS 1163, AS 1397 (excluding Grade G550, less than 0.9 mm in thickness), AS/NZS 1594, AS/NZS 1595 and AS/NZS 3678, as appropriate; and
- other steels, the properties and suitability of which are in accordance with Clause 1.5.1.4. The yield stress ( $f_y$ ) and tensile strength ( $f_u$ ) used in design shall be determined in accordance with Section 8 and AS 1391.

#### 1.5.1.2 Strength increase resulting from cold forming

Strength increase resulting from cold forming shall be permitted by substituting the average design yield stress ( $f_{ya}$ ) of the full section for  $f_y$ . Such increase shall be limited to Clauses 3.3 (excluding Clause 3.3.3.2), 3.4, 3.5, 3.6 and 4.4. The limitations and methods for determining  $f_{ya}$  shall be as follows:

- For axially loaded compression members and flexural members whose proportions are such that the quantity ( $\rho$ ) for load capacity is unity, as determined in accordance with Clause 2.2 for each of the component elements of the sections, the average design yield stress ( $f_{ya}$ ) shall be determined on the basis of one of the following:

- (i) Full section tensile tests (see Section 8).
- (ii) Stub column tests (see Section 8).
- (iii) The following calculation:

$$f_{ya} = Cf_{yc} + (1 - C)f_{yf} \leq f_{uv} \quad \dots 1.5.1.2(1)$$

where

$f_{ya}$  = average design yield stress of the steel in the full section of compression members or full flange sections of flexural members

$C$  = for compression members, ratio of the total bend cross-sectional area to the total cross-sectional area of the full section; and for flexural members, ratio of the total bend cross-sectional area of the controlling flange to the full cross-sectional area of the controlling flange

$f_{yc}$  = tensile yield stress of bends

$$= \frac{B_c f_{yv}}{(r_i / t)^m} \quad \dots 1.5.1.2(2)$$

Equation 1.5.1.2(2) is applicable only if  $f_{uv}/f_{yv}$  is greater than or equal to 1.2,  $r_i/t$  is less than or equal to 7 and the minimum included angle is less than or equal to 120°.

$B_c$  = constant

$$= 3.69 \left( \frac{f_{uv}}{f_{yv}} \right) - 0.819 \left( \frac{f_{uv}}{f_{yv}} \right)^2 - 1.79 \quad \dots 1.5.1.2(3)$$

$f_{yv}$  = tensile yield stress of unformed steel

$r_i$  = inside bend radius

$m$  = constant

$$= 0.192 \left( \frac{f_{uv}}{f_{yv}} \right) - 0.068 \quad \dots 1.5.1.2(4)$$

$f_{uv}$  = tensile strength of unformed steel

$f_{yf}$  = yield stress of the flat portions (see Clause 8.1.4); or yield stress of unformed steel if tests are not made

- (b) For axially loaded tension members,  $f_{ya}$  shall be determined by either Item (a)(i) or Item (a)(iii).

#### 1.5.1.3 Effect of welding

The effect of any welding on the mechanical properties of a member shall be determined on the basis of tests on specimens of the full section containing the weld within the gauge length. Any necessary allowance for such effect shall be made in the structural use of the member. Welded connections for all grades conforming with AS 1163 and grades G250, G300, G350 and G450 steel conforming with AS 1397, designed in accordance with Clause 5.2.3 for fillet welds and Clause 5.2.6 for flare welds do not require further testing.

#### 1.5.1.4 Ductility

Steels not listed in Clause 1.5.1.1 and used for forming structural members and connections shall comply with the following requirements:

- (a) The ratio of tensile strength to yield stress shall be not less than 1.08. The total elongation shall be not less than 10% for a 50 mm gauge length or 7% for a 200 mm gauge length standard specimen tested in accordance with AS 1391. If these requirements cannot be met, the following criteria shall be satisfied:
- (i) Local elongation in a 13 mm gauge length across the fracture shall be not less than 20%.
  - (ii) Uniform elongation outside the fracture shall be not less than 3%.

If the ductility of the material is determined on the basis of the local and uniform elongation criteria, the use of such material shall be restricted to the design of purlins and girts in accordance with Clauses 3.3.2.2, 3.3.3.2, 3.3.3.3 and 3.3.3.4. For purlins and girts subject to combined axial load and bending moment (see Clause 3.5),  $N^*/\phi R_u$  shall not exceed 0.15—

where

$N^*$  = design axial force

$\phi$  = capacity reduction factor

$R_u$  = nominal capacity

- (b) Steels conforming to AS 1397, Grade 550, less than 0.9 mm in thickness, which do not comply with Item (a) may be used provided—

- (i) the yield stress ( $f_y$ ) used in design in Sections 2, 3, 4 and 7, and the tensile strength ( $f_u$ ) used in design in Section 5 are taken as 90% of the corresponding specified values or 495 MPa, whichever is the lesser, and for steel less than 0.6 mm in thickness, the yield stress ( $f_y$ ) used in design in Sections 2, 3, 4 and 7, and the tensile strength ( $f_u$ ) used in design in Section 5 are taken as 75% of the corresponding specified values or 410 MPa, whichever is the lesser; or
- (ii) the suitability of such steel can be demonstrated by load test in accordance with Section 8.

#### 1.5.1.5 Acceptance of steels

Certified mill test reports, or test certificates issued by the mill, shall constitute sufficient evidence of compliance with the Standards referred to in this Standard.

The uncoated minimum steel thickness at any location of the cold-formed product, as delivered to the job site, shall be not less than 95% of the value used in its design. However, lesser thicknesses shall be permitted at bends (forming corners) due to cold-forming effects.

#### 1.5.1.6 Unidentified steel

If unidentified steel is used, it shall be free from surface imperfections and shall be used only where the particular physical properties of the steel and its weldability will not adversely affect the design capacities and serviceability of the structure. Unless a full test in accordance with AS 1391 is made, the yield stress of the steel used in design ( $f_y$ ) shall be 170 MPa or less, and the tensile strength used in design ( $f_u$ ) shall be 300 MPa or less.

#### 1.5.2 Design stresses

The minimum yield stress ( $f_y$ ) and tensile strength ( $f_u$ ) used in design shall not exceed the values given in Table 1.5 for the appropriate steel grade.

NOTE: Regardless of the closeness of yield stress and tensile strength of some steels, steel grades given in Table 1.5 are suitable for cold-forming provided that an appropriate inside bend radius ( $r_i$ ) is chosen.

A1

**TABLE 1.5**  
**MINIMUM STRENGTHS OF STEELS COMPLYING WITH**  
**AS 1163, AS 1397, AS/NZS 1594, AS/NZS 1595 AND AS/NZS 3678**

| Applicable Standard | Grade                                         | Yield stress ( $f_y$ ) MPa | Tensile strength ( $f_u$ ) MPa |
|---------------------|-----------------------------------------------|----------------------------|--------------------------------|
| AS 1163             | C250 and C250L0                               | 250                        | 320                            |
|                     | C350 and C350L0                               | 350                        | 430                            |
|                     | C450 and C450L0                               | 450                        | 500                            |
| AS 1397             | G250                                          | 250                        | 320                            |
|                     | G300                                          | 300                        | 340                            |
|                     | G350                                          | 350                        | 420                            |
|                     | G450*                                         | 450                        | 480                            |
|                     | G500†                                         | 500                        | 520                            |
|                     | G550‡                                         | 550                        | 550                            |
| AS/NZS 1594         | HA1                                           | (see Note)                 | (see Note)                     |
|                     | HA3                                           | 200                        | 300                            |
|                     | HA4N                                          | 170                        | 280                            |
|                     | HA200                                         | 200                        | 300                            |
|                     | HA250, HU250                                  | 250                        | 350                            |
|                     | HA250/1                                       | 250                        | 350                            |
|                     | HA300, HU300                                  | 300                        | 400                            |
|                     | HA300/1, HU300/1                              | 300                        | 430                            |
|                     | HW350                                         | 350                        | 430                            |
|                     | HW350                                         | 340                        | 450                            |
|                     | HA400                                         | 380                        | 460                            |
|                     | XF300                                         | 300                        | 440                            |
|                     | XF400                                         | 380                        | 460                            |
|                     | XF500                                         | 480                        | 570                            |
| AS/NZS 1595         | CA220                                         | 210                        | 340                            |
|                     | CA260                                         | 250                        | 350                            |
|                     | CW300                                         | 300                        | 450                            |
|                     | CA350                                         | 350                        | 430                            |
|                     | CA500                                         | 500                        | 510                            |
| AS/NZS 3678         | 200 ( $t \leq 8$ mm)                          | 200                        | 300                            |
|                     | 200 ( $8 \text{ mm} < t \leq 12$ mm)          | 200                        | 300                            |
|                     | 250, 250L15 ( $t \leq 8$ mm)                  | 280                        | 410                            |
|                     | 250, 250L15 ( $8 \text{ mm} < t \leq 12$ mm)  | 260                        | 410                            |
|                     | 250, 250L15 ( $12 \text{ mm} < t \leq 20$ mm) | 250                        | 410                            |
|                     | 250, 250L15 ( $20 \text{ mm} < t \leq 25$ mm) | 250                        | 410                            |
|                     | 300, 300L15 ( $t \leq 8$ mm)                  | 320                        | 430                            |
|                     | 300, 300L15 ( $8 \text{ mm} < t \leq 12$ mm)  | 310                        | 430                            |
|                     | 300, 300L15 ( $12 \text{ mm} < t \leq 20$ mm) | 300                        | 430                            |
|                     | 300, 300L15 ( $20 \text{ mm} < t \leq 25$ mm) | 280                        | 430                            |
|                     | 350, 350L15 ( $t \leq 8$ mm)                  | 360                        | 450                            |
|                     | 350, 350L15 ( $8 \text{ mm} < t \leq 12$ mm)  | 360                        | 450                            |
|                     | 350, 350L15 ( $12 \text{ mm} < t \leq 20$ mm) | 350                        | 450                            |
|                     | 350, 350L15 ( $20 \text{ mm} < t \leq 25$ mm) | 340                        | 450                            |
|                     | 400, 400L15 ( $t \leq 8$ mm)                  | 400                        | 480                            |
|                     | 400, 400L15 ( $8 \text{ mm} < t \leq 12$ mm)  | 400                        | 480                            |
|                     | 400, 400L15 ( $12 \text{ mm} < t \leq 20$ mm) | 380                        | 480                            |
|                     | 400, 400L15 ( $20 \text{ mm} < t \leq 25$ mm) | 360                        | 480                            |
|                     | 450, 450L15 ( $t \leq 8$ mm)                  | 450                        | 520                            |
|                     | 450, 450L15 ( $8 \text{ mm} < t \leq 12$ mm)  | 450                        | 520                            |
|                     | 450, 450L15 ( $12 \text{ mm} < t \leq 20$ mm) | 450                        | 520                            |
|                     | 450, 450L15 ( $20 \text{ mm} < t \leq 25$ mm) | 420                        | 500                            |

(continued)

**TABLE 1.5 (continued)**

| Applicable Standard | Grade                                             | Yield stress ( $f_y$ ) MPa | Tensile strength ( $f_u$ ) MPa |
|---------------------|---------------------------------------------------|----------------------------|--------------------------------|
|                     | WR350, WR350/L0 ( $t \leq 8$ mm)                  | 340                        | 450                            |
|                     | WR350, WR350/L0 ( $8 \text{ mm} < t \leq 12$ mm)  | 340                        | 450                            |
|                     | WR350, WR350/L0 ( $12 \text{ mm} < t \leq 20$ mm) | 340                        | 450                            |
|                     | WR350, WR350/L0 ( $20 \text{ mm} < t \leq 25$ mm) | 340                        | 450                            |

\* Applies to hard-rolled material of thickness greater than or equal to 1.5 mm

† Applies to hard-rolled material of thickness greater than 1.0 mm but less than 1.5 mm

‡ Applies to hard-rolled material of thickness less than or equal to 1.0 mm

NOTE: For design purposes, yield and tensile strengths approximate those of structural Grade HA200. For specific information contact the supplier.

### 1.5.3 Fasteners and electrodes

#### 1.5.3.1 Steel bolts, nuts and washers

Steel bolts, nuts and washers shall comply with AS 1110.1, AS 1111.1, AS 1112.1, AS 1112.2, AS 1112.3, AS 1112.4, AS/NZS 1252, AS/NZS 1559 and AS 4291.1 (ISO 898-1), as appropriate.

The use of high-strength fasteners, other than those complying with AS/NZS 1252, is permitted provided that evidence of their equivalence to high-strength bolts complying with AS/NZS 1252 is available.

#### 1.5.3.2 Welding consumables

All welding consumables shall comply with AS/NZS 1554.1, AS/NZS 1554.5 and AS/NZS 1554.7, as appropriate.

#### 1.5.3.3 Screws

Self-drilling screws shall comply with AS 3566.1 and AS 3566.2.

#### 1.5.3.4 Blind rivets

Blind rivets shall comply with the Industrial Fastener Institute document F114.

## 1.6 DESIGN REQUIREMENTS

### 1.6.1 Actions and combination of actions

A structure and its components shall be designed for the actions and combination of actions as specified in AS/NZS 1170.0.

### 1.6.2 Structural analysis and design

Structural analysis and design shall be in accordance with AS/NZS 1170.0.

NOTE: Guidance on the applicability of elastic structural analysis to continuous beams and frames is given in Appendix B.

### 1.6.3 Design capacity

The design capacity ( $R_d$ ) shall be determined by any one of the following:

- The nominal capacity ( $R_u$ ) in accordance with Sections 2 to 7 and the capacity reduction factor ( $\phi$ ) given in Table 1.6 as appropriate, i.e.,  $R_d = \phi R_u$ .
- Testing in accordance with Clause 8.2.3.

- (c) Where the composition or configuration of such components is such that Item (a) or (b) cannot be made in accordance with those provisions, structural performance shall be established from the design capacity or stiffness by rational engineering analysis based on appropriate theory, related testing if data is available and engineering judgement. Specifically, the design capacity shall be determined from the calculated nominal capacity by applying the following capacity reduction factors:
- (i) For members .....  $\phi = 0.80$ .
  - (ii) For connections .....  $\phi = 0.65$ .

#### 1.6.4 Earthquake design

##### 1.6.4.1 For Australia

All structures shall be designed for the actions and combination of actions specified in AS 1170.4. If cold-formed steel members are used as the primary earthquake resistance element, then the structural response factor ( $R_f$ ) shall be less than or equal to 2.0, unless specified otherwise.

##### 1.6.4.2 For New Zealand

###### 1.6.4.2.1 General

All structures shall be designed for the actions specified in NZS 1170.5 and combination of actions specified in AS/NZS 1170.0, subject to the limitations specified in Clauses 1.6.4.2.2 to 1.6.4.2.4.

###### 1.6.4.2.2 Structural ductility factor

For the ultimate limit state, the structural ductility factor ( $\mu$ ) shall be taken as follows:

- (a) For seismic-resisting systems involving an assemblage of elements acting as a single unit,  $\mu$  shall be less than or equal to 1.25.
- (b) For seismic-resisting systems using semi-rigid connections,  $\mu$  shall be less than or equal to 1.25.
- (c) Where a special study is undertaken (see Clause 1.6.4.2.3),  $\mu$  may be increased but shall not be greater than 4.0.
- (d) For all other earthquake-resisting systems,  $\mu = 1.0$ .

For the serviceability limit state,  $\mu = 1.0$ .

###### NOTES:

- 1 An example of an assemblage of elements is a braced wall panel, where the whole panel and its attachments at the top and base contribute to the earthquake resistance.
- 2 Earthquake resisting systems using semi-rigid connections cover frames with connections that are flexurally weaker than the members framing into the connection.

###### 1.6.4.2.3 Special studies

Where it is demonstrated by special study that  $\mu$  for a particular structural system is greater than 1.25, then—

- (a)  $\mu$  shall be based specifically from studies including the—
  - (i) structural form and configuration under consideration;
  - (ii) ductility of the material;
  - (iii) location of yielding regions of the structure;
  - (iv) structural damping characteristics involved in the structural system; and
  - (v) need to provide the structure with a small margin against collapse under the maximum considered event in accordance with NZS 1170.5.

- (b) where  $\mu$  greater than 1.25 is applicable to a design, then capacity design shall be used in order to protect elements of the earthquake resisting system from inelastic demands beyond their capability to dependably resist such demands; and
- (c) for buildings containing one or more suspended floors, capacity design principles shall be used to suppress inelastic demand in individual column members.

#### **1.6.4.2.4** *Structural performance factor*

When considering lateral stability of a whole structure, the structural performance factor ( $S_p$ ) shall be taken as 1.0.

For the ultimate limit state,  $S_p$  shall be taken as follows:

- (a) Where  $\mu$  is less than or equal to 2.0, but not less than 1.0—

$$S_p = 1.3 - 0.3\mu \quad \dots 1.6.4.2.3$$

- (b) Where  $\mu$  is greater than 2.0, then  $S_p = 0.70$ .

For the serviceability limit state,  $S_p = 0.70$ .

### **1.6.5 Durability**

#### **1.6.5.1** *General*

A structure shall be designed to perform its required functions during its expected life.

Where steelwork in a structure is to be exposed to a corrosive environment, the steelwork shall be given protection against corrosion. The degree of protection shall be determined after consideration has been given to the use of the structure, its maintenance and the climatic or other local conditions.

#### **1.6.5.2** *Corrosion protection*

NOTE: Corrosion protection should comply with AS/NZS 2311 and AS/NZS 2312, as appropriate. For further information, see Appendix C.



**TABLE 1.6**  
**CAPACITY REDUCTION FACTOR**

| Design capacity                                                                            | Clause reference | Capacity reduction factor ( $\phi$ ) |
|--------------------------------------------------------------------------------------------|------------------|--------------------------------------|
| (a) Stiffeners:                                                                            | 3.3.8            |                                      |
| Transverse stiffeners ( $\phi_c$ )                                                         | 3.3.8.1          | 0.85                                 |
| Bearing stiffeners ( $\phi_w$ )                                                            | 3.3.8.2          | 0.90                                 |
| Shear stiffeners ( $\phi_v$ )                                                              | 3.3.8.3          | 0.90                                 |
| (b) Members subject to axial tension ( $\phi_t$ )                                          | 3.2.1            | 0.90                                 |
| (c) Members subject to bending:                                                            | 3.3              |                                      |
| Section moment capacity—                                                                   | 3.3.2            |                                      |
| for sections with stiffened or partially stiffened compression flanges ( $\phi_b$ )        | 3.3.2            | 0.95                                 |
| for sections with unstiffened compression flanges ( $\phi_b$ )                             | 3.3.2            | 0.90                                 |
| Member moment capacity—                                                                    |                  |                                      |
| members subject to lateral buckling ( $\phi_b$ )                                           | 3.3.3.2          | 0.90                                 |
| members subject to distortional buckling ( $\phi_b$ )                                      | 3.3.3.3          | 0.90                                 |
| beams having one flange through-fastened to sheeting (channels or Z-sections) ( $\phi_b$ ) | 3.3.3.4          | 0.90                                 |
| Web design—                                                                                |                  |                                      |
| shear ( $\phi_v$ )                                                                         | 3.3.4            | 0.90                                 |
| Bearing ( $\phi_w$ )—                                                                      |                  |                                      |
| for built-up sections                                                                      | Table 3.3.6.2(A) | 0.75–0.90                            |
| for single web channel and channel-sections                                                | Table 3.3.6.2(B) | 0.75–0.90                            |
| for single web Z-sections                                                                  | Table 3.3.6.2(C) | 0.75–0.90                            |
| for single hat sections                                                                    | Table 3.3.6.2(D) | 0.75–0.90                            |
| for multiple web deck sections                                                             | Table 3.3.6.2(E) | 0.60–0.90                            |
| (d) Concentrically loaded compression members ( $\phi_c$ )                                 | 3.4              | 0.85                                 |
| (e) Combined axial load and bending:                                                       | 3.5              |                                      |
| Compression ( $\phi_c$ )                                                                   | 3.5.1            | 0.85                                 |
| Bending ( $\phi_b$ )—                                                                      | 3.5.1            |                                      |
| using Clause 3.3.2                                                                         |                  | 0.90 or 0.95                         |
| using Clause 3.3.3.1                                                                       |                  | 0.90                                 |
| (f) Cylindrical tubular members:                                                           | 3.6              |                                      |
| Bending ( $\phi_b$ )                                                                       | 3.6.2            | 0.95                                 |
| Compression ( $\phi_c$ )                                                                   | 3.6.3            | 0.85                                 |
| (g) Welded connections:                                                                    | 5.2              |                                      |
| Butt welds—                                                                                | 5.2.2            |                                      |
| tension or compression                                                                     | 5.2.2.1          | 0.90                                 |
| shear                                                                                      | 5.2.2.2(a)       | 0.80                                 |
| shear (base metal)                                                                         | 5.2.2.2(b)       | 0.90                                 |

(continued)

TABLE 1.6 (continued)

| Design capacity                             | Clause reference | Capacity reduction factor ( $\phi$ ) |
|---------------------------------------------|------------------|--------------------------------------|
| Fillet welds—                               | 5.2.3            |                                      |
| longitudinal loading                        | 5.2.3.2          | 0.55 or 0.60                         |
| transverse loading                          | 5.2.3.3          | 0.60                                 |
| Arc spot welds (puddle welds)—              | 5.2.4            |                                      |
| shear (welds)                               | 5.2.4.2(a)       | 0.60                                 |
| shear (connected part)                      | 5.2.4.2(b)       | 0.50 or 0.60                         |
| shear (minimum edge distance)               | 5.2.4.3          | 0.60 or 0.70                         |
| tension                                     | 5.2.4.4          | 0.65                                 |
| Arc seam welds—                             | 5.2.5            |                                      |
| shear (welds)                               | 5.2.5.2          | 0.60                                 |
| shear (connected part)                      | 5.2.5.2          | 0.60                                 |
| Flare welds—                                | 5.2.6            |                                      |
| transverse loading                          | 5.2.6.2(a)       | 0.55                                 |
| longitudinal loading                        | 5.2.6.2(b)       | 0.55                                 |
| Resistance welds—                           | 5.2.7            |                                      |
| spot welds                                  | 5.2.7(a)         | 0.65                                 |
| (h) Bolted connections:                     | 5.3              |                                      |
| Tearout                                     | 5.3.2            | 0.60 or 0.70                         |
| Net section tension:                        | 5.3.3            |                                      |
| With washers—                               | 5.3.3(a)         |                                      |
| double shear connection                     |                  | 0.65                                 |
| single shear connection                     |                  | 0.55                                 |
| Without washers                             | 5.3.3(b)         | 0.65                                 |
| Bearing                                     | 5.3.4            | 0.60 or 0.65                         |
| Bolts—                                      | 5.3.5            |                                      |
| bolt in shear                               | 5.3.5.1          | 0.80                                 |
| bolt in tension                             | 5.3.5.2          | 0.80                                 |
| (i) Screwed connections:                    | 5.4              |                                      |
| Screwed connections in shear—               | 5.4.2            |                                      |
| tension in the connected part               | 5.4.2.2          | 0.65                                 |
| tilting and hole bearing                    | 5.4.2.3          | 0.5                                  |
| tearout                                     | 5.4.2.4          | 0.60 or 0.70                         |
| Screwed connections in tension—             | 5.4.3            |                                      |
| pull-out of connected parts                 | 5.4.3.1          | 0.5                                  |
| pull-over (pull-through) of connected parts | 5.4.3.1          | 0.5                                  |

(continued)

**TABLE 1.6** *(continued)*

| <b>Design capacity</b>                   | <b>Clause reference</b> | <b>Capacity<br/>reduction factor<br/>(<math>\phi</math>)</b> |
|------------------------------------------|-------------------------|--------------------------------------------------------------|
| (j) Blind riveted connections:           | 5.5                     |                                                              |
| Riveted connections in shear—            | 5.5.2                   |                                                              |
| tension in the connected part            | 5.5.2.2                 | 0.65                                                         |
| tilting and hole bearing                 | 5.5.2.3                 | 0.50                                                         |
| tearout                                  | 5.5.2.4                 | 0.60 or 0.70                                                 |
| (k) Rupture:                             |                         |                                                              |
| Shear rupture                            | 5.6.1                   | 0.75                                                         |
| Block shear rupture (bolted connections) | 5.6.3                   | 0.65                                                         |

## SECTION 2      ELEMENTS

### 2.1 SECTION PROPERTIES

#### 2.1.1 General

Properties of sections, such as cross-sectional area, second moment of area, section modulus, radius of gyration, and centroid, shall be determined in accordance with conventional methods by division of the section shape into simple elements, including bends.

Properties shall be based on nominal dimensions and nominal base steel thickness (see Clause 1.5.1.6).

#### 2.1.2 Design procedures

##### 2.1.2.1 Full section properties

Properties of full, unreduced sections shall be based on the entire simplified shape with the flats and bends located along the element mid-lines unless the manufacturing process warrants consideration of a more accurate method.

To calculate the stability of members, a simplified shape where the bends are eliminated and the section is represented by straight mid-lines may be used when calculating the following properties:

- (a) Parameters for distortional buckling (see Appendix D).
- (b) Location of shear centre (see Paragraph E1 of Appendix E).
- (c) Warping constant (see Paragraph E1 of Appendix E).
- (d) Monosymmetry section constant (see Paragraph E2 of Appendix E).

##### 2.1.2.2 Effective section properties

For the design of cold-formed members with slender elements, the area of the sections shall be reduced at specified locations.

The reduction of the area is required to—

- (a) compensate for the effects of shear lag (see Clause 2.1.3.3); and
- (b) compensate for local instabilities of elements in compression (see Clauses 2.2 to 2.5).

##### 2.1.2.3 Location of reduced width

The location of reduced width shall be determined as follows:

- (a) For the design of uniformly compressed stiffened elements, the location of the lost portion shall be taken at the middle of the element (see Figures 2.2.1 and 2.4.2(b)).
- (b) For the design of stiffened elements under a stress gradient or where only a part of the element is in compression (e.g., the webs), the location of the lost portion shall be as shown in Figure 2.2.3.
- (c) For unstiffened elements, under either a stress gradient or uniform compression, the lost portion shall be taken at the unstiffened edge as shown in Figure 2.3.1. If the unstiffened element is subjected to both tension and compression across its width, the lost portion may be taken as specified in Clause 2.3.2.
- (d) For the design of elements with an edge stiffener, the location of the lost portion shall be as shown in Figure 2.4.2.

### 2.1.3 Dimensional limits

#### 2.1.3.1 Maximum flat-width-to-thickness ratios

The maximum overall flat-width-to-thickness ratios ( $b/t$ ), disregarding intermediate stiffeners and taking  $t$  as the nominal thickness of the element, shall be as follows:

- (a) For a stiffened compression element having one longitudinal edge connected to a web or flange element and the other stiffened by—
  - (i) simple lip .....60; and
  - (ii) any other kind of stiffener when—
    - (A)  $I_s < I_a$  .....60; and
    - (B)  $I_s \geq I_a$  ..... 90.
- (b) For a stiffened compression element with both longitudinal edges connected to other stiffened elements ..... 500.
- (c) For a unstiffened compression element..... 60.

NOTE: Unstiffened compression elements with  $b/t$  ratios greater than 30 and stiffened compression elements with  $b/t$  ratios greater than 250 are likely to develop noticeable deformation at the full design load, without affecting the ability of the member to carry the design load. Stiffened elements with  $b/t$  ratios greater than 500 can be used with adequate design capacity to sustain the design loads. However, substantial deformations of such elements usually will invalidate the design equations of this Standard.

#### 2.1.3.2 Flange curling

Where the flange of a flexural member is unusually wide and it is desired to limit the maximum amount of curling or movement of the flange toward the neutral axis, the maximum width ( $b_1$ ) of the compression and tension flanges, either stiffened or unstiffened projecting beyond the web for I-beams and similar sections or the maximum half distance ( $b_1$ ) between webs for box- or U-type beams, shall be determined from the following equation:

$$b_1 = \sqrt{\frac{0.061t_f d E}{f_{av}^*}} \sqrt[4]{\frac{100c_f}{d}} \quad \dots 2.1.3.2$$

where

$t_f$  = thickness of the flange

$d$  = depth of the section

$f_{av}^*$  = average design stress in the full, unreduced flange width (see Note 1)

$c_f$  = amount of curling (see Note 2)

NOTES:

- 1 Where members are designed by the effective design width procedure, the average stress equals the maximum stress multiplied by the ratio of the effective design width to the actual width.
- 2 The amount of curling that can be tolerated will vary with different kinds of sections and should be established by the designer. Amount of curling in the order of 5% of the depth of the section is usually not considered excessive.

#### 2.1.3.3 Shear lag effects (usually short spans supporting concentrated loads)

Where the span of the beam ( $l$ ) is less than  $30b_1$  and the beam carries one concentrated load, or several loads spaced greater than  $2b_1$ , the effective design width of any flange, whether in tension or compression, shall be limited to the values given in Table 2.1.3.3.

For flanges of I-beams and similar sections stiffened by lips at the outer edges,  $b_1$  shall be taken as the sum of the flange projection beyond the web and the depth of the lip.

**TABLE 2.1.3.3**  
**MAXIMUM RATIO OF EFFECTIVE**  
**DESIGN WIDTH TO ACTUAL WIDTH**  
**FOR SHORT WIDE FLANGE BEAMS**

| $l/b_1$ | Ratio | $l/b_1$ | Ratio |
|---------|-------|---------|-------|
| 30      | 1.00  | 14      | 0.82  |
| 25      | 0.96  | 12      | 0.78  |
| 20      | 0.91  | 10      | 0.73  |
| 18      | 0.89  | 8       | 0.67  |
| 16      | 0.86  | 6       | 0.55  |

NOTE:

$l$  = full span for simple beams; *or*  
distance between inflection points for continuous beams;  
*or*  
twice the length of cantilever beams

#### 2.1.3.4 Maximum web depth-to-thickness ratio

The maximum web depth-to-thickness ratio ( $d_1/t_w$ ) of flexural members shall not exceed the following:

- (a) For unreinforced webs:  $d_1/t_w$  ..... 200.
- (b) For webs with transverse stiffeners complying with Clause 3.3.8.1—
  - (i) if using bearing stiffeners only:  $d_1/t_w$  ..... 260; and
  - (ii) if using bearing stiffeners and intermediate stiffeners:  $d_1/t_w$  ..... 300;

where

$d_1$  = depth of the flat portion of the web measured along the plane of the web

$t_w$  = thickness of web

Where a web consists of two or more sheets, the ratio  $d_1/t_w$  shall be calculated for each sheet.

## 2.2 EFFECTIVE WIDTHS OF STIFFENED ELEMENTS

### 2.2.1 Uniformly compressed stiffened elements

#### 2.2.1.1 General

For uniformly compressed stiffened elements (see Figure 2.2.1), the effective widths for section or member capacity and deflection calculations shall be determined in accordance with Clauses 2.2.1.2 and 2.2.1.3 respectively.

#### 2.2.1.2 Effective width for capacity calculations

For determining the section or member capacity, the effective widths ( $b_e$ ) of uniformly compressed stiffened elements shall be determined from Equation 2.2.1.2(1) or Equation 2.2.1.2(2), as appropriate.

For  $\lambda \leq 0.673$ :  $b_e = b$  ..... 2.2.1.2(1)

For  $\lambda > 0.673$ :  $b_e = \rho b$  ..... 2.2.1.2(2)

where

$b$  = flat width of element excluding radii (see in Figure 2.2.1(a))

$\rho$  = effective width factor

$$= \frac{\left(1 - \frac{0.22}{\lambda}\right)}{\lambda} \leq 1.0 \quad \dots 2.2.1.2(3)$$

The slenderness ratio ( $\lambda$ ) shall be determined as follows:

$$\lambda = \left( \sqrt{\frac{f^*}{f_{cr}}} \right) \quad \dots 2.2.1.2(4)$$

where

$f^*$  = design stress in the compression element calculated on the basis of the effective design width (see Figure 2.2.1(b))

$f_{cr}$  = plate elastic buckling stress

$$= \left( \frac{k\pi^2 E}{12(1-\nu^2)} \right) \left( \frac{t}{b} \right)^2 \quad \dots 2.2.1.2(5)$$

$k$  = plate buckling coefficient

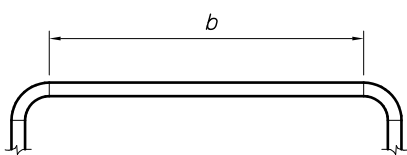
= 4 for stiffened elements supported by a web on each longitudinal edge ( $k$  values for different types of elements are given in the applicable clauses)

$E$  = Young's modulus of elasticity ( $200 \times 10^3$  MPa)

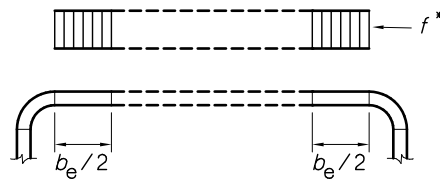
$\nu$  = Poisson's ratio = 0.3

$t$  = thickness of the uniformly compressed stiffened elements

Alternatively, the plate buckling coefficient ( $k$ ) for each flat element may be determined from a rational elastic buckling analysis of the whole section as a plate assemblage subjected to the longitudinal stress distribution in the section prior to buckling.



(a) Actual element



(b) Effective width ( $b_e$ ) of element and design stress ( $f^*$ ) on effective element

FIGURE 2.2.1 STIFFENED ELEMENTS WITH UNIFORM COMPRESSION

For determining the nominal section or member capacity of flexural members, the design stress ( $f^*$ ) shall be taken as follows:

- If the nominal section moment capacity ( $M_s$ ) is based on initiation of yielding as specified in Clause 3.3.2.2, and the initial yielding of the element being considered is in compression, then  $f^*$  shall be equal to  $f_y$ . If the initial yielding of the section is in tension, then  $f^*$  of the element being considered shall be determined on the basis of the effective section at  $M_y$  (moment causing initial yield).

- (b) If the nominal section moment capacity ( $M_s$ ) is based on inelastic reserve capacity as specified in Clause 3.3.2.3, then  $f^*$  shall be the stress of the element being considered at  $M_s$ . The effective section shall be used to determine  $M_s$ .
- (c) If the nominal member moment capacity ( $M_b$ ) is based on lateral buckling as specified in Clause 3.3.3.2 or on distortional buckling as specified in Clause 3.3.3.3, then  $f^*$  shall be equal to  $f_c$  as described in Clauses 3.3.3.2 and 3.3.3.3 in determining  $Z_c$ .

For determining the nominal section or member compression capacity,  $f^*$  shall be taken as follows:

- (i) If the nominal section capacity ( $N_s$ ) of the member in compression is based on initiation of yielding as specified in Clause 3.4, then  $f^*$  shall be equal to  $f_y$ .
- (ii) If the nominal member capacity ( $N_c$ ) of the member in compression is based on flexural, torsional or flexural-torsional buckling as specified in Clause 3.4, then  $f^*$  shall be equal to  $f_n$ , as specified in Clauses 3.4.1 and 3.4.6.

### 2.2.1.3 Effective width for deflection calculations

For determining the deflection, the effective widths ( $b_{ed}$ ) shall be determined from Equation 2.2.1.3(1) or Equation 2.2.1.3(2), as appropriate.

$$\text{For } \lambda \leq 0.673: b_{ed} = b \quad \dots 2.2.1.3(1)$$

$$\text{For } \lambda > 0.673: b_{ed} = \rho b \quad \dots 2.2.1.3(2)$$

The effective width factor ( $\rho$ ) shall be determined by either of the following two procedures:

- (a) *Procedure I* A low estimate of the effective width may be obtained from Equations 2.2.1.2(3) and 2.2.1.2(4), except that  $f_d^*$  is substituted for  $f^*$  where  $f_d^*$  is the design compressive stress in the element being considered based on the effective section at the load for which deflections are determined.
- (b) *Procedure II* For stiffened elements supported by a web on each longitudinal edge, an improved estimate of the effective width can be obtained by calculating  $\rho$  from Equations 2.2.1.3(3) to 2.2.1.3(5), as appropriate.

$$\text{A1} \quad \left| \quad \text{For } \lambda \leq 0.673: \quad \rho = 1 \quad \dots 2.2.1.3(3) \right.$$

$$\text{For } 0.673 < \lambda < \lambda_c: \quad \rho = \frac{1.358 - \frac{0.461}{\lambda}}{\lambda} \leq 1.0 \quad \dots 2.2.1.3(4)$$

$$\text{A1} \quad \left| \quad \text{For } \lambda \geq \lambda_c: \quad \rho = \frac{0.41 + 0.59 \sqrt{\frac{f_y}{f_d^*}} - \frac{0.22}{\lambda}}{\lambda} \leq 1.0 \quad \dots 2.2.1.3(5) \right.$$

$$\lambda_c = 0.256 + 0.328 \left( \frac{b}{t} \right) \sqrt{\frac{f_y}{E}} \quad \dots 2.2.1.3(6)$$

where  $\lambda$  shall be calculated from Equation 2.2.1.2(4) except that  $f_d^*$  is substituted for  $f^*$ .

## 2.2.2 Uniformly compressed stiffened elements with circular holes

### 2.2.2.1 General

For uniformly compressed stiffened elements with circular holes, the effective widths for section or member capacity and deflection calculations shall be determined in accordance with Clauses 2.2.2.2 and 2.2.2.3, respectively.



### 2.2.2.2 Effective width for capacity calculations

For determining the section or member capacity, where  $0.50 \geq d_h/b \geq 0$  and  $b/t \leq 70$ , and the centre-to-centre spacing of holes  $>0.5b$  and  $>3d_h$ , the effective width ( $b_e$ ) of uniformly compressed stiffened elements with circular holes shall be determined from Equation 2.2.2.1(1) or Equation 2.2.2.2(2), as appropriate.

$$\text{For } \lambda \leq 0.673: \quad b_e = b - d_h \quad \dots 2.2.2.2(1)$$

$$\text{For } \lambda > 0.673: \quad b_e = \frac{b \left( 1 - \frac{0.22}{\lambda} - \frac{0.8d_h}{b} \right)}{\lambda} \leq b - d_h \quad \dots 2.2.2.2(2)$$

where  $d_h$  is the diameter of holes and  $\lambda$  shall be calculated in accordance with Clause 2.2.1.2.

The value of  $b_e$  shall not exceed  $(b - d_h)$ .

### 2.2.2.3 Effective width for deflection calculations

For determining the deflection, the effective width ( $b_{ed}$ ) shall be equal to  $b_e$  determined in accordance with Procedure I of Clause 2.2.1.3 except that  $f_d^*$  is substituted for  $f^*$  where  $f_d^*$  is the design compressive stress of the element being considered, based on the effective section at the load for which deflections are determined.

## 2.2.3 Stiffened elements with stress gradient

### 2.2.3.1 General

For stiffened elements with stress gradient (see Figure 2.2.3), the effective widths for section or member capacity and deflection calculations shall be determined in accordance with Clauses 2.2.3.2 and 2.2.3.3, respectively.

### 2.2.3.2 Effective width for capacity calculations

For determining the section or member capacity, the effective width ( $b_{e1}$ ) (see Figure 2.2.3) shall be determined from the following:

$$b_{e1} = \frac{b_e}{3 - \psi} \quad \dots 2.2.3.2(1)$$

The effective width ( $b_{e2}$ ) (see Figure 2.2.3) shall be determined from Equation 2.2.3.2(2) or Equation 2.2.3.2(3), as appropriate.

$$\text{For } \psi \leq -0.236: \quad b_{e2} = \frac{b_e}{2} \quad \dots 2.2.3.2(2)$$

where  $(b_{e1} + b_{e2})$  shall not exceed the compression portion of the web calculated on the basis of effective section.

$$\text{For } \psi > -0.236: \quad b_{e2} = b_e - b_{e1} \quad \dots 2.2.3.2(3)$$

where

$b_e$  = effective width determined in accordance with Clause 2.2.1.2 with  $f_1^*$  substituted for  $f^*$  and with  $k$  determined as follows:

$$k = 4 + 2(1 - \psi)^3 + 2(1 - \psi) \quad \dots 2.2.3.2(4)$$

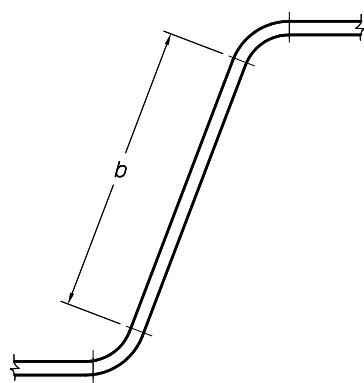
$$\psi = \frac{f_2^*}{f_1^*} \quad \dots 2.2.3.2(5)$$

$f_1^* / f_2^*$  = web stresses calculated on the basis of the effective section (see Figure 2.2.3)

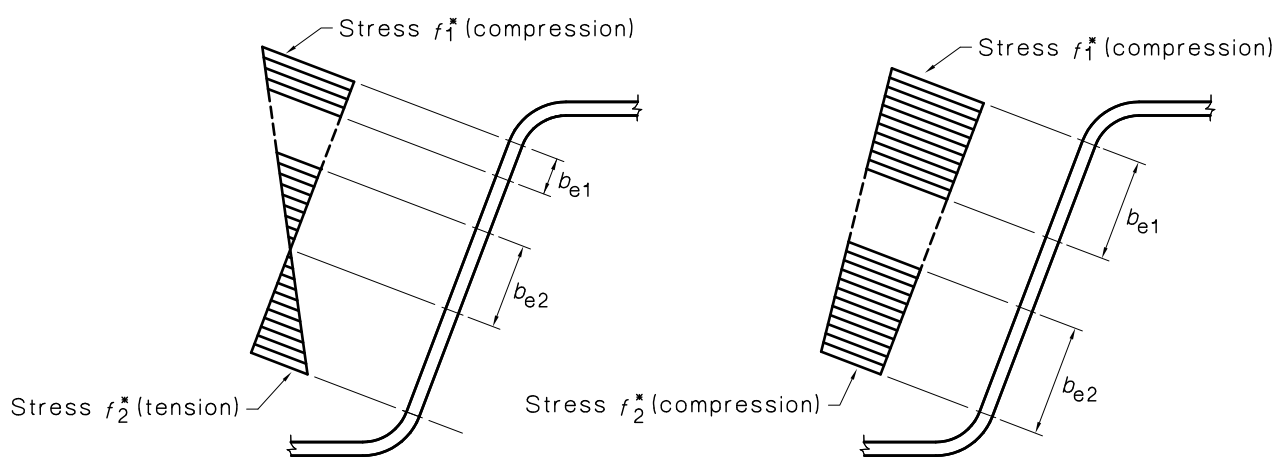
$f_1^*$  is compression (+) and  $f_2^*$  can be either tension (−) or compression (+). In case  $f_1^*$  and  $f_2^*$  are both compression,  $f_1^*$  shall be greater than or equal to  $f_2^*$ .

### 2.2.3.3 Effective width for deflection calculations

For determining the deflection, the effective widths ( $b_{e1}$ ) and ( $b_{e2}$ ) shall be determined in accordance with Clause 2.2.3.2 except that  $f_{d1}^*$  and  $f_{d2}^*$  are substituted for  $f_1^*$  and  $f_2^*$ . The calculated stresses  $f_1^*$  and  $f_2^*$  (see Figure 2.2.3) shall be used to determine  $f_{d1}^*$  and  $f_{d2}^*$ , respectively. Calculations shall be based on the effective section for the load for which deflections are determined.



(a) Actual element



(b) Effective element and stress on effective element

FIGURE 2.2.3 STIFFENED ELEMENTS AND WEBS WITH STRESS GRADIENT

## 2.2.4 Channel-section webs with holes and with stress gradient

### 2.2.4.1 General

The calculation of capacity and deflection for channel-section webs with holes and with stress gradient shall be applicable within the following limits:

- (a)  $d_{wh}/d_1 < 0.7$  . . . 2.2.4.1

where

$d_{wh}$  = depth of the web hole

$d_1$  = depth of the flat portion of the web measured along the plane of the web

- (b)  $d_1/t \leq 200$ .  
 (c) Holes centred at mid-depth of the web.  
 (d) Clear distance between holes is greater than or equal to 450 mm.  
 (e) Non-circular holes corner radii greater than or equal to  $2t$ .  
 (f) Non-circular holes with  $d_{wh} \leq 65$  mm and  $b \leq 115$  mm, where  $b$  is the length of the web hole.  
 (g) Circular hole diameters less than or equal to 150 mm.  
 (h)  $d_{wh} > 15$  mm.

### 2.2.4.2 Capacity calculations

When  $d_{wh}/d_1 < 0.38$ , the effective widths ( $b_1$ ) and ( $b_2$ ) shall be determined in accordance with Clause 2.2.3 by assuming no hole exists in the web.

When  $d_{wh}/d_1 \geq 0.38$ , the effective width shall be determined in accordance with Clause 2.3.1 assuming the compression portion of the web consists of an unstiffened element adjacent to the hole with  $f^* = f_1$  as shown in Figure 2.3.2.

### 2.2.4.3 Deflection calculations

The effective widths shall be determined in accordance with Clause 2.2.3 by assuming no hole exists in the web.

## 2.3 EFFECTIVE WIDTHS OF UNSTIFFENED ELEMENTS

### 2.3.1 Uniformly compressed unstiffened elements

#### 2.3.1.1 General

For uniformly compressed unstiffened elements (see Figure 2.3.1), the effective widths for section or member capacity and deflection calculations shall be determined in accordance with Clauses 2.3.1.2 and 2.3.1.3, respectively.

#### 2.3.1.2 Effective width for capacity calculations

For determining the section or member capacity, the effective widths ( $b_e$ ) of uniformly compressed unstiffened elements shall be determined in accordance with Clause 2.2.1.2 with the exception that  $k$  shall be taken as 0.43 and  $b$  shall be as shown in Figure 2.3.1.

#### 2.3.1.3 Effective width for deflection calculations

For determining the deflection, the effective widths ( $b_e$ ) shall be determined in accordance with Procedure I of Clause 2.2.1.3 except that  $f_d^*$  is substituted for  $f^*$  and  $k = 0.43$ .

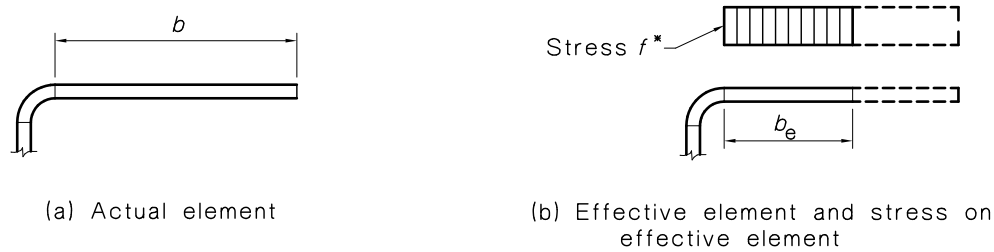


FIGURE 2.3.1 UNSTIFFENED ELEMENT WITH UNIFORM COMPRESSION

## 2.3.2 Unstiffened elements and edge stiffeners with stress gradient

### 2.3.2.1 General

For unstiffened elements and edge stiffeners with stress gradient, the effective widths for section or member capacity and deflection calculations shall be determined in accordance with Clauses 2.3.2.2 and 2.3.2.3, respectively.

### 2.3.2.2 Effective width for capacity calculations

For determining the section or member capacity, the effective widths ( $b_e$ ) measured from the supported edge of unstiffened compression elements and edge stiffeners with stress gradient shall be determined in accordance with Clause 2.2.1.2 with  $f^* = f_1^*$  and with  $k$  and  $\rho$  determined in accordance with this Clause.

$f_1^*, f_2^*$  = stresses shown in Figures 2.3.2(A) and (B) calculated on the basis of the gross section where  $f_1^*$  is compression (+) and  $f_2^*$  can be either tension (−) or compression (+). In the case where  $f_1^*$  and  $f_2^*$  are both in compression  $f_1^* \geq f_2^*$

$\psi$  = stress ratio

$$= f_2^* / f_1^* \quad \dots 2.3.2.2(1)$$

The effective width factor ( $\rho$ ) and the plate buckling coefficient ( $k$ ) shall be determined as follows:

- (a) For unstiffened elements with stress gradient causing compression at both longitudinal edges of the unstiffened element ( $f_1^*$  and  $f_2^*$ ) both in compression, as shown in Figure 2.3.2(A).

$\rho$  shall be determined using Equation 2.2.1.2(3) and  $\lambda$  shall be determined using Equation 2.2.1.2(4). The buckling coefficient ( $k$ ) in Equation 2.2.1.2(5) shall be determined as follows:

- (i) Where the stress decreases toward the unstiffened edge of the element as shown in Figure 2.3.2(A)(a),  $k$  shall be calculated as follows:

$$k = \frac{0.578}{\psi + 0.34} \quad \dots 2.3.2.2(2)$$

- (ii) Where the stress increases toward the unstiffened edge of the element as shown in Figure 2.3.2(A)(b),  $k$  shall be calculated as follows:

$$k = 0.57 - 0.21\psi + 0.07\psi^2 \quad \dots 2.3.2.2(3)$$

- (b) For unstiffened elements with stress gradient causing compression at one longitudinal edge and tension at the other longitudinal edge of the unstiffened element:

- (i) For  $f_1^*$  in compression at the unsupported edge and  $f_2^*$  in tension as shown in Figure 2.3.2(B)(a),  $\rho$  shall be calculated as follows:

$$\rho = 1 \quad \text{for } \lambda \leq 0.673(1 - \psi)$$

$$\rho = (1 - \psi) \frac{1 - \frac{0.22(1 - \psi)}{\lambda}}{\lambda} \quad \text{for } \lambda > 0.673(1 - \psi) \quad \dots 2.3.2.2(4)$$

$\lambda$  shall be determined using Equation 2.2.1.2(4)

$$k = 0.57 - 0.21\psi + 0.07\psi^2 \quad \dots 2.3.2.2(5)$$

- (ii) For  $f_1^*$  in compression at the supported edge and  $f_2^*$  in tension as shown in Figure 2.3.2(B)(b),  $\rho$  shall be calculated as follows:

$$\text{For } -1 \leq \psi < 0: \quad \rho = 1 \quad \text{for } \lambda \leq 0.673$$

$$\rho = (1 + \psi) \frac{\left(1 - \frac{0.22}{\lambda}\right)}{\lambda} - \psi \quad \text{for } \lambda > 0.673 \quad \dots 2.3.2.2(6)$$

$\lambda$  shall be determined using Equation 2.2.1.2(4)

$$k = 1.70 - 5\psi + 17.1\psi^2 \quad \dots 2.3.2.2(7)$$

$$\text{For } \psi \leq -1: \quad \rho = 1$$

Alternatively, the plate buckling coefficient ( $k$ ) in Equation 2.3.2.2(5) may be determined using Equation 2.3.2.2(8) for plain channels bent in the plane of symmetry with the unsupported edge of the unstiffened element in compression as follows:

$$k = 0.1451(b_2/b_1) + 1.256 \quad \dots 2.3.2.2(8)$$

where

$b_2$  = width of the unstiffened element

$b_1$  = width of the stiffened element

For other types of sections,  $k$  in Equations 2.3.2.2(2), 2.3.2.2(3), 2.3.2.2(5) and 2.3.2.2(7) for the unstiffened element and  $k$  for each remaining flat element of the section may be determined from a rational elastic buckling analysis of the whole section as a plate assemblage subjected to the longitudinal stress distribution in the section prior to buckling.

In calculating the effective section modulus ( $Z_e$ ) in Clause 3.3.2.2 or in Clause 3.3.3.2, the extreme compression fibre in Figures 2.3.2(A)(b) and 2.3.2(B)(a) is taken as the edge of the effective section (closer to the unsupported edge).

In calculating the effective section modulus ( $Z_e$ ) in Clause 3.3.2.2, the extreme tension fibre in Figure 2.3.2(B)(b) is taken as the edge of the effective section (closer to the unsupported edge).

A1

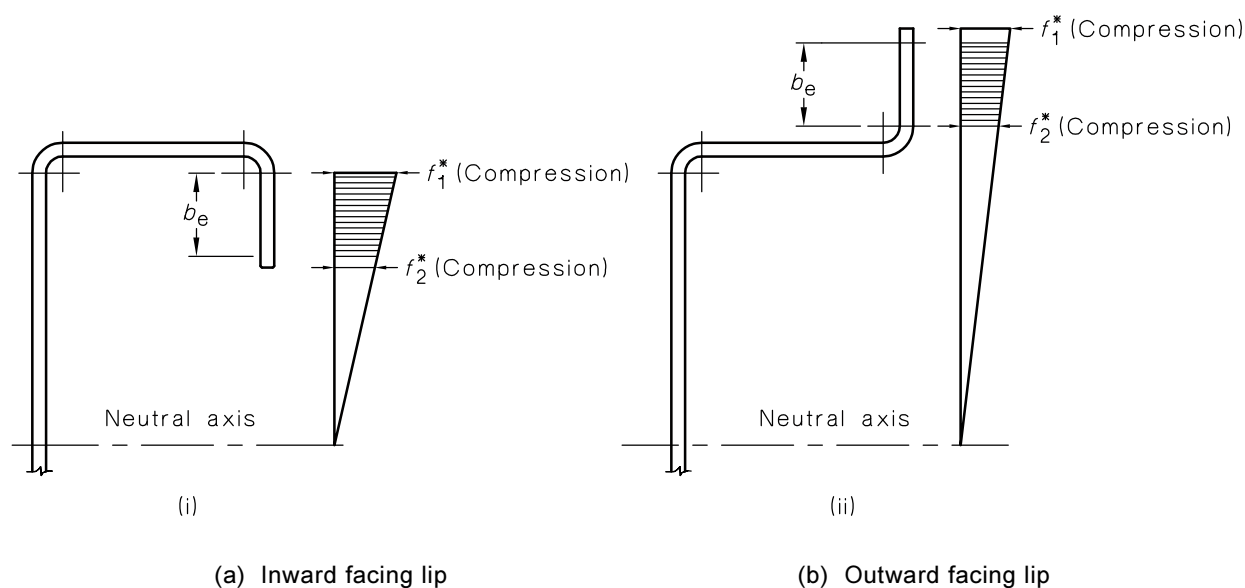


FIGURE 2.3.2(A) UNSTIFFENED ELEMENTS WITH STRESS GRADIENT—  
BOTH EDGES IN COMPRESSION

A1

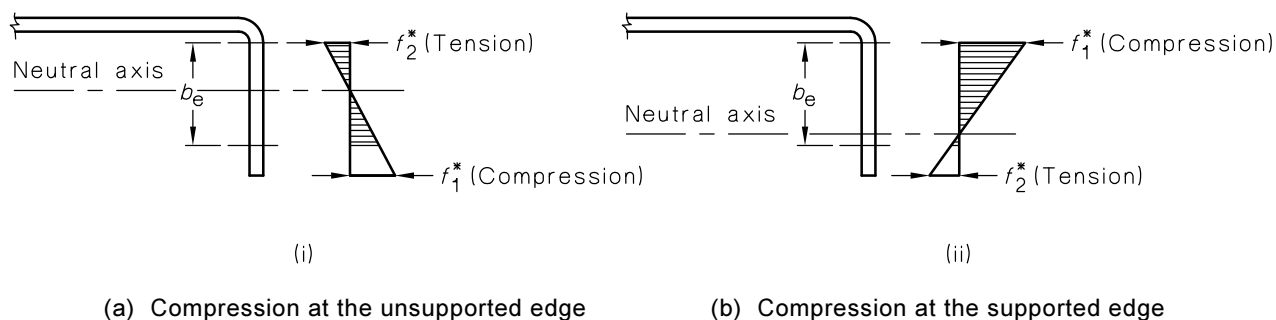


FIGURE 2.3.2(B) UNSTIFFENED ELEMENTS WITH STRESS GRADIENT—  
ONE EDGE IN COMPRESSION AND ONE EDGE IN TENSION

### 2.3.2.3 Effective width for deflection calculations

For determining the deflection, the effective widths ( $b_e$ ) of unstiffened elements and edge stiffeners with stress gradient shall be determined in accordance with Clause 2.3.2.2. except that  $f_{d1}^*$  and  $f_{d2}^*$  are substituted for  $f_1^*$  and  $f_2^*$ . The calculated stresses  $f_1^*$  and  $f_2^*$  (see Figures 2.3.2(A) and 2.3.2(B)) shall be used to determine  $f_{d1}^*$  and  $f_{d2}^*$  respectively. Calculations shall be based on the effective section for the load for which deflections are determined.

## 2.4 EFFECTIVE WIDTHS OF UNIFORMLY COMPRESSED ELEMENTS WITH AN EDGE STIFFENER

### 2.4.1 General

For uniformly compressed elements with an edge stiffener, the effective widths for section or member capacity and deflection calculations shall be determined in accordance with Clauses 2.4.2 and 2.4.3 respectively.

### 2.4.2 Effective width for capacity calculations

For determining the section or member capacity, the effective widths ( $b_e$ ) of uniformly compressed elements with an edge stiffener shall be determined as follows:

$$(a) \quad \frac{b}{t} \leq 0.328S$$

A1

$$I_a = 0 \text{ (no edge stiffener is required)}$$

= adequate second moment of area of the stiffener, so that each component element behaves as a stiffened element

$$b_e = b \quad \dots 2.4.2(1)$$

$$b_1 = b_2 = b/2 \quad \text{(see Figure 2.4.2)} \quad \dots 2.4.2(2)$$

$$d_s = d_{se} \quad \text{(for simple lip stiffener)} \quad \dots 2.4.2(3)$$

$$A_s = A_{se} \quad \text{(for other stiffener shapes)} \quad \dots 2.4.2(4)$$

$$(b) \quad \frac{b}{t} > 0.328S$$

$$b_1 = \frac{b_e}{2} \left( \frac{I_s}{I_a} \right) \quad \text{(see Figure 2.4.2)} \quad \dots 2.4.2(5)$$

$$b_2 = b_e - b_1 \quad \text{(see Figure 2.4.2)} \quad \dots 2.4.2(6)$$

$$d_s = d_{se} \left( \frac{I_s}{I_a} \right) \quad \text{(for simple lip stiffener)} \quad \dots 2.4.2(7)$$

$$A_s = A_{se} \left( \frac{I_s}{I_a} \right) \quad \text{(for other stiffener shapes)} \quad \dots 2.4.2(8)$$

A1

$$A_{se} = d_{se} t \quad \text{(for stiffener shown in Figure 2.4.2)} \quad \dots 2.4.2(9)$$

$$I_s = \frac{d^3 t \sin^2 \theta}{12} \quad \text{(for stiffener shown in Figure 2.4.2)} \quad \dots 2.4.2(10)$$

$$I_a = 399t^4 \left[ \left( \frac{b/t}{S} - 0.328 \right)^3 \leq t^4 \left[ 115 \frac{(b/t)}{S} + 5 \right] \right] \quad \dots 2.4.2(11)$$

If  $I_s$  is greater than or equal to  $I_a$ , then  $I_s$  is equal to  $I_a$  in Equation 2.4.2(5), (7), (8) and Table 2.4.2.

$$n = \left[ 0.582 - \frac{(b/t)}{4S} \right] \geq \frac{1}{3} \quad \dots 2.4.2(12)$$

$S$  = slenderness factor

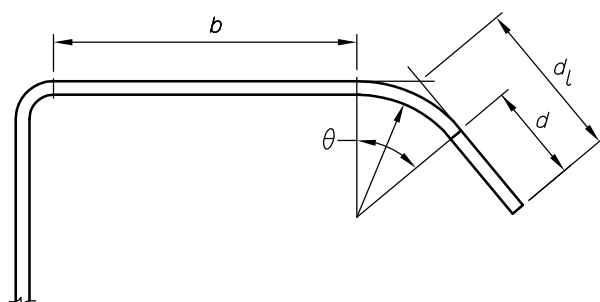
$$= 1.28 \sqrt{E / f^*} \quad \dots 2.4.2(13)$$

$f^*$  = stress (see Figure 2.4.2(b))

$b_e$  shall be calculated in accordance with Clause 2.2.1.2, where  $k$  shall be as given in Table 2.4.2.

**TABLE 2.4.2**  
**DETERMINATION OF PLATE BUCKLING COEFFICIENT ( $k$ )**

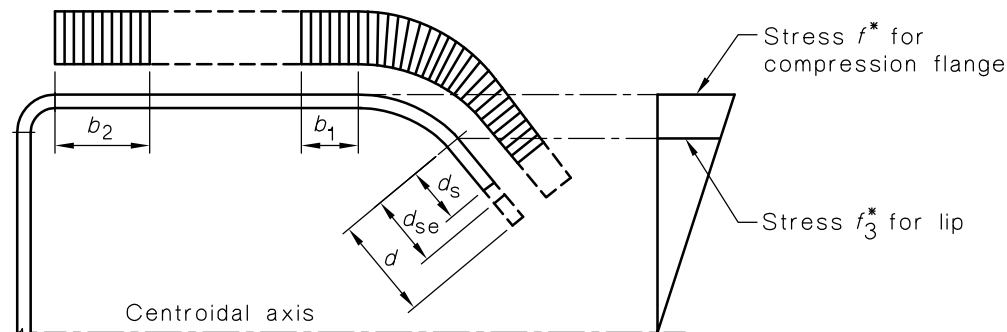
| Plate buckling coefficient ( $k$ )                                  |                                                                                       |                                                       |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|
| Simple lip edge stiffener ( $140^\circ \geq \theta \geq 40^\circ$ ) |                                                                                       | Other edge stiffener shapes                           |
| $d_l/b \leq 0.25$                                                   | $0.25 < d_l/b \leq 0.8$                                                               |                                                       |
| $3.57 \left( \frac{I_s}{I_a} \right)^n + 0.43 \leq 4$               | $\left( 4.82 - \frac{5d_l}{b} \right) \left( \frac{I_s}{I_a} \right)^n + 0.43 \leq 4$ | $3.57 \left( \frac{I_s}{I_a} \right)^n + 0.43 \leq 4$ |



where

$d_l, d$  = actual stiffener dimensions

(a) Actual and effective stiffener



where

$d_{se}$  = effective width of the stiffener calculated in accordance with Clause 2.3.2.2

$d_s$  = reduced effective width of stiffener

(b) Effective element and stress on effective element

**FIGURE 2.4.2 ELEMENTS WITH SIMPLE-LIP EDGE STIFFENER**

### 2.4.3 Effective width for deflection calculations

For determining the deflection, the effective widths ( $b_e$ ) shall be determined in accordance with Clause 2.4.2, except that  $f_d^*$  is substituted for  $f^*$ .



## 2.5 EFFECTIVE WIDTHS OF UNIFORMLY COMPRESSED STIFFENED ELEMENTS WITH ONE INTERMEDIATE STIFFENER

### 2.5.1 General

For uniformly compressed elements with an intermediate stiffener (see Figure 2.5.2), the effective widths for section or member capacity and deflection calculations shall be determined in accordance with Clauses 2.5.2 and 2.5.3 respectively.

### 2.5.2 Effective width for capacity calculations

For determining the section or member capacity, the effective widths ( $b_e$ ) of uniformly compressed elements with one intermediate stiffener shall be determined as follows:

$$(a) \quad \frac{b_2}{t} \leq S$$

$$I_a = 0 \text{ (no intermediate stiffener is required)}$$

= adequate second moment of area of the stiffener, so that each component element behaves as a stiffened element

$$b_e = b \quad \dots 2.5.2(1)$$

$b$  = flat width of element excluding corners or bends (see Figure 2.5.2)

$$A_s = \text{reduced area of the stiffener}$$

$$= A_{se} \quad \dots 2.5.2(2)$$

$A_{se}$  = effective area of the stiffener

$A_s$  shall be used in calculating the overall effective section properties. The centroid of the stiffener shall be considered located at the centroid of the full area of the stiffener, and the second moment of area of the stiffener about its own centroidal axis shall be that of the full section of the stiffener.

$$(b) \quad \frac{b_2}{t} > S$$

$$A_s = A_{se} \left( \frac{I_s}{I_a} \right) \quad \dots 2.5.2(3)$$

$n$  = exponent

$$= \left[ 0.583 - \frac{(b_2/t)}{12S} \right] \geq \frac{1}{3} \quad \dots 2.5.2(4)$$

$k$  = plate buckling coefficient

$$= 3 \left( \frac{I_s}{I_a} \right)^n + 1 \quad \dots 2.5.2(5)$$

$$I_a = t^4 \left[ 50 \frac{(b_2/t)}{S} - 50 \right] \quad \text{for } S < \frac{b_2}{t} < 3S \quad \dots 2.5.2(6)$$

$$I_a = t^4 \left[ \frac{128(b_2/t)}{S} - 285 \right] \quad \text{for } \frac{b_2}{t} \geq 3S \quad \dots 2.5.2(7)$$

where

$b_2$  = flat width of element with intermediate stiffener excluding radii (see Figure 2.5.2(a))

$I_s$  = second moment of area of the full stiffener about its own centroidal axis parallel to the element to be stiffened

$S$  = slenderness factor

$$= 1.28 \sqrt{\frac{E}{f^*}} \quad \dots 2.5.2(8)$$

If  $I_s$  is greater than or equal to  $I_a$ , then  $I_s = I_a$  in Equations 2.5.2(3) and (5).

The effective width  $b_e$  shall be calculated in accordance with Clause 2.2.1.2, where  $k$  shall be as specified in this Clause.

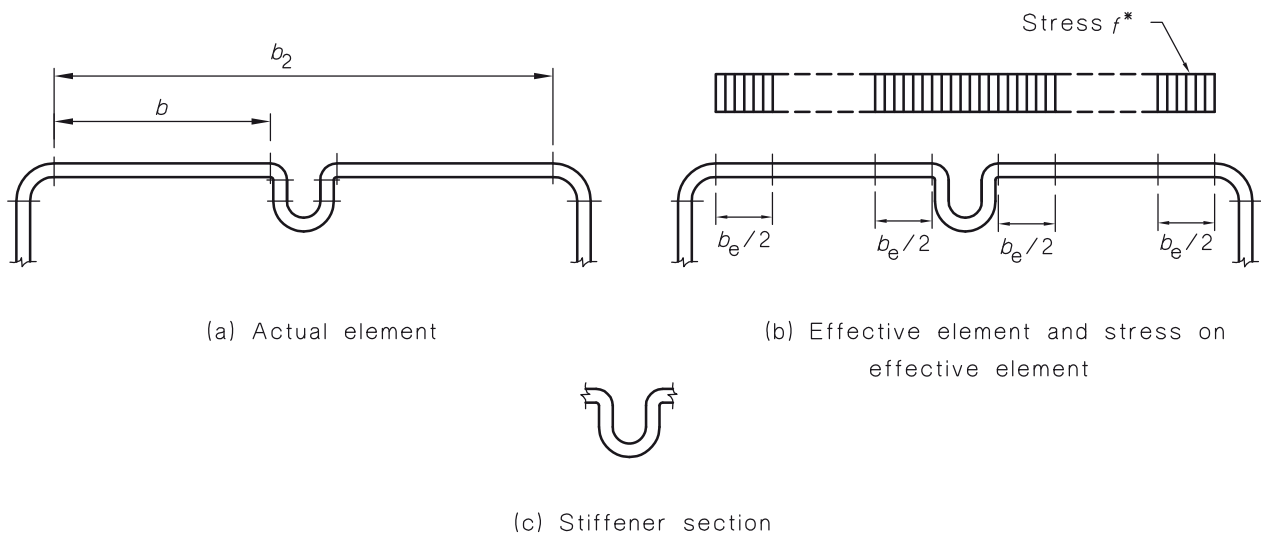


FIGURE 2.5.2 ELEMENTS WITH ONE INTERMEDIATE STIFFENER

The value of  $d_s$ , calculated in accordance with Clause 2.5.2, shall be used in calculating the overall effective section properties.

### 2.5.3 Effective width for deflection calculations

For determining the deflection, the effective widths ( $b_e$ ) shall be determined in accordance with Clause 2.5.2, except that  $f_d^*$  is substituted for  $f^*$ .

## 2.6 EFFECTIVE WIDTHS OF UNIFORMLY COMPRESSED STIFFENED ELEMENTS WITH MULTIPLE INTERMEDIATE STIFFENER

### 2.6.1 Determination of effective width

The effective width of the element shall be determined as follows:

$$b_e = \rho \left( \frac{A_g}{t} \right) \quad \dots 2.6.1(1)$$

where

$b_e$  = effective width of the element, located at the end of the element including stiffeners (see Figure 2.6.1(A))

$\rho$  = effective width factor

= 1 if  $\lambda \leq 0.673$

$$= \frac{\left(1 - \frac{0.22}{\lambda}\right)}{\lambda} \quad \text{if } \lambda > 0.673 \quad \dots 2.6.1(2)$$

$$\lambda = \frac{1.052}{\sqrt{k}} \left( \frac{b_o}{t} \right) \sqrt{\frac{f^*}{E}} \quad \dots 2.6.1(3)$$

$b_o$  = total flat width of the stiffened element (see Figure 2.6.1(B))

$A_g$  = gross area of the element including stiffeners

$t$  = thickness of element

The plate buckling coefficient ( $k$ ) shall be determined from the minimum of  $Rk_d$  and  $k_{loc}$ , determined in accordance with Clause 2.6.2 or Clause 2.6.3 as appropriate, where—

$$\begin{aligned} R &= \text{modification factor for the distortional plate buckling coefficient} \\ &= 2 \quad \text{if } b_o/d_1 < 1 \\ &= \frac{11 - (b_o/d_1)}{5} \geq \frac{1}{2} \quad \text{if } b_o/d_1 \geq 1 \end{aligned} \quad \dots 2.6.1(4)$$

$k_d$  = plate buckling coefficient for distortional buckling calculated using Equation 2.6.2.1(2)

$k_{loc}$  = plate buckling coefficient for local sub-element buckling calculated using Equation 2.6.2.1(1)

$d_1$  = width of elements adjoining the stiffened element, e.g., the depth of the web in a hat section with multiple intermediate stiffeners in the compression flange is equal to  $d_1$ ; if the adjoining elements have different widths, use the smallest one

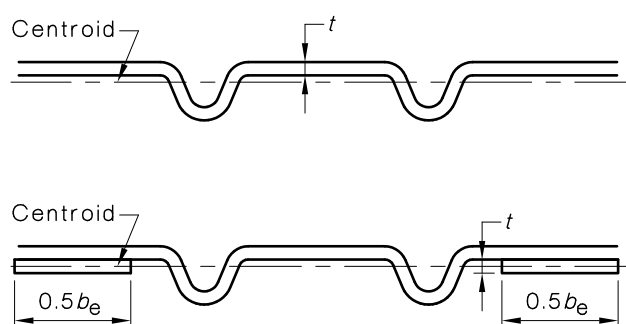


FIGURE 2.6.1(A) EFFECTIVE WIDTH LOCATIONS

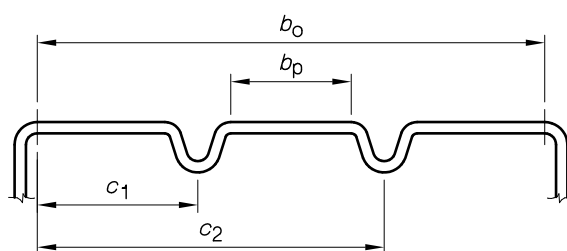


FIGURE 2.6.1(B) PLATE WIDTHS AND STIFFENER LOCATIONS

## 2.6.2 Specific case: 'n' identical stiffeners, equally spaced

### 2.6.2.1 Capacity calculation

$$k_{\text{loc}} = 4(n + 1)^2 \quad \dots 2.6.2.1(1)$$

$$k_d = \frac{(1 + \beta^2)^2 + \gamma(1 + n)}{\beta^2[1 + \delta(n + 1)]} \quad \dots 2.6.2.1(2)$$

$$\beta = [1 + \gamma(n + 1)]^{\frac{1}{4}} \quad \dots 2.6.2.1(3)$$

$$\gamma = \frac{10.92 I_{\text{sp}}}{b_o t^3} \quad \dots 2.6.2.1(4)$$

$$\delta = \frac{A_s}{b_o t} \quad \dots 2.6.2.1(5)$$

where

$\beta$  = coefficient

$\gamma$  = importance factor

$\delta$  = coefficient

$I_{\text{sp}}$  = second moment of area of a stiffener about the centre-line of the flat portion of the element. The radii which connect the stiffener to the flat may be included

$b_o$  = total flat width of the stiffened element (see Figure 2.6.1(B))

$A_s$  = gross area of the stiffener

If  $l_{\text{br}} < \beta b_a$  then  $l_{\text{br}}/b_o$  shall be permitted to be substituted for  $\beta$  to account for increased capacity due to bracing, where  $l_{\text{br}}$  is the unsupported length of bracing or other restraint that restricts distortional buckling of the element.

### 2.6.2.2 Deflection calculation

The effective width ( $b_e$ ) used in calculating deflection shall be determined in accordance with Clause 2.6.2.1, except that  $f_d^*$  shall be substituted for  $f^*$ , where  $f_d^*$  is the design compressive stress in the element being considered based on the effective section at the load for which deflections are determined.

## 2.6.3 General case: Arbitrary stiffener size, location and number

### 2.6.3.1 Capacity calculation

$$k_{\text{loc}} = 4 \left( \frac{b_o}{b_p} \right)^2 \quad \dots 2.6.3.1(1)$$

$$k_d = \frac{(1 + \beta^2)^2 + 2 \sum_{i=1}^n \gamma_i \omega_i}{\beta^2 \left( 1 + 2 \sum_{i=1}^n \gamma_i \omega_i \right)} \quad \dots 2.6.3.1(2)$$

$$\beta = \left( 2 \sum_{i=1}^n \gamma_i \omega_i + 1 \right)^{\frac{1}{4}} \quad \dots 2.6.3.1(3)$$

$$\gamma_i = \frac{10.92(I_{sp})_i}{b_o t^3} \quad \dots 2.6.3.1(4)$$

$$\omega_i = \sin^2 \left( \pi \frac{C_i}{b_o} \right) \quad \dots 2.6.3.1(5)$$

$$\delta_i = \frac{(A_s)_i}{b_o t} \quad \dots 2.6.3.1(6)$$

where

$b_p$  = greatest sub-element flat width (see Figure 2.6.1(B))

$\omega_i$  = coefficient

$C_i$  = horizontal distance from the edge of the element to the centre-line(s) of the stiffener(s) (see Figure 2.6.1(B))

$i$  = index for stiffener 'i'

If  $l_{br} < \beta b_o$  then  $l_{br}/b_o$  shall be permitted to be substituted for  $\beta$  to account for increased capacity due to bracing.

### 2.6.3.2 Deflection calculation

The effective width ( $b_e$ ) used in calculating deflection shall be determined in accordance with Clause 2.6.3.1, except that  $f_d^*$  shall be substituted for  $f^*$ , where  $f_d^*$  is the design compressive stress in the element being considered based on the effective section at the load for which deflections are determined.

## 2.7 EFFECTIVE WIDTHS OF UNIFORMLY COMPRESSED EDGE-STIFFENED ELEMENTS WITH INTERMEDIATE STIFFENERS

The effective width ( $b_e$ ) of uniformly compressed edge-stiffened elements with intermediate stiffeners shall be determined as follows:

- (a) If  $b_2/t \leq S/3$ , the element is fully effective and no local buckling reductions are required.
- (b) If  $b_2/t > S/3$ , the plate buckling coefficient ( $k$ ) shall be determined in accordance with this Clause but with  $b_2$  replacing  $b$  in all expressions,

where

$b_2$  = total flat width of the edge-stiffened element (see Figure 2.5.2)

$S$  = slenderness factor

If  $k$ , calculated in accordance with Clause 2.4.2, is less than 4 ( $k < 4$ ), the intermediate stiffeners shall be ignored and the provisions of Clause 2.4.2 shall be followed for the calculation of  $b_e$ .

If  $k$ , calculated in accordance with Clause 2.4.2, is equal to 4 ( $k = 4$ ),  $b_e$  of the edge-stiffened element shall be calculated in accordance with Clause 2.6, where the modification factor for the distortional plate buckling coefficient shall be less than or equal to 1.

## 2.8 ARCHED COMPRESSION ELEMENTS

A circular or parabolic arch-shaped compression element, stiffened at both ends, shall be considered self-stiffened and fully effective if the second moment of area of the arch about its own centroidal axis parallel to the base is not less than the minimum second moment of area ( $I_{\min.}$ ) specified in Clause 2.5. For the purpose of this Clause,  $b$  shall be taken as half the length of arch and the ratio  $b/t$  shall not exceed 60. For other conditions, the geometrical properties of sections shall be determined by load tests in accordance with Section 8.

## SECTION 3 MEMBERS

### 3.1 GENERAL

Section properties used for the determination of structural performance, moment capacity of beams or capacity of axial members in compression shall be those calculated in accordance with Section 2.

Both full and effective section properties, where applicable, shall be required. Full section properties shall be used for the determination of buckling moments or stresses. Effective section properties shall be used for the determination of section and member capacities.

### 3.2 MEMBERS SUBJECT TO AXIAL TENSION

#### 3.2.1 Design for axial tension

A member subject to a design axial tensile force ( $N^*$ ) shall satisfy—

$$N^* \leq \phi_t N_t \quad \dots 3.2.1$$

where

$\phi_t$  = capacity reduction factor for members in tension (see Table 1.6)

$N_t$  = nominal section capacity of the member in tension determined in accordance with Clause 3.2.2

#### 3.2.2 Nominal section capacity

The nominal section capacity of a member in tension shall be taken as the lesser of—

$$N_t = A_g f_y; \text{ and} \quad \dots 3.2.2(1)$$

$$N_t = 0.85 k_t A_n f_u \quad \dots 3.2.2(2)$$

where

$A_g$  = gross area of the cross-section

$f_y$  = yield stress used in design

$k_t$  = correction factor for distribution of forces determined in accordance with Clause 3.2.3.2

$A_n$  = net area of the cross-section, obtained by deducting from the gross area of the cross-section, the sectional area of all penetrations and holes, including fastener holes

$f_u$  = tensile strength used in design

#### 3.2.3 Distribution of forces

##### 3.2.3.1 End connections providing uniform force distribution

Where for design purposes it is assumed that the tensile force is distributed uniformly to a tension member, the end connections shall satisfy both the following:

- (a) The connections shall be made to each part of the member and shall be symmetrically placed about the centroidal axis of the member.
- (b) Each part of the connection shall be proportioned to transmit at least the maximum design force carried by the connected part of the member.

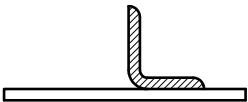
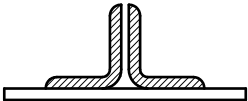
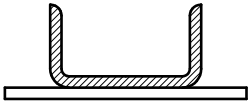
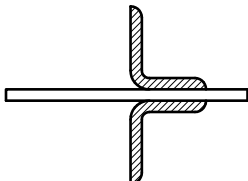
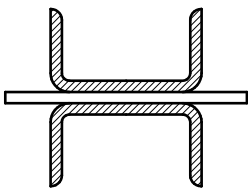
For connections satisfying these requirements, the value of  $k_t$  shall be taken as 1.0.

3.2.3.2 End connections providing non-uniform force distribution

If the end connections of a tension member do not comply with Clause 3.2.3.1, then the member shall be designed to comply with Clause 3.5 using  $k_t = 1.0$ , except that Clause 3.2.2 may be used for the following members:

- (a) *Eccentrically connected angles and channels* Eccentrically connected angles and channels may be designed in accordance with Clause 3.2.2, using the appropriate value of  $k_t$  given in Table 3.2.
- (b) *Channels connected by both flanges only* A symmetrical rolled or built-up member of channel-section connected by both flanges only may be designed in accordance with Clause 3.2.2 using a value of  $k_t$  equal to 0.85, provided—
  - (i) the length between the first and last rows of fasteners in the connection, or when the member is welded, the length of longitudinal weld provided to each side of the connected flanges is not less than the depth of the member; and
  - (ii) each flange connection is proportioned to transmit at least half of the maximum design force carried by the connected member.

TABLE 3.2  
CORRECTION FACTOR ( $k_t$ ) FOR SHADED ELEMENT

| Configuration case                                                                        | Correction factor ( $k_t$ )                                          |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| (i)    | 0.75 for unequal angles connected by the short leg<br>0.85 otherwise |
| (ii)   | As for case (i)                                                      |
| (iii)  | 0.85                                                                 |
| (iv)   | 1.0                                                                  |
| (v)    | 1.0                                                                  |

3.3 MEMBERS SUBJECT TO BENDING

3.3.1 Bending moment

The design bending moment ( $M^*$ ) of a flexural member shall satisfy the following:

- (a)  $M^* \leq \phi_b M_s$  . . . 3.3.1(1)
- (b)  $M^* \leq \phi_b M_b$  . . . 3.3.1(2)



where

$\phi_b$  = capacity reduction factor for bending (see Table 1.6)

$M_s$  = nominal section moment capacity calculated in accordance with Clause 3.3.2

$M_b$  = nominal member moment capacity calculated in accordance with Clause 3.3.3

### 3.3.2 Nominal section moment capacity

#### 3.3.2.1 General

The nominal section moment capacity ( $M_s$ ) shall be calculated either on the basis of initiation of yielding in the effective section specified in Clause 3.3.2.2 or on the basis of the inelastic reserve capacity specified in Clause 3.3.2.3.

#### 3.3.2.2 Based on initiation of yielding

The nominal section moment capacity ( $M_s$ ) shall be determined as follows:

$$M_s = Z_e f_y \quad \dots 3.3.2.2$$

where  $Z_e$  is the effective section modulus calculated with the extreme compression or tension fibre at  $f_y$ .

#### 3.3.2.3 Based on inelastic reserve capacity

The inelastic flexural reserve capacity may be used if the following conditions are met:

- (a) The member is not subject to twisting or to lateral, torsional, distortional or flexural-torsional buckling.
- (b) The effect of cold-forming is not included in determining the yield stress ( $f_y$ ).
- (c) For Item (i) (below), the ratio of the depth of the compressed portion of the web ( $d_w$ ) to its thickness ( $t_w$ ) does not exceed the slenderness ratio ( $\lambda_1$ ).
- (d) The design shear force ( $V^*$ ) does not exceed  $0.35 f_y$  times the web area ( $d_1 t_w$ ) for Item (i) (below) and ( $bt$ ) for Item (ii) (below).
- (e) The angle between any web and a perpendicular to the flange does not exceed  $30^\circ$ .

The nominal section moment capacity ( $M_s$ ) shall not exceed either  $1.25 Z_e f_y$ , where  $Z_e f_y$  shall be determined in accordance with Clause 3.3.2.2 or that causing a maximum compression strain of  $C_y e_y$ ,

where

$C_y$  = compression strain factor

$e_y$  = yield strain

$$= \frac{f_y}{E} \quad \dots 3.3.2.3(1)$$

$E$  = Young's modulus of elasticity ( $200 \times 10^3$  MPa)

NOTE: There is no limit for the maximum tensile strain.

The compression strain factor ( $C_y$ ) shall be determined as follows:

- (i) For stiffened compression elements without intermediate stiffeners:

$$\text{For } b/t \leq \lambda_1: \quad C_y = 3 \quad \dots 3.3.2.3(2)$$

$$\text{For } \lambda_1 < b/t < \lambda_2: \quad C_y = 3 - 2[(b/t) - \lambda_1] / (\lambda_2 - \lambda_1) \quad \dots 3.3.2.3(3)$$

$$\text{For } b/t \geq \lambda_2: \quad C_y = 1 \quad \dots 3.3.2.3(4)$$

$$\lambda_1 = \frac{1.11}{\sqrt{f_y / E}} \quad \dots 3.3.2.3(5)$$

$$\lambda_2 = \frac{1.28}{\sqrt{f_y / E}} \quad \dots 3.3.2.3(6)$$

(ii) For unstiffened compression elements:

(A) Under stress gradient causing compression at one longitudinal edge and tension at the other longitudinal edge of the unstiffened element.

$$\text{For } \lambda \leq \lambda_3: \quad C_y = 3$$

$$\text{For } \lambda_3 < \lambda \leq \lambda_4: \quad C_y = 3 - 2[(\lambda - \lambda_3)/(\lambda_4 - \lambda_3)] \quad \dots 3.3.2.3(7)$$

$$\text{For } \lambda \geq \lambda_4: \quad C_y = 1$$

$$\lambda_3 = 0.43$$

$$\lambda_4 = 0.673(1 - \psi) \quad \dots 3.3.2.3(8)$$

where  $\lambda$  and  $\psi$  shall be determined in accordance with Clause 2.3.2.2

(B) Under stress gradient causing compression at both longitudinal edges of the unstiffened element.

$$C_y = 1$$

(iii) For multiple-stiffened compression elements and compression elements with edge stiffeners:

$$C_y = 1$$

If applicable, effective design widths shall be used in calculating section properties.  $M_s$  shall be calculated considering equilibrium of stresses, assuming an ideally elastic-plastic stress-strain curve that is the same in tension as in compression, small deformation and that plane sections remain plane during bending. Combined bending and web crippling shall be in accordance with Clause 3.3.7.

### 3.3.3 Nominal member moment capacity

#### 3.3.3.1 General

The nominal member moment capacity ( $M_b$ ) shall be the lesser of  $M_s$  and the values calculated in accordance with Clauses 3.3.3.2 and 3.3.3.3. Clause 3.3.3.4 may be used in lieu of Clause 3.3.3.2 where appropriate.

#### 3.3.3.2 Members subject to lateral buckling

##### 3.3.3.2.1 Open section members

This Clause does not apply to multiple-web deck, U-box and curved or arch members subject to lateral buckling. It does not apply to members whose cross-sections distort laterally, such as those otherwise laterally stable members whose unbraced compression flanges buckle laterally.

For channel- and Z-purlins in which the tension flange is attached to sheeting, see Clause 3.3.3.4.

The nominal member moment capacity ( $M_b$ ) of the laterally unbraced segments of singly-, doubly- and point-symmetric sections subjected to lateral buckling shall be calculated as follows:

$$M_b = Z_c f_c \quad \dots 3.3.3.2(1)$$

where

$Z_c$  = effective section modulus calculated at a stress  $f_c$  in the extreme compression fibre

$$f_c = M_c / Z_f \quad \dots 3.3.3.2(2)$$

$M_c$  = critical moment

$Z_f$  = full unreduced section modulus for the extreme compression fibre

The critical moment ( $M_c$ ) shall be calculated as follows:

$$\text{For } \lambda_b \leq 0.60: \quad M_c = M_y \quad \dots 3.3.3.2(3)$$

$$\text{For } 0.60 < \lambda_b < 1.336: \quad M_c = 1.11 M_y \left[ 1 - \left( \frac{10 \lambda_b^2}{36} \right) \right] \quad \dots 3.3.3.2(4)$$

$$\text{For } \lambda_b \geq 1.336: \quad M_c = M_y \left( \frac{1}{\lambda_b^2} \right) \quad \dots 3.3.3.2(5)$$

where

$\lambda_b$  = non-dimensional slenderness ratio used to determine  $M_c$  for members subject to lateral buckling

$$= \sqrt{\frac{M_y}{M_o}} \quad \dots 3.3.3.2(6)$$

$M_y$  = moment causing initial yield at the extreme compression fibre of the full section

$$= Z_f f_y \quad \dots 3.3.3.2(7)$$

$M_o$  = elastic buckling moment

$M_o$  shall be determined as follows:

(a) *For singly-, doubly- and point-symmetric sections (see Figures 1.5(a), (b) and (c))* For singly-symmetric sections,  $x$ -axis is the axis of symmetry oriented such that the shear centre has a negative  $x$ -coordinate and  $y_o$  is zero.

(i) For singly-symmetric sections bent about the symmetry axis, for doubly-symmetric sections bent about the  $x$ -axis and for  $Z$ -sections bent about an axis perpendicular to the web,  $M_o$  shall be calculated as follows:

$$M_o = C_b A r_{ol} \sqrt{f_{oy} f_{oz}} \quad \dots 3.3.3.2(8)$$

where

$C_b$  = coefficient depending on moment distribution in the laterally unbraced segment

$$= \frac{12.5 M_{\max.}}{2.5 M_{\max.} + 3 M_3 + 4 M_4 + 3 M_5} \quad \dots 3.3.3.2(9)$$

$M_{\max.}$  = absolute value of the maximum moment in the unbraced segment

$M_3$  = absolute value of the moment at quarter point of the unbraced segment

$M_4$  = absolute value of the moment at mid-point of the unbraced segment

$M_5$  = absolute value of the moment at three-quarter point of the unbraced segment

$C_b$  is permitted to be taken as unity for all cases. For cantilevers or overhangs where the free end is unbraced,  $C_b$  shall be taken as unity.

Alternatively,  $C_b$  may be computed from Table 3.3.3.2.

$A$  = area of the full cross-section

$r_{o1}$  = polar radius of gyration of the cross-section about the shear centre

$$= \sqrt{r_x^2 + r_y^2 + x_o^2 + y_o^2} \quad \dots 3.3.3.2(10)$$

$r_x, r_y$  = radii of gyration of the cross-section about the  $x$ - and  $y$ -axes, respectively

$x_o, y_o$  = coordinates of the shear centre of the cross-section

$f_{oy}$  = elastic buckling stress in an axially loaded compression member for flexural buckling about the  $y$ -axis

$$= \frac{\pi^2 E}{(l_{ey} / r_y)^2} \quad \dots 3.3.3.2(11)$$

$f_{oz}$  = elastic buckling stress in an axially loaded compression member for torsional buckling

$$= \frac{GJ}{Ar_{o1}^2} \left( 1 + \frac{\pi^2 EI_w}{GJl_{ez}^2} \right) \quad \dots 3.3.3.2(12)$$

$l_{ex}, l_{ey}, l_{ez}$  = effective lengths for buckling about the  $x$ - and  $y$ -axes, and for twisting, respectively

$G$  = shear modulus of elasticity ( $80 \times 10^3$  MPa)

$J$  = torsion constant for a cross-section

$I_w$  = warping constant for a cross-section

The value of  $I_y$  to be used in the calculation of  $f_{oy}$  for Z-sections shall be the value calculated about the inclined minor principal axis. Alternatively, for Z-sections restrained by sheeting against lateral movement effectively bracing the tension flange in accordance with Clause 4.3.2.1, the values of  $I_y$  and  $I_w$  shall be those for an equivalent channel where the direction of the flange of the Z-section attached to the sheeting is reversed.

For a channel- or Z-section that is intermediately braced in accordance with Clause 4.3.2.3, the bracing interval ( $a$ ) shall be used instead of the lengths ( $l_{ey}, l_{ez}$ ) in the calculation of  $M_o$ .

Values of the bracing interval ( $a$ ) and coefficient ( $C_b$ ) for uniformly distributed loads, applied within the span of intermediately braced simply supported beams, are given in Table 3.3.3.2.

Alternatively,  $M_o$  can be calculated using Equation 3.3.3.2(17) for point-symmetric Z-sections.

A1

- (ii) For singly-symmetric sections bent about the centroidal axis perpendicular to the symmetry axis,  $M_o$  shall be calculated as follows:

$$M_o = \frac{C_s A f_{ox} \left[ (\beta_y / 2) + C_s \sqrt{(\beta_y / 2)^2 + r_{ol}^2 (f_{oz} / f_{ox})} \right]}{C_{TF}} \quad \dots 3.3.3.2(13)$$

where

$f_{ox}$  = elastic buckling stress in an axially loaded compression member for flexural buckling about the  $x$ -axis

$$= \frac{\pi^2 E}{(l_{ex} / r_x)^2} \quad \dots 3.3.3.2(14)$$

$C_{TF}$  = coefficient for unequal end moment

$$= 0.6 - 0.4 \left( \frac{M_1}{M_2} \right) \quad \dots 3.3.3.2(15)$$

$M_1$  is the smaller and  $M_2$  the larger bending moment at the ends of the unbraced length. The ratio of end moments ( $M_1/M_2$ ) is positive if  $M_1$  and  $M_2$  have the same sign (reverse curvature bending) and negative if they are of opposite sign (single curvature bending). If the bending moment at any point within an unbraced length is larger than that at both ends of this length,  $C_{TF}$  shall be taken as unity.

$C_s$  = coefficient

= +1, for moment causing compression on the shear centre side of the centroid (see Figure E1 of Appendix E)

= -1, for moment causing tension on the shear centre side of the centroid (see Figure E1 of Appendix E)

$\beta_y$  = monosymmetry section constant about the  $y$ -axis (see Paragraph E2 of Appendix E)

$$= \frac{I}{I_y} \left( \int_A xy^2 dA + \int_A x^3 dA \right) - 2x_o \quad \dots 3.3.3.2(16)$$

$I_y$  = second moment of area of the cross-section about the  $y$ -axis

$x, y$  = principal axes of the cross-section

- (b) *For point-symmetric Z-sections* For point-symmetric Z-sections,  $M_o$  shall be calculated as follows:

$$M_o = \frac{\pi^2 E C_b d I_{yc}}{2 l^2} \quad \dots 3.3.3.2(17)$$

where

$I_{yc}$  = second moment of area of the compression portion of the section about the centroidal axis of the full section parallel to the web, using the full unreduced section

$l$  = unbraced length of the member

Alternatively, the value of  $M_o$  may be determined by a rational flexural-torsional buckling analysis.

**TABLE 3.3.3.2**  
**COEFFICIENTS ( $C_b$ ) FOR SIMPLY SUPPORTED SINGLE SPAN BEAMS**  
**WITH UNIFORMLY DISTRIBUTED LOADS WITHIN THE SPAN**

| Load position      | Coefficient ( $C_b$ )                     |                                     |                                                        |
|--------------------|-------------------------------------------|-------------------------------------|--------------------------------------------------------|
|                    | No bracing<br>( $a = l$ )<br>(see Note 1) | One central brace<br>( $a = 0.5l$ ) | Third point bracing<br>( $a = 0.33l$ )<br>(see Note 2) |
| Tension flange     | 1.92                                      | 1.59                                | 1.47                                                   |
| Shear centre       | 1.22                                      | 1.37                                | 1.37                                                   |
| Compression flange | 0.77                                      | 1.19                                | 1.28                                                   |

NOTES:

- 1 Channel and Z-beams without intermediate bracing may show noticeable twisting even when torsionally restrained by sheeting.
- 2  $C_b$  applies to the central section.

### 3.3.3.2.2 Closed box members

For closed box members, the nominal member moment capacity ( $M_b$ ) shall be determined as follows:

- (a) If the unbraced length of the member is less than or equal to  $l_u$ ,  $M_b$  shall be determined in accordance with Clause 3.3.3.2.1,

where

$$l_u = \text{limit of unbraced length by which lateral-torsional buckling is not considered}$$

$$= \frac{0.36C_b\pi}{f_y Z_f} \sqrt{EGJ_y} \quad \dots 3.3.3.2(18)$$

- (b) If the laterally unbraced length of a member is greater than  $l_u$ ,  $M_b$  shall be determined in accordance with Clause 3.3.3.2.1, where the elastic buckling moment ( $M_o$ ) shall be calculated as follows:

$$M_o = \frac{C_b\pi}{l} \sqrt{EGJ_y} \quad \dots 3.3.3.2(19)$$

where

$$l = \text{laterally unbraced length of the member}$$

### 3.3.3.3 Members subject to distortional buckling

The nominal member moment capacity ( $M_b$ ) of sections subject to distortional buckling shall be calculated as follows:

$$M_b = Z_c f_c \quad \dots 3.3.3.3(1)$$

The following cases, as appropriate, shall be considered:

- (a) *Where distortional buckling involves rotation of a flange and lip about the flange/web junction of a channel- or Z-section*  $Z_c$  is the full section modulus except that when  $k_\phi$  as given by Equation D3(2) is negative then  $Z_c$  is the effective section modulus calculated at a stress ( $f_c$ ) in the extreme compression fibre using  $k = 4.0$  for the compressive flange in Equation 2.2.1.2(4) and ignoring Clause 2.4.1, where  $f_c$  shall be calculated as follows:

$$f_c = M_c / Z_f \quad \dots 3.3.3.3(2)$$

where

$M_c$  = critical moment

$Z_f$  = full section modulus

The critical moment ( $M_c$ ) shall be calculated as follows:

$$\text{For } \lambda_d \leq 0.674: M_c = M_y \quad \dots 3.3.3.3(3)$$

$$\text{For } \lambda_d > 0.674: M_c = \frac{M_y}{\lambda_d} \left( 1 - \frac{0.22}{\lambda_d} \right) \quad \dots 3.3.3.3(4)$$

- (b) *Where distortional buckling involves transverse bending of a vertical web with lateral displacement of the compression flange*  $Z_c$  is the effective section modulus calculated at a stress ( $f_c$ ) in the extreme compression fibre, where  $f_c$  shall be calculated using Equation 3.3.3.3(2).

The critical moment ( $M_c$ ) shall be calculated as follows:

$$A1 \quad \left| \quad \text{For } \lambda_d \leq 0.59: M_c = M_y \quad \dots 3.3.3.3(5) \right.$$

$$\text{For } 0.59 < \lambda_d \leq 1.70: M_c = M_y \left( \frac{0.59}{\lambda_d} \right) \quad \dots 3.3.3.3(6)$$

$$A1 \quad \left| \quad \text{For } \lambda_d > 1.70: M_c = M_y \left( \frac{1}{\lambda_d^2} \right) \quad \dots 3.3.3.3(7) \right.$$

where

$M_y$  = moment causing initial yield at the extreme compression fibre of the full section

$\lambda_d$  = non-dimensional slenderness used to determine  $M_c$  for member subject to distortional buckling

$$= \sqrt{\frac{M_y}{M_{od}}} \quad \dots 3.3.3.3(8)$$

$M_{od}$  = elastic buckling moment in the distortional mode

$$= Z_f f_{od} \quad \dots 3.3.3.3(9)$$

$f_{od}$  = elastic distortional buckling stress

NOTE:  $f_{od}$  may be calculated using the appropriate equations given in Appendix D or a rational elastic buckling analysis.

### 3.3.3.4 Beams having one flange through-fastened to sheeting

The nominal member moment capacity ( $M_b$ ) of a channel- or Z-section loaded in a plane parallel to the web, with the tension flange attached to sheeting and with compression flange laterally unbraced, shall be calculated as follows:

$$M_b = RZ_e f_y \quad \dots 3.3.3.4$$

where  $R$  is the reduction factor.

The reduction factor ( $R$ ) shall be taken as follows:

- (a) *Uplift loading* For continuous lapped purlins with three or more spans using Z-sections, and simple spans using channel- and Z-sections with cyclone washers, the  $R$  factor shall be as follows:
- (i) No bridging ..... 0.75.
  - (ii) One row of bridging in end and interior spans ..... 0.85.
  - (iii) Two rows of bridging in end span and one or more rows  
in interior spans of continuous lapped purlins ..... 0.95.
  - (iv) Two rows of bridging in simple span ..... 1.00.

The combined bending and shear at the end of the lap shall be considered for the case with two rows of bridging.

For double spans using Z-section, the  $R$  factor shall be as follows:

- (A) No bridging ..... 0.60.
- (B) One row of bridging per span ..... 0.70.
- (C) Two rows of bridging per span ..... 0.80.

NOTE: For simple spans without cyclone washers, the values recommended in the AISI Specification should be used.

- (b) *Downwards loading* For continuous lapped purlins with three or more spans using Z-section, without bridging or any other configuration, the  $R$  factor shall be equal to 0.85.

The combined bending and shear at the end of the lap need not be considered separately for this case.

The reduction factor ( $R$ ) shall be limited to roof and wall systems complying with the following:

- (i) Member depth shall be less than or equal to 300 mm.
- (ii) Flanges shall be edge-stiffened compression elements with the lip perpendicular to the stiffened flanges.
- (iii)  $75 < \text{depth/thickness} < 135$ .
- (iv)  $2.3 < \text{depth/flange width} < 3.2$ .
- (v)  $25 < \text{flat width/thickness of flange} < 44$ .
- (vi) For continuous span systems, the total lap length at each interior support in each direction (distance between centre-line of bolts at each end of lap) shall be not less than—
  - (A) 13% of span for triple spans; and
  - (B) 15% of span for double spans, such that the support bolts are located at the centre of the lap.



- (vii) Member span length shall be not greater than 10.5 m.
- (viii) For continuous span systems, the longest member span shall be not more than 20% greater than the shortest span.
- (ix) Cleat plates shall be used at the supports.
- (x) Roof or wall panels shall be steel sheets, minimum of 0.42 mm base metal thickness, having a minimum rib depth of 27 mm, at a maximum spacing of 200 mm on centres and attached in such a manner as to effectively inhibit relative movement between the panel and purlin flange.
- (xi) Insulation shall not be used between the roof sheeting and purlins.
- (xii) Fastener type shall be minimum No. 12 self-drilling or self-tapping sheet metal screws for triple and double spans, and No. 12 screws with load-spreading washers for simple spans. Side lap fasteners shall be used between the sheets.
- (xiii) Screws shall be crest-fastened.
- (xiv) Fasteners shall be located at every crest.
- (xv) Bridging shall be of a type that effectively prevents lateral and torsional deformations at support points.

If variables fall outside any of the requirements in Items (i) to (xv), full-scale tests shall be made in accordance with Section 6, or another rational analysis procedure shall be applied. In any case, it is permitted to perform tests, in accordance with Section 6, as an alternative to the procedure described in this Clause.

### 3.3.3.5 *Beams having one flange fastened to a standing seam roof or clip-fixed deck system*

The nominal section moment capacity ( $M_s$ ) of a channel- or Z-section, added in a plane parallel to the web with the top flange supporting a standing seam roof system shall be determined using a discrete point bracing and the provisions of Clause 3.3.3.2.1, or shall be calculated as follows:

$$M_s = RS_e f_y \quad \dots 3.3.3.5$$

$$\phi_b = 0.9$$

where

$R$  = reduction factor determined by testing in accordance with Section 8

$S_e$  = elastic section modulus of the effective section calculated with extreme compression or tension fibre at  $f_y$

### 3.3.4 **Shear**

#### 3.3.4.1 *Shear capacity of webs without holes*

The design shear force ( $V^*$ ) at any cross-section shall satisfy—

$$V^* \leq \phi_v V_v$$

where

$\phi_v$  = capacity reduction factor for shear (see Table 1.6)

$V_v$  = nominal shear capacity of the web

The nominal shear capacity ( $V_v$ ) of a web shall be calculated from the following equations, as appropriate.

$$\text{For } d_1/t_w \leq \sqrt{Ek_v / f_y} : \quad V_v = 0.64f_y d_1 t_w \quad \dots 3.3.4(1)$$

$$\text{For } \sqrt{Ek_v / f_y} < d_1/t_w \leq 1.415 \sqrt{Ek_v / f_y} : V_v = 0.64t_w^2 \sqrt{Ek_v f_y} \quad \dots 3.3.4(2)$$

A1

$$\text{For } d_1/t_w > 1.415 \sqrt{Ek_v / f_y} : V_v = \frac{0.905Ek_v t_w^3}{d_1} \quad \dots 3.3.4(3)$$

where

$d_1$  = depth of the flat portion of the web measured along the plane of the web

$t_w$  = thickness of web

$k_v$  = shear buckling coefficient determined as follows:

(i) For unstiffened webs:  $k_v = 5.34$

(ii) For beam webs with transverse stiffeners complying with Clause 2.7—

$$\text{for } a/d_1 \leq 1.0: k_v = 4.00 + [5.34/(a/d_1)^2] \quad \dots 3.3.4(4)$$

$$\text{for } a/d_1 > 1.0: k_v = 5.34 + [4.00/(a/d_1)^2] \quad \dots 3.3.4(5)$$

$a$  = shear panel length for unstiffened web element; *or*  
distance between transverse stiffeners for stiffened web elements

For a web consisting of two or more sheets, each sheet shall be considered as a separate element carrying its share of the shear force.

### 3.3.4.2 Shear capacity of channel-section webs with holes

Shear capacity of channel-section webs with holes shall be applicable within the following limits:

(a)  $d_{wh}/d_1 < 0.7$ ,

where

$d_{wh}$  = depth of the web hole

$d_1$  = depth of the flat portion of the web measured along the plane of the web

(b)  $d_{wh}/t \leq 200$ .

(c) Holes centred at mid-depth of the web.

(d) Clear distance between holes is greater than or equal to 450 mm.

(e) Non-circular holes corner radii greater than or equal to  $2t$ .

(f) Non-circular holes with  $d_o \leq 65$  mm and  $b \leq 115$  mm, where  $b$  is the length of the web hole.

(g) Circular hole diameters less than or equal to 150 mm.

(h)  $d_o > 15$  mm.

The nominal shear capacity ( $V_v$ ) determined in accordance with Clause 3.3.4.1 shall be multiplied by  $q_s$ , where—

$$q_s = 1.0 \text{ when } \frac{c}{t} \geq 54 \quad \dots 3.3.4.2(1)$$

$$= \frac{c}{54t} \text{ when } 5 \leq \frac{c}{t} < 54 \quad \dots 3.3.4.2(2)$$

$$c = \frac{d_1}{2} - \frac{d_{wh}}{2.83} \quad \text{for circular holes} \quad \dots 3.3.4.2(3)$$

$$= \frac{d_1}{2} - \frac{d_{wh}}{2} \quad \text{for non-circular holes} \quad \dots 3.3.4.2(4)$$

$d_{wh}$  = depth of web hole

$d_1$  = depth of flat portion of the web measured along the plane of the web

### 3.3.5 Combined bending and shear

For beams with unstiffened webs, the design bending moment ( $M^*$ ) and the design shear force ( $V^*$ ) shall satisfy—

$$\left( \frac{M^*}{\phi_b M_s} \right)^2 + \left( \frac{V^*}{\phi_v V_v} \right)^2 \leq 1.0 \quad \dots 3.3.5(1)$$

For beams with transverse web stiffeners, the design bending moment ( $M^*$ ) shall satisfy—

$$M^* \leq \phi_b M_b \quad \dots 3.3.5(2)$$

The design shear force ( $V^*$ ) shall satisfy—

$$V^* \leq \phi_v V_v \quad \dots 3.3.5(3)$$

If  $\frac{M^*}{\phi_b M_s} > 0.5$ ; and

$$\frac{V^*}{\phi_v V_v} > 0.7$$

then  $M^*$  and  $V^*$  shall satisfy—

$$0.6 \left( \frac{M^*}{\phi_b M_s} \right) + \left( \frac{V^*}{\phi_v V_v} \right) \leq 1.3 \quad \dots 3.3.5(4)$$

where

$M_s$  = nominal section moment capacity about the centroidal axes determined in accordance with Clause 3.3.2

$V_v$  = nominal shear capacity when shear alone exists determined in accordance with Clause 3.3.4

$M_b$  = nominal member moment capacity when bending alone exists determined in accordance with Clause 3.3.3

### 3.3.6 Bearing

#### 3.3.6.1 Design for bearing

A member subject to bearing ( $R_b^*$ ) shall satisfy—

$$R_b^* \leq \phi_w R_b \quad \dots 3.3.6.1$$

where

$\phi_w$  = capacity reduction factor for bearing (see Table 1.6)

$R_b$  = nominal capacity for concentrated load or reaction for one solid web connecting top and bottom flanges

### 3.3.6.2 Bearing without holes

The nominal capacity for concentrated load or reaction for one solid web connecting top and bottom flanges ( $R_b$ ) shall be determined as follows:

$$R_b = C t_w^2 f_y \sin \theta \left( 1 - C_r \sqrt{\frac{r_i}{t_w}} \right) \left( 1 + C_l \sqrt{\frac{l_b}{t_w}} \right) \left( 1 - C_w \sqrt{\frac{d_1}{t_w}} \right) \quad \dots 3.3.6.2$$

where

$C$  = coefficient (see Tables 3.3.6.2(A) to (E))

$t_w$  = thickness of the web

$\theta$  = angle between the plane of the web and the plane of the bearing surface.  $\theta$  shall be within the following limits:

$$90^\circ \geq \theta \geq 45^\circ$$

$C_r$  = coefficient of inside bent radius (see Tables 3.3.6.2(A) to (E))

$r_i$  = inside bent radius

$C_l$  = coefficient of bearing length (see Tables 3.3.6.2(A) to (E))

$l_b$  = actual bearing length. For the case of two equal and opposite concentrated loads distributed over unequal bearing lengths, the smaller value of  $l_b$  shall be taken

$C_w$  = coefficient of web slenderness (see Tables 3.3.6.2(A) to (E))

$d_1$  = depth of the flat portion of the web measured along the plane of the web

Webs of members in bending for which  $d_1/t_w$  is greater than 200 shall be provided with adequate means of transmitting concentrated actions or reactions directly into the web(s).

$R_b$  is the nominal capacity for load or reaction for one solid web connecting top and bottom flanges. For webs consisting of two or more such sheets,  $R_b$  shall be calculated for each individual sheet and the results added to obtain the nominal load or reaction for the full section.

A1 | One-flange loading or reaction occurs when the clear distance between the bearing edges of adjacent opposite concentrated actions or reactions is greater than  $1.5d_1$ .

Two-flange loading or reaction occurs when the clear distance between the bearing edges of adjacent opposite concentrated actions or reactions is less than or equal to  $1.5d_1$ .

End loading or reaction occurs when the distance from the edge of the bearing to the end of the member is less than or equal to  $1.5d_1$ .

Interior loading or reaction occurs when the distance from the edge of the bearing to the end of the member is greater than  $1.5d_1$ .

A1 | The capacity reduction factors shall be as given in Tables 3.3.6.2(A) to 3.3.6.2(E).

**TABLE 3.3.6.2(A)**  
**BACK-TO-BACK CHANNEL-SECTIONS**

| Support and flange conditions |                                          | Load cases                     |          | $C$  | $C_r$ | $C_l$ | $C_w$ | $\phi_w$ | Limits           |
|-------------------------------|------------------------------------------|--------------------------------|----------|------|-------|-------|-------|----------|------------------|
| Fastened to support           | Stiffened or partially stiffened flanges | One-flange loading or reaction | End      | 10   | 0.14  | 0.28  | 0.001 | 0.75     | $r_1/t_w \leq 5$ |
|                               |                                          |                                | Interior | 20   | 0.15  | 0.05  | 0.003 | 0.90     | $r_1/t_w \leq 5$ |
| Unfastened                    | Stiffened or partially stiffened flanges | One-flange loading or reaction | End      | 10   | 0.14  | 0.28  | 0.001 | 0.75     | $r_1/t_w \leq 5$ |
|                               |                                          |                                | Interior | 20.5 | 0.17  | 0.11  | 0.001 | 0.85     | $r_1/t_w \leq 3$ |
|                               |                                          | Two-flange loading or reaction | End      | 15.5 | 0.09  | 0.08  | 0.04  | 0.75     | $r_1/t_w \leq 3$ |
|                               |                                          |                                | Interior | 36   | 0.14  | 0.08  | 0.04  | 0.75     |                  |
|                               | Unstiffened flanges                      | One-flange loading or reaction | End      | 10   | 0.14  | 0.28  | 0.001 | 0.75     | $r_1/t_w \leq 5$ |
|                               |                                          |                                | Interior | 20.5 | 0.17  | 0.11  | 0.001 | 0.85     | $r_1/t_w \leq 3$ |

NOTES:

- 1 Table 3.3.6.2(A) applies to I-beams made from two channels connected back to back.
- 2 The coefficients in Table 3.3.6.2(A) apply if  $l_b/t_w \leq 210$ ,  $l_b/d_1 \leq 1.0$  and  $\theta = 90^\circ$ .

**TABLE 3.3.6.2(B)**  
**SINGLE WEB CHANNEL-SECTIONS AND C-SECTIONS**

| Support and flange conditions |                                          | Load cases                     |          | $C$ | $C_r$ | $C_l$ | $C_w$ | $\phi_w$ | Limits            |
|-------------------------------|------------------------------------------|--------------------------------|----------|-----|-------|-------|-------|----------|-------------------|
| Fastened to support           | Stiffened or partially stiffened flanges | One-flange loading or reaction | End      | 4   | 0.14  | 0.35  | 0.02  | 0.85     | $r_1/t_w \leq 9$  |
|                               |                                          |                                | Interior | 13  | 0.23  | 0.14  | 0.01  | 0.90     | $r_1/t_w \leq 5$  |
|                               |                                          | Two-flange loading or reaction | End      | 7.5 | 0.08  | 0.12  | 0.048 | 0.85     | $r_1/t_w \leq 12$ |
|                               |                                          |                                | Interior | 20  | 0.10  | 0.08  | 0.031 | 0.85     | $r_1/t_w \leq 12$ |
| Unfastened                    | Stiffened or partially stiffened flanges | One-flange loading or reaction | End      | 4   | 0.14  | 0.35  | 0.02  | 0.80     | $r_1/t_w \leq 5$  |
|                               |                                          |                                | Interior | 13  | 0.23  | 0.14  | 0.01  | 0.90     |                   |
|                               |                                          | Two-flange loading or reaction | End      | 13  | 0.32  | 0.05  | 0.04  | 0.90     | $r_1/t_w \leq 3$  |
|                               |                                          |                                | Interior | 24  | 0.52  | 0.15  | 0.001 | 0.80     |                   |
|                               | Unstiffened flanges                      | One-flange loading or reaction | End      | 4   | 0.40  | 0.60  | 0.03  | 0.85     | $r_1/t_w \leq 2$  |
|                               |                                          |                                | Interior | 13  | 0.32  | 0.10  | 0.01  | 0.85     | $r_1/t_w \leq 1$  |
|                               |                                          | Two-flange loading or reaction | End      | 2   | 0.11  | 0.37  | 0.01  | 0.75     | $r_1/t_w \leq 1$  |
|                               |                                          |                                | Interior | 13  | 0.47  | 0.25  | 0.04  | 0.80     |                   |

NOTE: The coefficients in Table 3.3.6.2(B) apply if  $d_1/t_w \leq 200$ ,  $l_b/t_w \leq 210$ ,  $l_b/d_1 \leq 2.0$  and  $\theta = 90^\circ$ .

**TABLE 3.3.6.2(C)**  
**SINGLE WEB Z-SECTIONS**

| Support and flange conditions |                                          | Load cases                     |          | $C$ | $C_r$ | $C_l$ | $C_w$ | $\phi_w$ | Limits            |
|-------------------------------|------------------------------------------|--------------------------------|----------|-----|-------|-------|-------|----------|-------------------|
| Fastened to support           | Stiffened or partially stiffened flanges | One-flange loading or reaction | End      | 4   | 0.14  | 0.35  | 0.02  | 0.85     | $r_i/t_w \leq 9$  |
|                               |                                          |                                | Interior | 13  | 0.23  | 0.14  | 0.01  | 0.90     | $r_i/t_w \leq 5$  |
|                               |                                          | Two-flange loading or reaction | End      | 9   | 0.05  | 0.16  | 0.052 | 0.85     | $r_i/t_w \leq 12$ |
|                               |                                          |                                | Interior | 24  | 0.07  | 0.07  | 0.04  | 0.80     | $r_i/t_w \leq 12$ |
| Unfastened                    | Stiffened or partially stiffened flanges | One-flange loading or reaction | End      | 5   | 0.09  | 0.02  | 0.001 | 0.85     | $r_i/t_w \leq 5$  |
|                               |                                          |                                | Interior | 13  | 0.23  | 0.14  | 0.01  | 0.90     |                   |
|                               |                                          | Two-flange loading or reaction | End      | 13  | 0.32  | 0.05  | 0.04  | 0.90     | $r_i/t_w \leq 3$  |
|                               |                                          |                                | Interior | 24  | 0.52  | 0.15  | 0.001 | 0.80     |                   |
|                               | Unstiffened flanges                      | One-flange loading or reaction | End      | 4   | 0.40  | 0.60  | 0.03  | 0.85     | $r_i/t_w \leq 2$  |
|                               |                                          |                                | Interior | 13  | 0.32  | 0.10  | 0.01  | 0.85     | $r_i/t_w \leq 1$  |
|                               |                                          | Two-flange loading or reaction | End      | 2   | 0.11  | 0.37  | 0.01  | 0.75     | $r_i/t_w \leq 1$  |
|                               |                                          |                                | Interior | 13  | 0.47  | 0.25  | 0.04  | 0.80     |                   |

NOTE: The coefficients in Table 3.3.6.2(C) apply if  $d_1/t_w \leq 200$ ,  $l_b/t_w \leq 210$ ,  $l_b/d_1 \leq 2.0$  and  $\theta = 90^\circ$ .

**TABLE 3.3.6.2(D)**  
**SINGLE HAT SECTIONS**

| Support conditions  | Load cases                     |          | $C$ | $C_r$ | $C_l$ | $C_w$ | $\phi_w$ | Limits            |
|---------------------|--------------------------------|----------|-----|-------|-------|-------|----------|-------------------|
| Fastened to support | One-flange loading or reaction | End      | 4   | 0.25  | 0.68  | 0.04  | 0.75     | $r_i/t_w \leq 5$  |
|                     |                                | Interior | 17  | 0.13  | 0.13  | 0.04  | 0.80     | $r_i/t_w \leq 10$ |
|                     | Two-flange loading or reaction | End      | 9   | 0.10  | 0.07  | 0.03  | 0.85     | $r_i/t_w \leq 10$ |
|                     |                                | Interior | 10  | 0.14  | 0.22  | 0.02  | 0.85     |                   |
| Unfastened          | One-flange loading or reaction | End      | 4   | 0.25  | 0.68  | 0.04  | 0.75     | $r_i/t_w \leq 4$  |
|                     |                                | Interior | 17  | 0.13  | 0.13  | 0.04  | 0.90     | $r_i/t_w \leq 4$  |

NOTE: The coefficients in Table 3.3.6.2(D) apply if  $d_1/t_w \leq 200$ ,  $l_b/t_w \leq 200$ ,  $l_b/d_1 \leq 2.0$  and  $\theta = 90^\circ$ .

**TABLE 3.3.6.2(E)**  
**MULTI-WEB DECK SECTIONS**

| Support conditions  | Load cases                     |          | $C$ | $C_r$ | $C_l$ | $C_w$ | $\phi_w$ | Limits            |
|---------------------|--------------------------------|----------|-----|-------|-------|-------|----------|-------------------|
| Fastened to support | One-flange loading or reaction | End      | 4   | 0.04  | 0.25  | 0.025 | 0.90     | $r_i/t_w \leq 20$ |
|                     |                                | Interior | 8   | 0.10  | 0.17  | 0.004 | 0.85     | $r_i/t_w \leq 10$ |
|                     | Two-flange loading or reaction | End      | 9   | 0.12  | 0.14  | 0.040 | 0.85     | $r_i/t_w \leq 10$ |
|                     |                                | Interior | 10  | 0.11  | 0.21  | 0.020 | 0.85     |                   |
| Unfastened          | One-flange loading or reaction | End      | 3   | 0.04  | 0.29  | 0.028 | 0.60     | $r_i/t_w \leq 20$ |
|                     |                                | Interior | 8   | 0.10  | 0.17  | 0.004 | 0.85     |                   |
|                     |                                | End      | 6   | 0.16  | 0.15  | 0.050 | 0.90     | $r_i/t_w \leq 5$  |
|                     |                                | Interior | 17  | 0.10  | 0.10  | 0.046 | 0.90     |                   |

NOTE: The coefficients in Table 3.3.6.2(E) apply if  $d_1/t_w \leq 200$ ,  $l_b/t_w \leq 210$ ,  $l_b/d_1 \leq 3.0$  and  $45^\circ < \theta \leq 90^\circ$ .

### 3.3.6.3 Web crippling strength of channel-section webs with holes

When a web hole is within the bearing length ( $l_b$ ), a bearing stiffener shall be used. For beam webs with holes, the web crippling strength shall be calculated in accordance with Clause 3.3.6.2 multiplied by the reduction factor ( $R_c$ ), given in this Clause.

Web crippling strength of channel-section webs with holes, as determined by Clause 3.3.6.2, shall be applicable within the following limits:

- (a)  $d_{wh}/d_1 < 0.7$ .
- (b)  $d_1/t \leq 200$ .
- (c) Holes centred at mid-depth of the web.
- (d) Clear distance between holes is greater than or equal to 450 mm.
- (e) Distance between the end of the member and the edge of the hole is greater than or equal to  $d$ .
- (f) Non-circular holes corner radii greater than or equal to  $2t$ .
- (g) Non-circular holes with  $d_{wh} \leq 65$  mm and  $b \leq 115$  mm.
- (h) Circular hole diameters less than or equal to 150 mm.
- (i)  $d_{wh} > 15$  mm.

When a web hole is not within the bearing length ( $l_b \geq 25$  mm):

$$R_c = 1.01 - \frac{0.325d_{wh}}{d_1} + \frac{0.083x}{d_1} \leq 1.0 \quad \dots 3.3.6.3(1)$$

When any portion of a web hole is not within the bearing length ( $l_b \geq 75$  mm):

$$R_c = 0.90 - \frac{0.047d_{wh}}{d_1} + \frac{0.053x}{d_1} \leq 1.0 \quad \dots 3.3.6.3(2)$$

where

$l_b$  = bearing length

$d$  = depth of cross-section

$x$  = nearest distance between the web hole and the edge bearing

### 3.3.7 Combined bending and bearing

Unstiffened flat webs of shapes subjected to a combination of bending and reaction or concentrated load shall be designed as follows:

- (a) *Shapes having single unstiffened webs* Shapes having single unstiffened webs shall satisfy—

$$1.07 \left( \frac{R^*}{\phi_w R_b} \right) + \left( \frac{M^*}{\phi_b M_s} \right) \leq 1.42 \quad \dots 3.3.7(1)$$

At the interior supports of continuous spans, the above interaction is not applicable to deck or beams with two or more single webs, where the compression edges of adjacent webs are laterally supported in the negative moment region by continuous or intermittently connected flange elements, rigid cladding, or lateral bracing, and the spacing between adjacent webs does not exceed 250 mm.

- (b) *Back-to-back channel beams and beams with restraint against web rotation* Back-to-back channel beams and beams with restraint against web rotation, such as I-sections made by welding two angles to a channel, shall satisfy—

$$0.82 \left( \frac{R^*}{\phi_w R_b} \right) + \left( \frac{M^*}{\phi_b M_s} \right) \leq 1.32 \quad \dots 3.3.7(2)$$

If  $d_1/t_w \leq 2.33/\sqrt{(f_y/E)}$  and  $\lambda \leq 0.673$ , the nominal concentrated load or reaction strength may be determined in accordance with Clause 3.3.6.

In Items (a) and (b), the following applies:

$R^*$  = design concentrated load or reaction in the presence of bending moment

$R_b$  = nominal capacity for concentrated load or reaction in the absence of bending moment assuming single web interior one flange loading for the nested Z-sections, i.e., the sum of the two webs evaluated individually determined in accordance with Clause 3.3.6

A1 |  $\phi_b$  = capacity reduction factor for bending

$\phi_w$  = capacity reduction factor for bearing

$M^*$  = design bending moment at, or immediately adjacent to, the point of application of the design concentrated load or reaction ( $R^*$ )

A1 |  $M_s$  = nominal section moment capacity about the centroidal axes determined in accordance with Clause 3.3.1

$t_w$  = thickness of the web

$\lambda$  = slenderness ratio (see Clause 2.2.1.2)

- (c) *Two nested Z-sections* Two nested Z-sections shall satisfy—

$$0.85 \left( \frac{R^*}{\phi R_b} \right) + \left( \frac{M^*}{\phi M_s} \right) \leq 1.65 \quad \dots 3.3.7(3)$$

where

$R^*$  = design concentrated action or reaction in the presence of bending moment

$R_b$  = nominal capacity for concentrated action or reaction in the absence of bending moment assuming single web interior one flange loading for the nested Z-sections, i.e., the sum of the two webs evaluated individually determined in accordance with Clause 3.3.6

$\phi$  = 0.9

$M^*$  = design bending moment at the section under consideration to, the point of application of the design concentrated load or reaction ( $R^*$ )

$M_s$  = nominal section moment capacity for the two nested Z-sections, i.e., the sum of the two sections evaluated individually, determined in accordance with Clause 3.3.1

In addition,  $M^*$  and  $R^*$  shall satisfy—

$$M^* \leq \phi M_s \quad \dots 3.3.7(4)$$

$$R^* \leq \phi R_b \quad \dots 3.3.7(5)$$



Equation 3.3.7(3) applies, if—

- (i)  $d_1/t \leq 150$ ;
- (ii)  $I_b/t_w \leq 140$ ;
- (iii)  $f_y \leq 480$  MPa; and
- (iv)  $r_i/t \leq 5.5$ .

The following conditions shall be satisfied:

- (A) The ends of each section shall be connected to the other section by a minimum of two 12.0 mm diameter A307 bolts through the web.
- (B) The combined section shall be connected to the support by a minimum of two 12.0 mm A307 bolts through the flanges.
- (C) The webs of the two sections shall be in contact.
- (D) The ratio of the thicker to the thinner part shall not exceed 1.3.

### 3.3.8 Stiffeners

#### 3.3.8.1 Transverse stiffeners

Transverse stiffeners attached to beam webs at points of concentrated loads or reactions shall be designed as compression members. Concentrated loads or reactions shall be applied directly into the stiffeners, or each stiffener shall be fitted accurately to the flat portion of the flange to provide direct load-bearing into the end of the stiffener. Means for shear transfer between the stiffener and the web shall be provided in accordance with Clause 3.3.8.2. The design concentrated loads or reactions ( $N^*$ ) shall satisfy the following:

- (a)  $N^* \leq \phi_c N_s$  . . . 3.3.8.1(1)
- (b)  $N^* \leq \phi_c N_c$  . . . 3.3.8.1(2)

where

- $\phi_c$  = capacity reduction factor for members in compression (see Table 1.6)
- $N_s$  = nominal section capacity of a member in compression (see Clause 3.4)  
 $= f_{wy} A_{s1}$
- $N_c$  = nominal member capacity of a member in compression (see Clause 3.4)  
 $= f_n A_{s2}$
- $f_{wy}$  = lower yield stress value of the beam web ( $f_y$ ) or of the stiffener section ( $f_{ys}$ )
- $f_n$  = critical stress (see Clause 3.4)
- $A_{s1}, A_{s2}$  = area of a member in compression consisting of the transverse stiffeners and a portion of the web
- $A_{s1} = 18t^2 + A_s$  . . . 3.3.8.1(3)  
 (for transverse stiffeners at interior support and under concentrated load)
- $= 10t^2 + A_s$  . . . 3.3.8.1(4)  
 (for transverse stiffeners at end support)
- $A_{s2} = b_1 t + A_s$  . . . 3.3.8.1(5)  
 (for transverse stiffeners at interior support and under concentrated load)

$$= b_2 t + A_s \quad \dots 3.3.8.1(6)$$

(for transverse stiffeners at end support)

$t$  = base thickness of beam web

$A_s$  = cross-sectional area of transverse stiffeners

$$b_1 = 25t \left[ 0.0024 \left( \frac{l_{st}}{t} \right) + 0.72 \right] \leq 25t \quad \dots 3.3.8.1(7)$$

$$b_2 = 25t \left[ 0.0044 \left( \frac{l_{st}}{t} \right) + 0.83 \right] \leq 12t \quad \dots 3.3.8.1(8)$$

$l_{st}$  = length of transverse stiffener

The  $b/t_s$  ratio for the stiffened and unstiffened elements of cold-formed steel transverse stiffeners shall not exceed  $1.28\sqrt{(E/f_{ys})}$  and  $0.37\sqrt{(E/f_{ys})}$ , respectively, where  $f_{ys}$  is the yield stress and  $t_s$  is the thickness of the stiffener steel.

### 3.3.8.2 Bearing stiffeners in channel-section flexural members

For two-flange loading of channel-section flexural members with bearing stiffeners that do not meet the requirements of Clause 3.3.8.1, the design bearing capacity ( $\phi_w R_b$ ) shall be determined from—

$$R_b = 0.7(R_{wc} + A_e f_y) \geq R_{wc} \quad \dots 3.3.8.2$$

where

$\phi_w$  = capacity reduction factor for bearing stiffeners (see Table 1.6)

$R_{wc}$  = web crippling capacity for channel-section flexural member calculated in accordance with Equation 3.3.6.1 for single web members, at the end or interior locations

$A_e$  = effective area of the bearing stiffener subjected to uniform compressive stress, calculated at yield point

$f_y$  = yield point of the bearing stiffener steel

Equation 3.3.8.2 applies within the following limits:

- Full bearing of the stiffener is required. If the bearing width is narrower than the stiffener such that one of the stiffener flanges is unsupported,  $R_b$  shall be reduced by 50%.
- Stiffeners shall be channel-section stud or track members with a minimum web depth of 90 mm and a minimum base steel thickness of 0.85 mm.
- The stiffener shall be attached to the flexural member web with at least three fasteners (screws or bolts).
- The distance from the flexural member flanges to the first fastener(s) shall be not less than  $d/8$ , where  $d$  is the overall depth of the flexural member.
- The length of the stiffener shall be not less than the depth of the flexural member minus 10 mm.
- The bearing width shall be not less than 40 mm.

### 3.3.8.3 Shear stiffeners

Where shear stiffeners are required, the spacing shall be such that the design shear force shall not exceed the design shear capacity ( $\phi_v V_v$ ) specified in Clause 3.3.4, and the ratio  $a/d_1$  shall exceed neither  $[260/(d_1/t)]^2$  nor 3.0.

The actual second moment of area ( $I_{s,min.}$ ) of a pair of attached shear stiffeners, or of a single shear stiffener, with reference to an axis in the plane of the web, shall have a minimum value as follows:

$$I_{s,min.} = 5d_1 t^3 \left[ \frac{d_1}{a} - 0.7 \left( \frac{a}{d_1} \right) \right] \geq \left( \frac{d_1}{50} \right)^4 \quad \dots 3.3.8.3(1)$$

The gross area of shear stiffeners ( $A_{st}$ ) shall be not less than—

$$A_{st} = \left( \frac{1 - k_s}{2} \right) \left\{ \frac{a}{d_1} - \frac{\left( \frac{a}{d_1} \right)^2}{\left[ \frac{a}{d_1} + \sqrt{1 + \left( \frac{a}{d_1} \right)^2} \right]} \right\} \psi k_{st} d_1 t \quad \dots 3.3.8.3(2)$$

where

$$\begin{aligned} k_s &= \text{shear stiffener coefficient} \\ &= \frac{1.53 E k_v}{f_y \left( \frac{d_1}{t} \right)^2} \text{ if } k_s \leq 0.8 \end{aligned} \quad \dots 3.3.8.3(3)$$

$$= \frac{1.11}{\left( \frac{d_1}{t} \right)} \left( \sqrt{\frac{k_v}{f_y}} \right) \text{ if } k_s > 0.8 \quad \dots 3.3.8.3(4)$$

$$\psi = \frac{\text{yield stress of web}}{\text{yield stress of stiffener}}$$

$$\begin{aligned} k_{st} &= \text{stiffener type coefficient} \\ &= 1.0 \text{ for stiffeners in pairs} \\ &= 1.8 \text{ for single-angle stiffeners} \\ &= 2.4 \text{ for single-plate stiffeners} \end{aligned}$$

$$\begin{aligned} k_v &= \text{shear buckling coefficient} \\ &= 4.00 + \frac{5.34}{\left( \frac{a}{d_1} \right)^2} \text{ if } \frac{a}{d_1} \leq 1.0 \text{ if } d_1 \leq 1.0 \end{aligned} \quad \dots 3.3.8.3(5)$$

$$= 5.34 + \frac{5.34}{\left( \frac{a}{d_1} \right)^2} \text{ if } \frac{a}{d_1} > 1.0 \text{ if } \frac{a}{d_1} > 1.0 \quad \dots 3.3.8.3(6)$$

$$a = \text{distance between transverse stiffeners}$$

### 3.3.8.4 Non-conforming stiffeners

The design capacities of members with transverse stiffeners that do not comply with Clauses 3.3.8.1 or 3.3.8.2, such as stamped or rolled-in transverse stiffeners, shall be determined by tests in accordance with Section 8.

## 3.4 CONCENTRICALLY LOADED COMPRESSION MEMBERS

### 3.4.1 General

This Clause applies to members in which the resultant of all loads acting on the member is an axial load passing through the centroid of the effective section calculated at the critical stress ( $f_n$ ). The design compressive axial force ( $N^*$ ) shall satisfy the following:

(a)  $N^* \leq \phi_c N_s$

(b)  $N^* \leq \phi_c N_c$

where

$\phi_c$  = capacity reduction factor for members in compression (see Table 1.6)

$N_s$  = nominal section capacity of the member in compression

$$= A_e f_y \quad \dots 3.4.1(1)$$

$A_e$  = effective area at yield stress ( $f_y$ )

$N_c$  = nominal member capacity of the member in compression

$$= A_e f_n \quad \dots 3.4.1(2)$$

$A_e$  = effective area at the critical stress ( $f_n$ ). For sections with circular holes,  $A_e$  shall be determined in accordance with Clause 2.2.2.2. If the product of the number of holes in the effective length region and the hole diameter divided by the effective length does not exceed 0.015,  $A_e$  can be determined ignoring the holes

$f_n$  = critical stress, and shall be determined from Equation 3.4.1(3) or Equation 3.4.1(4), as appropriate

$$\text{For } \lambda_c \leq 1.5: f_n = (0.658^{\lambda_c^2}) f_y \quad \dots 3.4.1(3)$$

$$\text{For } \lambda_c > 1.5: f_n = (0.877 / \lambda_c^2) f_y \quad \dots 3.4.1(4)$$

where

$\lambda_c$  = non-dimensional slenderness used to determine  $f_n$

$$= \sqrt{\frac{f_y}{f_{oc}}} \quad \dots 3.4.1(5)$$

$f_{oc}$  = least of the elastic flexural, torsional and flexural-torsional buckling stress determined in accordance with Clauses 3.4.2 to 3.4.4, or a rational elastic buckling analysis

Concentrically loaded angle sections shall be designed for an additional bending moment as specified in the definition of  $M_y^*$  specified in Clause 3.5.1.

NOTE: The slenderness ratio ( $l_e/r$ ) of all compression members should not exceed 200, except that only during construction  $l_e/r$  should not exceed 300.

### 3.4.2 Sections not subject to torsional or flexural-torsional buckling

For doubly-symmetric sections, closed cross-sections and any other sections that can be shown not to be subject to torsional or flexural-torsional buckling, the elastic flexural buckling stress ( $f_{oc}$ ) shall be determined as follows:

$$f_{oc} = \frac{\pi^2 E}{(l_e / r)^2} \quad \dots 3.4.2(1)$$

where

$l_e$  = effective length of member

$r$  = radius of gyration of the full, unreduced cross-section

For G550 steel to AS 1397 less than 0.9 mm in thickness, a reduced radius of gyration  $r$  shall be used in Equation 3.4.2(1) when the value of the effective length ( $l_e$ ) is less than  $1.1l_o$ ,

where

$$l_o = \pi r \sqrt{\frac{E}{f_{cr}}} \quad \dots 3.4.2(2)$$

$f_{cr}$  = plate elastic buckling stress

$$\gamma = 0.65 + \left( \frac{0.35l_e}{1.1l_o} \right) \quad \dots 3.4.2(3)$$

NOTES:

- 1 In frames where lateral stability is provided by diagonal bracing, shear walls, attachment to an adjacent structure having adequate lateral stability, or floor slabs or roof decks secured horizontally by walls or bracing systems parallel to the plane of the frame, and in trusses, the effective length ( $l_e$ ) for compression members that do not depend upon their own bending stiffness for lateral stability of the frame or truss should be taken as equal to the unbraced length ( $l$ ), unless analysis shows that a smaller value may be used.
- 2 In a frame that depends upon its own bending stiffness for lateral stability, the effective length ( $l_e$ ) of the compression members should be determined by a rational method and should be not less than the actual unbraced length.

### 3.4.3 Doubly- or singly-symmetric sections (see Figures 1.5(a) and (c)) subject to torsional or flexural-torsional buckling

For sections subject to torsional or flexural-torsional buckling,  $f_{oc}$  shall be taken as the smaller of  $f_{oc}$  calculated using Equation 3.4.2(1) with  $r = r_y$  and  $f_{oxz}$  calculated as follows:

$$f_{oxz} = \frac{1}{2\beta} \left[ (f_{ox} + f_{oz}) - \sqrt{(f_{ox} + f_{oz})^2 - 4\beta f_{ox} f_{oz}} \right] \quad \dots 3.4.3(1)$$

where  $f_{ox}$  and  $f_{oz}$  are specified in Clause 3.3.3.2.1(a).

Alternatively, a conservative estimate of  $f_{oxz}$  can be obtained from Equation 3.4.3(2) as follows:

$$f_{oxz} = f_{oz} f_{ox} / (f_{oz} + f_{ox}) \quad \dots 3.4.3(2)$$

where  $f_{ox}$  and  $f_{oz}$  are specified in Clause 3.3.3.2.1(a).

$$\beta = 1 - (x_o / r_{ol})^2 \quad \dots 3.4.3(3)$$

$r_{ol}$  = polar radius of gyration of the cross-section about the shear centre

For singly-symmetric sections, the  $x$ -axis shall be assumed to be the axis of symmetry.

For doubly-symmetric sections subject to torsional buckling,  $f_{oc}$  shall be taken as the smaller of  $f_{oc}$  calculated in accordance with Clause 3.4.2 and  $f_{oc} = f_{oz}$ , where  $f_{oz}$  is specified in Clause 3.3.3.2.1.

For singly-symmetric unstiffened angle sections for which the effective area ( $A_e$ ) at stress ( $f_y$ ) is equal to the full unreduced cross-sectional area ( $A$ ),  $f_{oc}$  shall be calculated using Equation 3.4.2 where  $r$  is the least radius of gyration.

#### 3.4.4 Point-symmetric sections (see Figure 1.5(b))

For point-symmetric sections subject to flexural or torsional buckling,  $f_{oc}$  shall be taken as the smaller of  $f_{oc}$  calculated in accordance with Clause 3.4.2 and  $f_{oz}$  specified in Clause 3.3.3.2(a).

#### 3.4.5 Non-symmetric sections (see Figure 1.5(d))

For shapes whose cross-sections do not have any symmetry, either about an axis or about a point,  $f_{oc}$  shall be taken as the smallest value which shall satisfy cubic Equation 3.4.5, as follows:

$$f_{oc}^3 (r_{o1}^2 - x_o^2 y_o^2) - f_{oc}^2 [r_{o1}^2 (f_{ox} + f_{oy} + f_{oz}) - (f_{oy} x_o^2 + f_{ox} y_o^2)] + f_{oc} r_{o1}^2 (f_{ox} f_{oy} + f_{oy} f_{oz} + f_{ox} f_{oz}) - (f_{ox} f_{oy} f_{oz} r_{o1}^2) = 0 \quad \dots 3.4.5$$

Alternatively, compression members composed of such shapes may be tested in accordance with Clause 6.2.

#### 3.4.6 Singly-symmetric sections (see Figure 1.5(c)) subject to distortional buckling

For monosymmetric sections subject to distortional buckling, such as lipped channels with additional rear flanges, the value of  $N_c$  in Equation 3.4.1(2) shall be the lesser of the following:

(a)  $A_e f_n$  calculated in accordance with Equations 3.4.1(3) and 3.4.1(4).

(b) For  $f_{od} > \frac{f_y}{2}$ :  $A f_n = A f_y \left( 1 - \frac{f_y}{4 f_{od}} \right) \quad \dots 3.4.6(1)$

(c) For  $\frac{f_y}{13} \leq f_{od} \leq \frac{f_y}{2}$ :  $A f_n = A f_y \left[ 0.055 \left( \sqrt{\frac{f_y}{f_{od}}} - 3.6 \right)^2 + 0.237 \right] \quad \dots 3.4.6(2)$

$f_{od}$  shall be calculated using either the appropriate equations given in Appendix D or a rational elastic buckling analysis.  $A$  is the area of the full cross-section.

#### 3.4.7 Columns with one flange through-fastened to sheeting

This Clause applies to channel- or Z-sections concentrically loaded along their longitudinal axis, with only one flange attached to sheeting by screw fasteners.

The nominal member capacity ( $N_c$ ) in axial compression of simple span or continuous channels or Z-sections shall be determined from Equation 3.4.7 as follows:

(a) For weak axis:

$$N_c = (0.79s + 0.54)(0.046t + 0.93)(0.1b_f - 0.064d + 22.8) \frac{EA}{29500} \quad \dots 3.4.7$$

where

$s$  = fastener distance from the centre-line of the web divided by the flange width for Z-sections; or

flange width minus the fastener distance from the centre-line of the web  
divided by the flange width for channel-sections

$t$  = thickness of the channel- or Z-section

$b_f$  = flange width of the channel- or Z-section

$d$  = depth of the channel- or Z-section

$A$  = gross cross-sectional area of the channel- or Z-section

NOTE: Units of  $t$ ,  $b_f$  and  $d$  in Equation 3.4.7 are in millimetres, since Equation 3.4.7 is not non-dimensional.

Equation 3.4.7 shall be limited to roof and wall systems complying with the following:

- (i) Channel- and Z-sections not exceeding 3.2 mm in thickness.
- (ii) Channel- and Z-sections with depths of 150 to 300 mm.
- (iii) Flanges are edge-stiffened compression elements.
- (iv)  $70 \leq \text{depth/thickness} \leq 170$ .
- (v)  $2.8 \leq \text{depth/flange width} < 5$ .
- (vi)  $16 \leq \text{flat width/thickness of flange} < 50$ .
- (vii) Both flanges prevented from moving laterally at the supports.
- (viii) Roof or wall panels with fasteners spaced 300 mm on centre or less and having a minimum rotational lateral stiffness of  $10.3 \text{ kN/mm}^2$  [fastener at mid-flange width as determined by the relevant AISI test procedure (see Note).]  
NOTE: See the test procedure titled 'Rational-lateral stiffness test method for beam-to-panel assemblies' in Part IV of the AISI Cold-Formed Design Manual, 2002 Edition.
- (ix) Minimum yield stress of 230 MPa.
- (x) Span lengths from 4.5 to 9 m.

(b) *For strong axis* The equations given in Clauses 3.5.1 and 3.5.2 shall be used.

### 3.5 COMBINED AXIAL COMPRESSION OR TENSION, AND BENDING

#### 3.5.1 Combined axial compression and bending

The design axial compression ( $N^*$ ), and the design bending moments ( $M_x^*$  and  $M_y^*$ ) about the  $x$ - and  $y$ -axes of the effective section, respectively, shall satisfy the following:

$$(a) \quad \frac{N^*}{\phi_c N_c} + \frac{C_{mx} M_x^*}{\phi_b M_{bx} \alpha_{nx}} + \frac{C_{my} M_y^*}{\phi_b M_{by} \alpha_{ny}} \leq 1.0 \quad \dots 3.5.1(1)$$

$$(b) \quad \frac{N^*}{\phi_c N_s} + \frac{M_x^*}{\phi_b M_{bx}} + \frac{M_y^*}{\phi_b M_{by}} \leq 1.0 \quad \dots 3.5.1(2)$$

If  $N^*/\phi_c N_c \leq 0.15$ , the following interaction may be used in lieu of Items (a) and (b):

$$\frac{N^*}{\phi_c N_c} + \frac{M_x^*}{\phi_b M_{bx}} + \frac{M_y^*}{\phi_b M_{by}} \leq 1.0 \quad \dots 3.5.1(3)$$

where

$N_c$  = nominal member capacity of the member in compression determined in accordance with Clause 3.4

$C_{mx}, C_{my}$  = coefficients for unequal end moment whose value shall be taken as follows:

- (i) For compression members in frames subject to joint translation (side-sway):

$$C_m = 0.85$$

- (ii) For restrained compression members in frames braced against joint translation and not subject to transverse loading between their supports in the plane of bending:

$$C_m = 0.6 - 0.4(M_1/M_2) \quad \dots 3.5.1(4)$$

$M_1/M_2$  is the ratio of the smaller to the larger moment at the ends of that portion of the member under consideration which is unbraced in the plane of bending.  $M_1/M_2$  is positive if the member is bent in reverse curvature and negative if it is bent in single curvature.

- (iii) For compression members in frames braced against joint translation in the plane of loading and subject to transverse loading between their supports, the value of  $C_m$  may be determined by rational analysis. However, in lieu of such analysis, the following values may be used:

- (A) For members whose ends are restrained

$$C_m = 0.85$$

- (B) For members whose ends are unrestrained

$$C_m = 1.0$$

$M_x^*, M_y^*$  = design bending moment about the  $x$ - and  $y$ -axes of the effective section, respectively, determined for the design axial force alone

For singly-symmetric unstiffened angle sections with unreduced effective area,  $M_y^*$  shall be permitted to be taken as the design bending moment only. For other angle sections or singly-symmetric unstiffened angles for which the effective area ( $A_e$ ) at stress ( $f_y$ ) is less than the full unreduced cross-sectional area ( $A$ ),  $M_y^*$  shall be taken either as the design bending moment, or the required flexural moment plus  $N^*/1000$ , whichever results in a lower value of  $N_c$ .

$M_{bx}, M_{by}$  = nominal member moment capacity about the  $x$ - and  $y$ -axes, respectively, determined in accordance with Clause 3.3.3

$\phi_b$  = capacity reduction factor for bending  
 = 0.95 and 0.90 for bending strength (see Table 1.6); or  
 0.90 for laterally unbraced beam (see Table 1.6)

$\phi_c$  = capacity reduction factor for members in compression  
 = 0.85

$N_s$  = nominal section capacity of the member in compression determined in accordance with Clause 3.4, with  $f_n$  equal to  $f_y$

$\alpha_{nx}, \alpha_{ny}$  = moment amplification factors



$$= 1 - \left( \frac{N^*}{N_e} \right) \quad \dots 3.5.1(5)$$

$N_e$  = elastic buckling load

$$= \frac{\pi^2 EI_b}{(l_{eb})^2} \quad \dots 3.5.1(6)$$

$I_b$  = second moment of area of the full, unreduced cross-section about the bending axis

$l_{eb}$  = effective length in the plane of bending

In addition, each individual ratio in Equations 3.5.1(1), 3.5.1(2) and 3.5.1(3) shall not exceed unity.

### 3.5.2 Combined axial tension and bending

The design axial tension ( $N^*$ ), and the required bending moments ( $M_x^*$ ) and ( $M_y^*$ ) about the  $x$ - and  $y$ -axes of the effective section, respectively, shall satisfy the following:

$$(a) \quad \frac{M_x^*}{\phi_b M_{bx}} + \frac{M_y^*}{\phi_b M_{by}} - \frac{N^*}{\phi_t N_t} \leq 1.0 \quad \dots 3.5.2(1)$$

$$(b) \quad \frac{N^*}{\phi_t N_t} + \frac{M_x^*}{\phi_b M_{sxf}} + \frac{M_y^*}{\phi_b M_{syf}} \leq 1.0 \quad \dots 3.5.2(2)$$

where

$N_t$  = nominal section capacity of the member in tension determined in accordance with Clause 3.2

$M_{sxf}, M_{syf}$  = nominal section yield moment capacity of the full section about the  $x$ -axis and  $y$ -axis, respectively

$$= Z_{ft} f_y \quad \dots 3.5.2(3)$$

$Z_{ft}$  = section modulus of the full unreduced section for the extreme tension fibre about the appropriate axis

$M_{bx}, M_{by}$  = nominal member moment capacity about the  $x$ - and  $y$ -axes, respectively, of the effective section

## 3.6 CYLINDRICAL TUBULAR MEMBERS

### 3.6.1 General

This Clause applies to cylindrical tubular members having a ratio of outside diameter to wall thickness ( $d_o/t$ ) not greater than  $0.441E/f_y$ .

### 3.6.2 Bending

For flexural members, the design bending moment ( $M^*$ ) uncoupled from axial load, shear and local concentrated forces or reactions shall satisfy—

$$M^* \leq \phi_b M_b$$

where  $M_b$  is the nominal member moment capacity and shall be calculated from Equations 3.6.2(1) to 3.6.2(3), as appropriate.

$$\text{For } d_o/t \leq 0.0714E/f_y: \quad M_b = 1.25f_y Z_f \quad \dots 3.6.2(1)$$

$$\text{For } 0.0714E/f_y < d_o/t \leq 0.318E/f_y: \quad M_b = [0.970 + 0.020(E/f_y)/(d_o/t)]f_y Z_f \quad \dots 3.6.2(2)$$

$$A1 \quad \left| \quad \text{For } 0.318E/f_y < d_o/t \leq 0.441E/f_y, \quad M_b = 0.328E/(d_o/t)Z_f \quad \dots 3.6.2(3) \right.$$

where  $d_o$  is the outside diameter of the tubular member.

### 3.6.3 Compression

This Clause applies to members in which the resultant of all design loads and design bending moments acting on the member is equivalent to a single force in the direction of the member axis passing through the centroid of the section.

The design axial load ( $N^*$ ) shall satisfy—

$$N^* \leq \phi_c N_c$$

where

$$N_c = f_n A_e \quad \dots 3.6.3(1)$$

$f_n$  = critical stress and shall be calculated from Equation 3.6.3(2) or Equation 3.6.3(3), as appropriate

$$= (0.658 \lambda_c^2) f_y \text{ for } \lambda_c \leq 1.5 \quad \dots 3.6.3(2)$$

$$= \left( \frac{0.877}{\lambda_c^2} \right) f_y \text{ for } \lambda_c > 1.5 \quad \dots 3.6.3(3)$$

$\lambda_c$  = slenderness factor

$$= \sqrt{\frac{f_y}{f_{oc}}} \quad \dots 3.6.3(4)$$

$f_{oc}$  = elastic flexural buckling stress determined in accordance with Clause 3.4.1

$A_e$  = effective area at the critical stress ( $f_n$ )

$$= A_o + R(A - A_o) \quad \dots 3.6.3(5)$$

$A_o$  = reduced area due to local buckling

$$= \left[ \left( \frac{0.037}{d_o f_y / t E} \right) + 0.667 \right] A \leq A \text{ for } \frac{d_o}{t} \leq 0.441 \left( \frac{E}{f_y} \right) \quad \dots 3.6.3(6)$$

$R$  = reduction factor

$$= \frac{f_y}{2 f_e} \leq 1.0 \quad \dots 3.6.3(7)$$

$A$  = area of the full, unreduced cross-section

### 3.6.4 Combined bending and compression

Combined bending and compression shall be in accordance with Clause 3.5.

## SECTION 4 STRUCTURAL ASSEMBLIES

### 4.1 BUILT-UP SECTIONS

#### 4.1.1 I-sections composed of two channels

The maximum longitudinal spacing of welds or other connectors ( $s_{\max.}$ ) joining two channels to form an I-section shall be determined as follows:

(a) For compression members—

$$s_{\max.} = \frac{lr_{cy}}{2r_1} \quad \dots 4.1.1(1)$$

where

$l$  = unbraced length of the member in compression

$r_{cy}$  = radius of gyration of one channel about its centroidal axis parallel to the web

$r_1$  = radius of gyration of the I-section about the axis perpendicular to the direction in which buckling occurs for the given conditions of end support and intermediate bracing

(b) For flexural members—

$$s_{\max.} = \frac{l}{6} \quad \dots 4.1.1(2)$$

$$\leq \frac{2s_g N^*}{mq}$$

where

$l$  = span of beam

$s_g$  = vertical distance between two rows of connections nearest to the top and bottom flanges

$N^*$  = design tensile force of the connection

$q$  = intensity of the design action on the beam

$m$  = distance from the shear centre of one channel to the mid-plane of its web (see Table E1 of Appendix E)

The intensity of the design load ( $q$ ) shall be obtained by dividing the magnitude of the design concentrated actions or reactions by the length of bearing. For beams designed for a uniformly distributed load,  $q$  shall be equal to three times the intensity of the uniformly distributed design action. If the length of bearing of a concentrated action or reaction is less than the weld spacing ( $s_w$ ), the design tensile force of the welds or connections closest to the load or reaction shall be determined as follows:

$$N^* = \frac{mR_b^*}{2s_g} \quad \dots 4.1.1(3)$$

where  $R_b^*$  is the design concentrated action or reaction.

The maximum spacing of connections ( $s_{\max}$ ) depends upon the intensity of the action applied directly at the connection. Therefore, if uniform spacing of connections is used over the whole length of the beam, it shall be determined at the point of maximum local load intensity. In cases where this procedure may result in uneconomically close spacing, either of the following methods may be adopted:

- (i) The connection spacing may be varied along the beam in accordance with the variation of the load intensity.
- (ii) The reinforcing cover plates may be welded to the flanges at points where concentrated loads occur. The design shear force of the connections joining these plates to the flanges shall then be used for  $N^*$ , and  $s_g$  shall be taken as the depth of the beam.

#### 4.1.2 Cover plates, sheets or non-integral stiffeners in compression

The spacing ( $s$ ) in the line of stress of welds, bolts or rivets connecting a cover plate, sheet, or a non-integral stiffener in compression to another element shall not exceed—

- (a) that which is required to transmit the shear between the connected parts on the basis of the design shear force per connection specified in this Clause;
- (b)  $1.16t\sqrt{(E/f_c)}$ , where  $t$  is the thickness of the cover plate or sheet, and  $f_c$  is the stress at unfactored load in the cover plate or sheet; and
- (c) three times the flat width ( $b$ ) of the narrowest unstiffened compression element in that portion of the cover plate or sheet that is tributary to the connections, but not less than  $1.11t\sqrt{(E/f_y)}$  if  $b/t < 0.50\sqrt{(E/f_y)}$ , or  $1.33t\sqrt{(E/f_y)}$  if  $b/t \geq 0.50\sqrt{(E/f_y)}$  unless closer spacing is required by Item (a) or (b).

In the case of intermittent fillet welds parallel to the direction of stress, the spacing shall be taken as the clear distance between welds plus 12 mm. In all other cases, the spacing shall be taken as the centre-to-centre distance between connections.

This Clause does not apply to cover sheets that act only as sheeting material and shall not be considered as load-carrying elements.

## 4.2 MIXED SYSTEMS

The design of members in mixed systems using cold-formed steel components in conjunction with other materials shall conform to this Standard and to the relevant material Standard.

## 4.3 LATERAL RESTRAINTS

### 4.3.1 General

Lateral restraints required to restrain lateral bending or twisting of a loaded beam or column shall be in accordance with Clauses 4.3.2 and 4.3.3. Local buckling at the points of attachment of the restraints shall be avoided.

### 4.3.2 Symmetrical beams and columns

#### 4.3.2.1 General

Restraints and restraining systems, including connections, shall be designed in accordance with strength and stiffness requirements.

#### 4.3.2.2 *Restraint against lateral deflection*

The lateral restraint at any cross-section of the member being restrained shall be designed to transfer a force acting at the critical flange equal to 0.025 times the maximum force in the critical flanges of the adjacent segments or sub-segments, except where the restraints are more closely spaced than is required to ensure that  $M^* = \phi_b M_b$ .

If the restraints are more closely spaced, then the restraint may be designed for a lesser force. The actual arrangement of restraints shall be assumed to be equivalent to a set of restraints that will ensure that  $M^* = \phi_b M_b$ . Each equivalent restraint shall correspond to an appropriate group of actual restraints. This group shall then be designed as a whole to transfer the force of 0.025 times the maximum force in the critical flanges of the equivalent adjacent segments or sub-segments.

#### 4.3.2.3 *Restraint against twist rotation*

A torsional restraint at a cross-section of the member being restrained may be deemed to provide effective restraint against twist rotation if it is designed to transfer a force equal to 0.025 times the maximum force in the critical flange from any unrestrained flange to the lateral restraint.

#### 4.3.2.4 *Parallel restrained members*

If a series of parallel members is restrained by a line of restraints, each restraining element shall be designed to transfer a force equal to the sum of 0.025 times the flange force from the connected member and 0.0125 times the sum of the flange forces in the connected members beyond, except that no more than seven members need be considered.

#### 4.3.2.5 *Restraint against lateral rotation*

A rotational restraint at a cross-section of the member being restrained may be deemed to provide restraint against lateral rotation out of the plane of bending, provided its flexural stiffness in the plane of rotation is comparable with the corresponding stiffness of the restrained member.

### 4.3.3 **Channel- and Z-section beams**

#### 4.3.3.1 *General*

The requirements for bracing to restrain twisting of channel- and Z-sections used as beams and loaded in the plane of the web, apply only if—

- (a) the top flange is connected to the deck or sheeting material in accordance with Clauses 4.3.3.2 and 4.3.3.3 so as to effectively restrain lateral deflection of the connected flange; or
- (b) neither flange is connected.

If both flanges are connected, further bracing is not required.

#### 4.3.3.2 *One flange connected to sheeting and subjected to wind uplift*

Channel- and Z-sections, used to support attached covering material and loaded in a plane parallel to the web, shall be designed taking into account the restraining effects of covering materials and fasteners. Provisions shall be made for the forces, from each beam, that accumulate in the covering material. These forces shall be transferred from the covering material to a member or assembly of sufficient strength and stiffness to resist these forces.

NOTE: This may be achieved by one of the following means:

- (a) A system of bridging or bracing members sufficiently strong to carry the forces to a stiff member.
- (b) Arranging equally loaded alternating members to oppose each other.
- (c) A diaphragm with sufficient rigidity to transfer the forces to a stiff perimeter member, coupled with devices (e.g., cleats), which restrain rotation of the beams at their supports.

- (d) Direct axial stress in the roof sheets. The forces in this case may be taken out at the roof where equal and opposite forces meet.
- (e) Other designs in which the forces might be transferred to stiff members at the eaves, such as the eaves struts in a shed roof.

For beam systems that satisfy the cleat and screw-fastening requirements of Clause 3.3.3.4, Items (ix) to (xiv), the bracing does not need to be connected to a stiff member but shall be capable of preventing torsional deformation of the beam at the point of attachment.

#### 4.3.3.3 Bracing for roof systems under gravity load

For channel- and Z-sections, designed in accordance with Clause 3.3.3 and having the sheeting fastened directly to the top flanges in such a manner that inhibits relative movement between the sheeting and the purlin flange, provisions shall be made to restrain the flanges so that the maximum top flange lateral displacements with respect to the purlin reaction points shall not exceed the span length divided by 360. If the top flanges of all purlins face in the same direction, anchorage of the restraint system shall comply with Items (a) and (b) of this Clause. If the top flanges of adjacent lines of purlins face in opposite directions, a restraint system shall be provided to resist the downward component of the total gravity load.

Anchored braces may be connected to only one line of purlins in each purlin bay of each roof slope, if provision is made to transmit forces from other purlin lines through the sheeting and its fastening system. Anchored braces shall be as close as possible to the flange that is connected to the sheeting. Anchored braces shall be provided for each purlin bay.

For bracing arrangements other than those specified in Items (a) and (b), tests in accordance with Section 6 shall be performed so that the type and spacing of braces selected are such that the test capacity of the braced purlin assembly is greater than or equal to its nominal flexural capacity, instead of that required by Section 6.

For roof systems using channel- and Z-sections, the following shall be considered:

- (a) *Channel-sections* For roof systems using channel-sections for purlins with all compression flanges facing in the same direction, a system possessing the following restraint force  $N_{ib}^*$ , in addition to other loads, shall be provided:

$$N_{ib}^* = (0.05\alpha \cos\theta - \sin\theta)F_p^* \quad \dots 4.3.3.3(1)$$

where

$N_{ib}^*$  = design force to be resisted by intermediate beam brace

$\alpha$  = coefficient

= +1 for purlin facing upward direction

= -1 for purlin facing downward direction

$\theta$  = angle between the vertical and the plane of the web of the channel-section

$F_p^*$  = total vertical design load supported by all purlin lines being restraint. Where more than one brace is used at a purlin,  $N_{ib}^*$  shall be divided equally between all braces

NOTE: A positive value for  $N_{ib}^*$  means that restraint is required to prevent movement of the purlin flanges in the upward roof slope direction, and a negative value means that restraint is required to prevent movement of purlin flanges in the downward slope direction.

- (b) *Z-sections* For roof systems with a diaphragm stiffness of at least 350 kN/m, and 4 to 20 Z-purlin lines with all top flanges facing in the direction of the upward roof slope, and with restraint braces at the purlin supports, midspan or one-third points, each brace shall be designed to resist a design force ( $N_{ib}^*$ ) determined as follows:

- (i) Single-span system with restraint at the supports:

$$N_{ib}^* = 0.5 \left( \frac{0.22b_f^{1.50} \cos \theta}{n_p^{0.72} d^{0.90} t^{0.60}} - \sin \theta \right) F_p^* \quad \dots 4.3.3.3(2)$$

- (ii) Single-span system with third-point restraints:

$$N_{ib}^* = 0.5 \left( \frac{0.474b_f^{1.22} \cos \theta}{n_p^{0.57} d^{0.89} t^{0.33}} - \sin \theta \right) F_p^* \quad \dots 4.3.3.3(3)$$

- (iii) Single-span system with midspan restraint:

$$N_{ib}^* = \left( \frac{0.22b_f^{1.32} \cos \theta}{n_p^{0.65} d^{0.83} t^{0.50}} - \sin \theta \right) F_p^* \quad \dots 4.3.3.3(4)$$

- (iv) Multiple-span system with restraints at the supports:

$$N_{ib}^* = C_{tr} \left( \frac{0.053b_f^{1.88} l^{0.13} \cos \theta}{n_p^{0.95} d^{0.94}} - \sin \theta \right) F_p^* \quad \dots 4.3.3.3(5)$$

$C_{tr}$  = coefficient used to determine  $N_{ib}^*$  for multiple-span system with restraints at the supports

= 0.63 for braces at end supports of multiple-span systems

= 0.87 for braces at the first interior supports

= 0.81 for all other braces

- (v) Multiple-span system with third-point restraints:

$$N_{ib}^* = C_{th} \left( \frac{0.181b_f^{1.15} l^{0.25} \cos \theta}{n_p^{0.54} d^{1.11} t^{0.29}} - \sin \theta \right) F_p^* \quad \dots 4.3.3.3(6)$$

$C_{th}$  = coefficient used to determine  $N_{ib}^*$  for multiple-span system with third-point restraints

= 0.57 for outer braces in exterior spans

= 0.48 for all other braces

- (vi) Multiple-span system with midspan restraints:

$$N_{ib}^* = C_{ms} \left( \frac{0.116b_f^{1.32} l^{0.18} \cos \theta}{n_p^{0.70} d^{1.00} t^{0.50}} - \sin \theta \right) F_p^* \quad \dots 4.3.3.3(7)$$

$C_{ms}$  = coefficient used to determine  $N_{ib}^*$  for multiple-span system with midspan restraints

= 1.05 for braces in exterior spans

= 0.90 for all other braces

where

$b_f$  = flat width of flange

$n_p$  = number of parallel purlin lines

$t$  = thickness of the section

$\theta$  = angle between the vertical and the plane of the web of the Z-section

$l$  = length of the member

$F_p^*$  = total design load supported by the purlin lines between adjacent supports

The design force ( $N_{ib}^*$ ) is positive if restraint is required to prevent movement of the purlin flanges in the upward roof slope direction.

For systems with less than four purlin lines, the brace force shall be determined by taking 1.1 times the force calculated using Equations 4.3.3.3(2) to 4.3.3.3(7), with  $n_p = 4$ . For systems with more than 20 purlin lines, the brace force shall be determined using Equations 4.3.3.3(2) to 4.3.3.3(7) with  $n_p = 20$  and  $F_p^*$  based on the total number of purlins.

#### 4.3.3.4 Neither flange connected to sheeting or connected to sheeting with concealed fasteners

Each intermediate brace, at the top and bottom flanges, shall be designed to resist a horizontal design force ( $N_{ib}^*$ ) determined as follows:

A1

- (a) For uniformly distributed loads,  $N_{ib}^* = 1.5k'$  times the design load within a distance  $0.5l_b$  each side of the brace, where  $l_b$  is the distance between centre-line of braces.
- (b) For concentrated loads,  $N_{ib}^* = 1.0k'$  times each design concentrated load within a distance  $0.3l_b$  each side of the brace, plus  $1.4k'(1 - m/l_b)$  times each design concentrated load located farther than  $0.3a$  but not farther than  $1.0a$  from the brace, where  $m$  is the distance from the concentrated load to the brace and  $a$  is the distance between centre-lines of braces.

For channel-sections:

$$k' = \frac{m}{d} \quad \dots 4.3.3.4(1)$$

For Z-sections:

$$k' = \frac{I_{x'y'}}{I_{x'}} \quad \dots 4.3.3.4(2)$$

where

$k'$  = coefficient used to determine  $N_{ib}^*$  where neither flange is connected to the sheeting or connected to the sheeting with concealed fasteners

$I_{x'y'}$  = product of second moment of area of the full section about its major and minor principal axes parallel and perpendicular to the web

$I_{x'}$  = second moment of area of the cross-section about its centroidal axis perpendicular to the web

Braces shall be designed to avoid local buckling at the points of attachment to the member.



Braces shall be attached in such a manner as to prevent lateral deflection of either flange in either direction at intermediate braces. If one-third or more of the total design load on the beam is concentrated over a length of one-twelfth or less of the span of the beam, an additional brace shall be placed at or near the centre of this loaded length.

Other braces are not required if all loads and reactions on a beam are transmitted through members that frame into the section in such a manner as to effectively restrain the section against rotation and lateral displacement.

#### **4.4 WALL STUDS AND WALL STUD ASSEMBLIES**

The design capacity of a stud may be calculated in accordance with Section 3 (neglecting sheeting and using steel only) or on the basis that sheeting (attached to one or both sides of the stud) produces lateral and rotational support to the stud in the plane of the wall, provided that the stud, sheeting, and attachments comply with the following:

- (a) Both ends of the stud shall be braced to restrain rotation about the longitudinal stud axis and horizontal displacement perpendicular to the stud axis. However, the ends may or may not be free to rotate about both axes perpendicular to the stud axis. The sheeting shall be connected to the top and bottom members of the wall assembly to enhance the restraint provided to the stud and stabilize the overall assembly.

If intermediate braces such as noggings (dwangs) are used for stability at points along the wall stud for systems either with or without sheeting, they shall be connected to the stud so as to resist lateral and torsional deformation of the stud at the points of connection.

- (b) If sheeting is used for stability of the wall studs, the sheeting shall retain its capacity and stiffness for the expected service life of the wall and additional bracing shall be provided for the required structural integrity during construction and in the completed structure.

## SECTION 5 CONNECTIONS

## 5.1 GENERAL

Any suitable fastening system, such as welding, bolting, screwing, riveting, clinching, nailing, structural adhesive or other mechanical means, may be used to join component parts.

Connection elements consist of members, connection components (cleats, gusset plates, brackets, connecting plates) and connectors (welds, bolts, screws, rivets, clinches, nails, adhesives). The connections in a structure shall be proportioned so as to be consistent with the assumptions made in the analysis of the structure and to comply with this Section. Connections shall be capable of transmitting the design action effects [design action] calculated from this analysis.

Design capacities of specific connections may be obtained by prototype testing in accordance with Section 8.

## 5.2 WELDED CONNECTIONS

## 5.2.1 General

This Clause applies to welded connections for cold-formed steel structural members in which the weld is produced by the arc welding or resistance welding processes.

Arc-welded connections, where at least one of the connected parts is less than 3 mm thick, or less than 2.5 mm thick for fillet welds, shall be in accordance with AS/NZS 1554.7. The arc weld design capacities shall be determined in accordance with Clauses 5.2.2 to 5.2.6, as appropriate.

Arc-welded connections, where each connected part is greater than or equal to 3 mm thick, or greater than or equal to 2.5 mm thick for fillet welds, shall be in accordance with AS/NZS 1554.1 or AS/NZS 1554.2, as appropriate. The arc weld design capacities shall be determined in accordance with AS 4100 or NZS 3404, as appropriate.

Resistance welds shall be in accordance with AWS C1.1 or AWS C1.3, as appropriate. The resistance weld design capacities shall be determined in accordance with Clause 5.2.7.

## NOTES:

- 1 For high-strength cold-rolled steel, reference should be made to Clause 1.5.1.4 for design strength reduction near welds.
- 2 AS/NZS 1554.1 requires designers to specify the required weld category, these being either GP (general purpose) or SP (structural purpose), and any associated non-destructive examination requirements.

## 5.2.2 Butt welds

## 5.2.2.1 Tension or compression

The design tensile or compressive force ( $N_w^*$ ) normal to the area of a butt weld shall satisfy—

$$N_w^* \leq \phi N_w \quad \dots 5.2.2.1(1)$$

where

$\phi$  = capacity reduction factor of a butt weld in tension or compression (see Table 1.6)

$N_w$  = nominal tensile or compressive capacity of a butt weld

The nominal tensile or compressive capacity of a butt weld, welded from one or both sides, shall be determined as follows:

$$N_w = l_w t_f f_y \quad \dots 5.2.2.1(2)$$

where

$l_w$  = length of the full size of the weld

$t_t$  = design throat thickness of a butt weld as defined in AS/NZS 1554.1

$f_y$  = yield stress used in design for the lower strength base steel

### 5.2.2.2 Shear

The design shear force ( $V_w^*$ ) on a butt weld shall satisfy—

$$V_w^* \leq \phi V_w \quad \dots 5.2.2.2(1)$$

where

$\phi$  = capacity reduction factor of a butt weld in shear (see Table 1.6)

$V_w$  = nominal shear capacity of a butt weld

The design shear capacity ( $\phi V_w$ ) of a butt weld shall be the lesser of the following:

(a)  $\phi = 0.80$

$$V_w = l_w t_t (0.6 f_{uw}) \quad \dots 5.2.2.2(2)$$

(b)  $\phi = 0.90$

$$V_w = l_w t_t \left( \frac{f_y}{\sqrt{3}} \right) \quad \dots 5.2.2.2(3)$$

where  $f_{uw}$  is the nominal tensile strength of the weld metal.

### 5.2.3 Fillet welds

#### 5.2.3.1 General

A fillet weld subject to a design shear force ( $V_w^*$ ) shall satisfy—

$$V_w^* \leq \phi V_w \quad \dots 5.2.3.1$$

where

$\phi$  = capacity reduction factor of a fillet weld (see Table 1.6)

$V_w$  = nominal shear capacity of a fillet weld

The design shear capacity ( $\phi V_w$ ) of a fillet weld shall satisfy Clauses 5.2.3.2, 5.2.3.3 and 5.2.3.4.

#### 5.2.3.2 Longitudinal loading

For longitudinal loading,  $\phi V_w$  shall be determined as follows from the lesser of Items (a)(i) and (b)(i), or the lesser of Items (a)(ii) and (b)(ii), as applicable, as follows:

(a) (i) For  $\frac{l_w}{t_1} < 25$ :

$$\phi = 0.60$$

$$V_w = \left[ 1 - \frac{0.01 l_w}{t_1} \right] t_1 l_w f_{ul} \quad \dots 5.2.3.2(1)$$

(ii) For  $\frac{l_w}{t_2} < 25$ :

$$\phi = 0.60$$

$$V_w = \left[ 1 - \frac{0.01l_w}{t_2} \right] t_2 l_w f_{u2} \quad \dots 5.2.3.2(2)$$

(b) (i) For  $\frac{l_w}{t_1} \geq 25$ :

$$\phi = 0.55$$

$$V_w = 0.75 t_1 l_w f_{u1} \quad \dots 5.2.3.2(3)$$

(ii) For  $\frac{l_w}{t_2} \geq 25$ :

$$\phi = 0.55$$

$$V_w = 0.75 t_2 l_w f_{u2} \quad \dots 5.2.3.2(4)$$

For Grade G450 steel specified in AS 1397, the capacity factor of 0.55 shall be used throughout Clause 5.2.3.2.

### 5.2.3.3 Transverse loading

For transverse loading,  $\phi V_w$  shall be determined as follows:

$$\phi = 0.60$$

$$V_w = t_1 l_w f_{u1}; \text{ or} \quad \dots 5.2.3.3(1)$$

$$= t_2 l_w f_{u2} \quad \dots 5.2.3.3(2)$$

whichever is the lesser

where

$t_1, t_2$  = thickness of the two connecting plates of the tensile strengths  $f_{u1}$  and  $f_{u2}$ , respectively (see Figures 5.2.3(a) and (b))

$l_w$  = length of fillet weld

$f_{u1}, f_{u2}$  = tensile strength used in the design of the two connecting plates of the thicknesses  $t_1$  and  $t_2$ , respectively

Where inclination failure can occur, a reduced capacity factor of 0.55 shall be used.

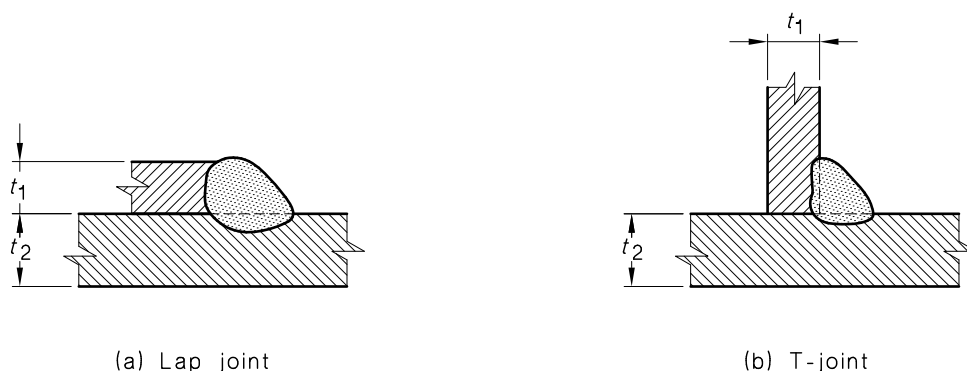


FIGURE 5.2.3 FILLET WELDS

#### 5.2.3.4 Longitudinal and transverse loading

For longitudinal and transverse loading, or both,  $\phi V_w$  shall be determined as follows:

For  $t \geq 2.5$  mm

$$\phi = 0.60$$

$$V_w = 0.75 t_t l_w f_{uw} \quad \dots 5.2.3.4(1)$$

where

$$t_t = \text{design throat thickness of the weld (see Figure 5.2.3)} \\ = 0.707 t_{w1} \text{ or } 0.707 t_{w2}, \text{ whichever is the smaller} \quad \dots 5.2.3.4(2)$$

$t_{w1}, t_{w2}$  = leg lengths of fillet weld

$l_w$  = length of fillet weld

$f_{uw}$  = nominal tensile strength of weld metal

A larger design throat thickness than those calculated using Equation 5.2.3.4(2) shall be permitted if measurement shows that the welding procedure used consistently yields the larger value.

#### 5.2.4 Arc spot welds (puddle welds)

##### 5.2.4.1 General

Arc spot welds (see Figure 5.2.4(A)) apply to welding sheet steel to thicker supporting members in the flat position or sheet to sheet in the flat position. Arc spot welds shall not be made on steel where the thinnest connected part is greater than 3 mm thick, nor through a combination of steel sheets having a total thickness greater than 3 mm.

Weld washers (see Figures 5.2.4(B)) shall be used if the thickness of the sheet is less than 0.7 mm. Weld washers shall have a thickness between 1.3 and 2.0 mm with a minimum prepunched hole of 10 mm diameter.

Sheet to sheet welds do not require weld washers.

Arc spot welds shall be specified by the minimum effective diameter of the fused area ( $d_e$ ) (see Figure 5.2.4(A)). The minimum effective diameter shall be 10 mm.

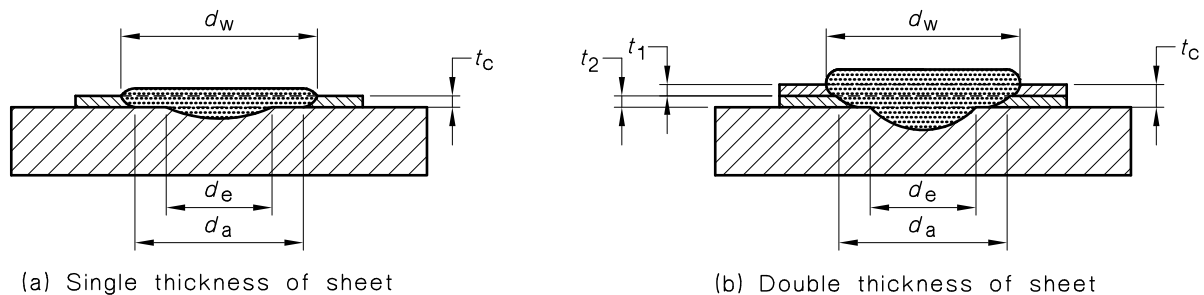
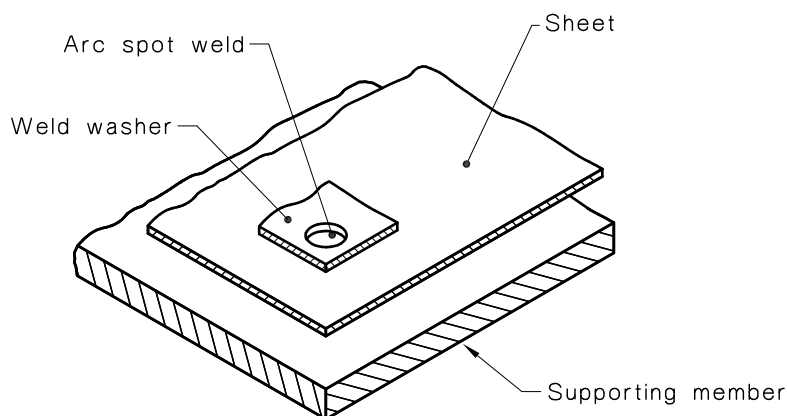
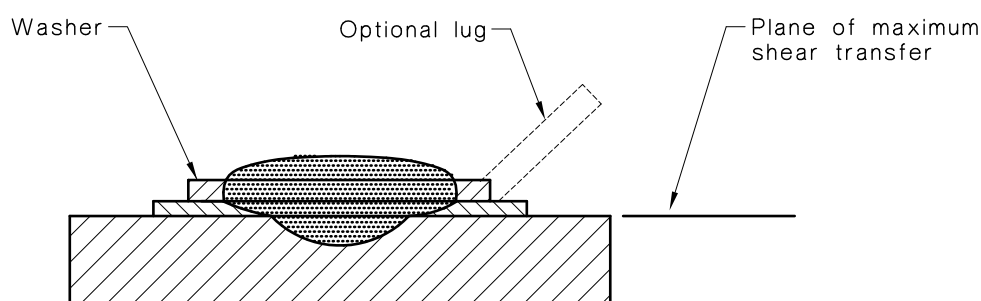


FIGURE 5.2.4(A) ARC SPOT WELDS



(a) Typical weld washer



(b) Arc spot weld using washer

FIGURE 5.2.4(B) WELD WASHER FOR ARC SPOT WELDS

**5.2.4.2 Shear**

The design shear force ( $V_w^*$ ) on an arc spot weld shall satisfy—

$$V_w^* \leq \phi V_w \quad \dots 5.2.4.2(1)$$

where

$\phi$  = capacity reduction factor of an arc spot weld in shear (see Table 1.6)

$V_w$  = nominal shear capacity of an arc spot weld

The design shear capacity ( $\phi V_w$ ) of each arc spot weld between sheet or sheets, and supporting member shall be the lesser of Item (a) and whichever is applicable of Items (b)(i), (ii) and (iii), as follows:

(a)  $\phi = 0.60$

$$V_w = 0.589 d_e^2 f_{uw} \quad \dots 5.2.4.2(2)$$

(b) (i) For  $\frac{d_a}{t_c} \leq 0.815 \sqrt{\frac{E}{f_u}}$ :

$$\phi = 0.60$$

$$V_w = 2.20 t_c d_a f_u \quad \dots 5.2.4.2(3)$$

$$(ii) \quad \text{For } 0.815 \sqrt{\frac{E}{f_u}} < \left( \frac{d_a}{t_c} \right) < 1.397 \sqrt{\frac{E}{f_u}} :$$

$$\phi = 0.50$$

$$V_w = 0.280 \left[ 1 + 5.59 \frac{\sqrt{\frac{E}{f_u}}}{\frac{d_a}{t_c}} \right] t_c d_a f_u \quad \dots 5.2.4.2(4)$$

$$(iii) \quad \text{For } \frac{d_a}{t_c} \geq 1.397 \sqrt{\frac{E}{f_u}} :$$

$$\phi = 0.50$$

$$V_w = 1.40 t_c d_a f_u \quad \dots 5.2.4.2(5)$$

where

$$\begin{aligned} d_e &= \text{effective diameter of fused area (see Figures 5.2.4(A))} \\ &= (0.7d_w - 1.5t_c) \leq 0.55d_w \quad \dots 5.2.4.2(6) \end{aligned}$$

$d_w$  = visible diameter of outer surface of arc spot weld

$E$  = Young's modulus of elasticity ( $200 \times 10^3$  MPa)

$d_a$  = average diameter of an arc spot weld at mid-thickness of  $t_c$  (see Figure 5.2.4(A))

$$= (d_w - t_c) \quad \text{for a single sheet} \quad \dots 5.2.4.2(7)$$

$$= (d_w - 2t_c) \quad \text{for multiple sheets} \quad \dots 5.2.4.2(8)$$

(not more than four lapped sheets over a supporting member)

$t_c$  = total combined base steel thickness of sheets (exclusive of coatings) involved in shear transfer above the plane of maximum shear transfer (see Figure 5.2.4(B)(b))

NOTE: If it can be shown by measurement that a given weld procedure will consistently give a larger effective diameter ( $d_e$ ) or average diameter ( $d_a$ ), as applicable, this larger diameter may be used provided the particular welding procedure used for making those welds is followed.

### 5.2.4.3 Tearout

A connected part shall have a spacing between arc spot welds and an edge distance ( $e$ ) from an arc spot weld such that the design shear force ( $V_w^*$ ) transmitted by the weld satisfies—

$$V_w^* \leq \phi V_w \quad \dots 5.2.4.3(1)$$

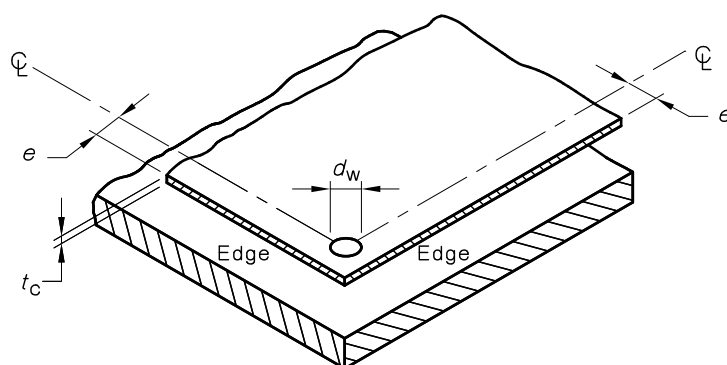
where

$\phi$  = capacity reduction factor of the connected part of an arc spot weld in shear (see Table 1.6)

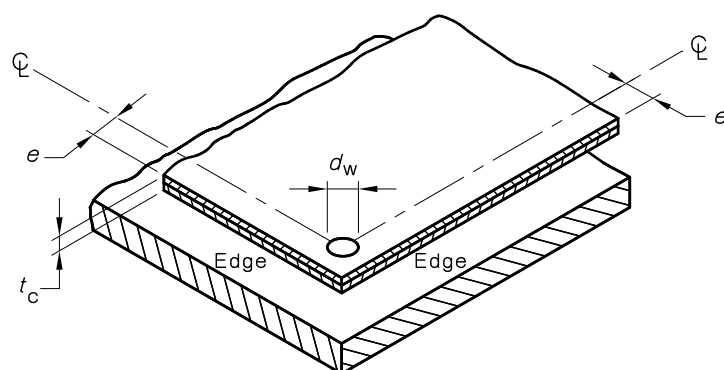
$$= 0.70 \quad \text{for } \frac{f_u}{f_y} \geq 1.08$$

$$= 0.60 \quad \text{for } \frac{f_u}{f_y} < 1.08$$

- $V_w$  = nominal shear force transmitted by the weld  
 $= tef_u$  . . . 5.2.4.3(2)  
 $t$  = thickness of the thinnest connected sheet  
 $e$  = edge distance measured in the line of the force from centre-line of an arc spot weld to the nearest edge of an adjacent weld or to the end of the connected part toward which the force is directed (see Figures 5.2.4(C)(a) and (b))



(a) Single sheet



(b) Double sheet

FIGURE 5.2.4(C) EDGE DISTANCE FOR ARC SPOT WELDS

In addition, the edge distance ( $e$ ) from the centre-line of any weld to the end or boundary of the connected member shall be not less than  $1.5d_w$ . In no case shall the clear distance of welds and the end of member be less than  $1.0d_w$ .

#### 5.2.4.4 Tension

The design tensile force ( $N_w^*$ ) on an arc spot weld shall satisfy—

$$N_w^* \leq \phi N_w \quad \dots 5.2.4.4(1)$$

The design tensile capacity ( $\phi N_w$ ) of each arc spot weld between sheet and supporting member shall be determined as follows:

$$\phi = 0.65$$

$N_w$  = nominal tensile capacity of an arc spot weld



$$= 0.7td_a f_u \quad \dots 5.2.4.4(2)$$

The following additional limitations for use in Equation 5.2.4.3 shall apply:

$$e \geq d_w$$

$$f_{uw} \geq 410 \text{ MPa}$$

$$f_u \leq 410 \text{ MPa}$$

$$t \leq 0.7 \text{ mm}$$

NOTE: If it can be shown by measurement that a given weld procedure will consistently give a larger average diameter ( $d_a$ ), this larger diameter may be used provided the particular welding procedure used for making those welds is followed.

The effects of any eccentric loading on an arc spot weld subject to uplift tension load, e.g., an arc spot weld on the perimeter of a roof or floor system, shall be evaluated and considered within the design of the weld.

## 5.2.5 Arc seam welds

### 5.2.5.1 General

Arc seam welds (see Figure 5.2.5.1) apply only to the following connections:

- Sheet to thicker supporting member welded in the flat position.
- Sheet to sheet welded in the horizontal or flat position.

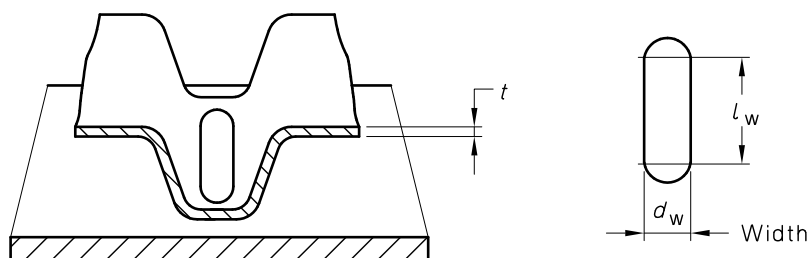


FIGURE 5.2.5.1 ARC SEAM WELD—SHEET TO SUPPORTING MEMBER IN FLAT POSITION

### 5.2.5.2 Shear

The design shear force ( $V_n^*$ ) on an arc seam weld shall satisfy—

$$V_n^* \leq \phi V_n \quad \dots 5.2.5.2(1)$$

where

$\phi$  = capacity reduction factor of an arc seam weld in shear (see Table 1.6)

$V_n$  = nominal shear capacity of an arc seam weld

The design shear capacity ( $\phi V_n$ ) of an arc seam weld shall be the lesser of the following:

- $\phi = 0.60$

$$V_n = \left[ \frac{\pi d_e^2}{4} + l_w d_e \right] 0.75 f_{uw} \quad \dots 5.2.5.2(2)$$

- $\phi = 0.60$

$$V_n = 2.5 t f_u (0.25 l_w + 0.96 d_a) \quad \dots 5.2.5.2(3)$$

where

$$\begin{aligned} d_e &= \text{effective width of an arc seam weld at fused surfaces} \\ &= 0.7d_w - 1.5t \end{aligned} \quad \dots 5.2.5.2(4)$$

$d_w$  = width of an arc seam weld

$l_w$  = length of the full size of the weld not including the circular ends. For calculation purposes,  $l_w$  shall not exceed  $3d_w$

$t$  = thickness of the thinnest connected part

$d_a$  = average width of arc seam weld

$$= d_w - t \quad (\text{for single sheet}) \quad \dots 5.2.5.2(5)$$

$$= d_w - 2t \quad (\text{for double sheet}) \quad \dots 5.2.5.2(6)$$

### 5.2.5.3 Tearout

The design tearout capacity ( $\phi V_w$ ) of the connected part based on the edge distance ( $e$ ) (see Figure 5.2.5.3) shall be determined as for the arc spot weld specified in Clause 5.2.4.3.

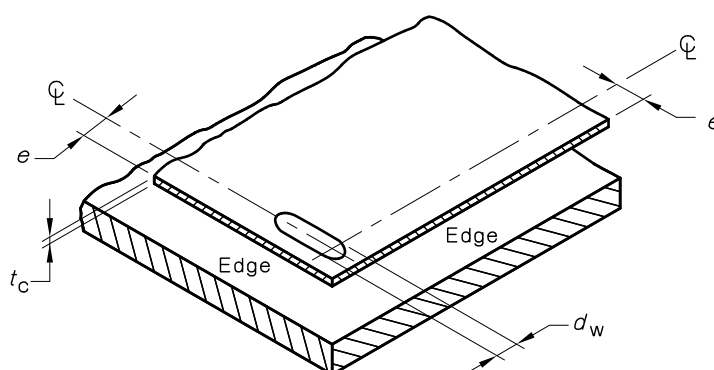


FIGURE 5.2.5.3 EDGE DISTANCE FOR ARC SEAM WELD

## 5.2.6 Flare welds

### 5.2.6.1 General

Flare welds (see Figure 5.2.6(a), (b) and (c)) apply only to the following connections welded in any position:

- (a) Sheet to sheet for flare V-welds.
- (b) Sheet to sheet for flare-bevel welds.
- (c) Sheet to thicker steel member for flare-bevel welds.

### 5.2.6.2 Shear

The design shear force ( $V_w^*$ ) on a flare weld shall satisfy—

$$V_w^* \leq \phi V_w \quad \dots 5.2.6.2(1)$$

where

$\phi$  = capacity reduction factor of flare welds subject to transverse and longitudinal loading (see Table 1.6)

$V_w$  = nominal shear capacity of a flare weld

The design shear capacity ( $\phi V_w$ ) of a flare weld shall be the least of the following values:

- (a) For flare-bevel welds, subject to transverse loading (see Figure 5.2.6(a)):

$$\phi = 0.55$$

$$V_w = 0.833t_l f_u \quad \dots 5.2.6.2(2)$$

- (b) For flare welds, subject to longitudinal loading (see Figure 5.2.6(b), (c), (d), (e), (f) and (g)):

- (i) For  $t \leq t_w < 2t$  or if the lip height is less than  $l_w$ :

$$\phi = 0.55$$

$$V_w = 0.75t_l f_u \quad \dots 5.2.6.2(3)$$

- (ii) For  $t_w \geq 2t$  and the lip height is greater than or equal to  $l_w$ :

$$\phi = 0.55$$

$$V_w = 1.5t_l f_u \quad \dots 5.2.6.2(4)$$

- (c) For longitudinal and transverse loading:

For  $t \geq 2.5$  mm:

$$\phi = 0.60$$

$$V_w = 0.75t_w l_{uw} f_{uw} \quad \dots 5.2.6.2(5)$$

where

$$t_w = \text{design throat thickness of the weld [see Figure 5.2.6(d), (e), (f) and (g)]}$$

$$= (5/16)R \quad \text{for flare bevel weld filled flush to the surface} \quad \dots 5.2.6.2(6)$$

$$= (1/2)R \text{ or } (3/8)R \quad \text{if } R > 12.0 \text{ mm for flare V-weld filled flush to the surface; or} \quad \dots 5.2.6.2(7)$$

$$= \text{effective throat thickness of flare weld not filled-flush to surface}$$

$$= 0.707w_1 \text{ or } 0.707w_2, \text{ whichever is smaller}$$

$$R = \text{radius of outside bend surface}$$

### 5.2.7 Resistance welds

The design shear force ( $V_w^*$ ) on a resistance weld shall satisfy—

$$V_w^* \leq \phi V_w \quad \dots 5.2.7(1)$$

where

$\phi$  = capacity reduction factor of the resistance weld (see Table 1.6)

$V_w$  = nominal shear capacity of the resistance weld

The design shear capacity ( $\phi V_w$ ) shall be determined as follows:

- (a) For spot welds, the design shear capacity ( $\phi V_w$ ) shall be as follows:

$$\phi = 0.65$$

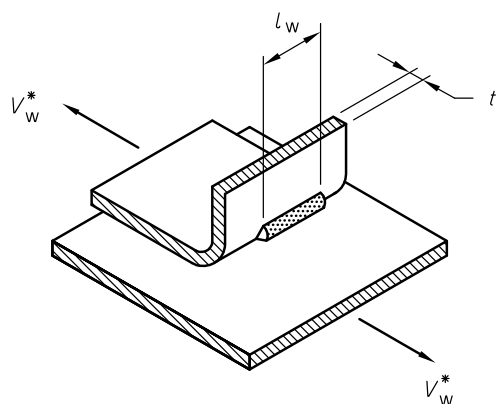
$$V_w = 5.51t^{1.47} \text{ (kN)} \quad \text{for } 0.25 \text{ mm} \leq t < 3.56 \text{ mm} \quad \dots 5.2.7(2)$$

$$7.6t + 8.57 \text{ (kN)} \quad \text{for } 3.56 \text{ mm} \leq t < 4.57 \text{ mm} \quad \dots 5.2.7(3)$$

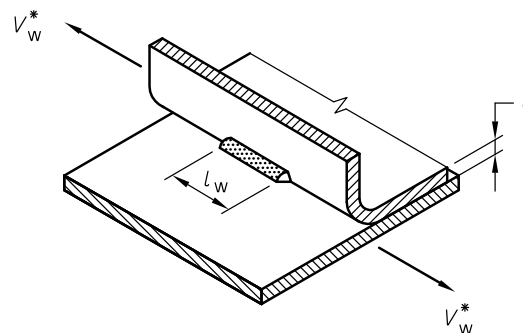
$t$  = thickness of thinnest outside sheet, in millimetres

- (b) For seam welds, the design shear capacity ( $\phi V_w$ ) shall be determined by testing in accordance with Section 6.

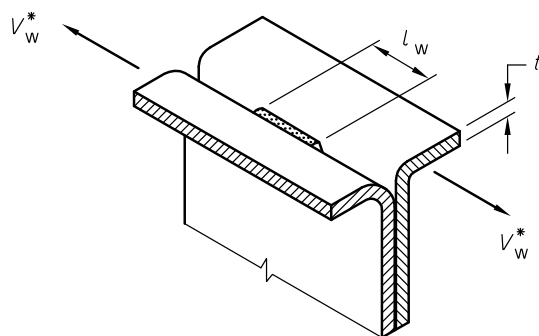
A1



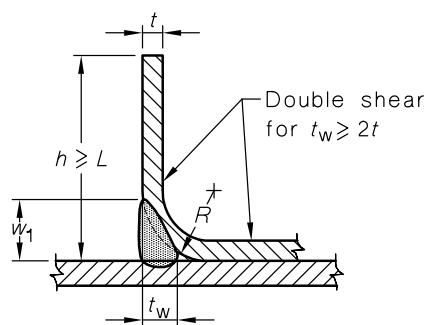
(a) Flare-bevel weld subject to transverse loading



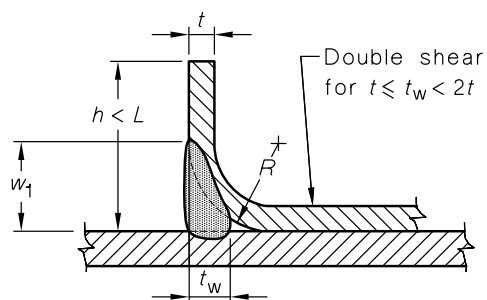
(b) Flare-bevel weld subject to longitudinal loading



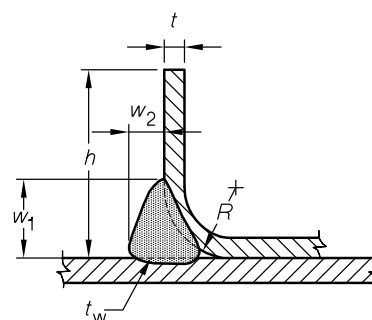
(c) Flare V-weld subject to longitudinal loading



(d) Flare-bevel weld  
(Filled flush to surface,  $w_1 = R$ )



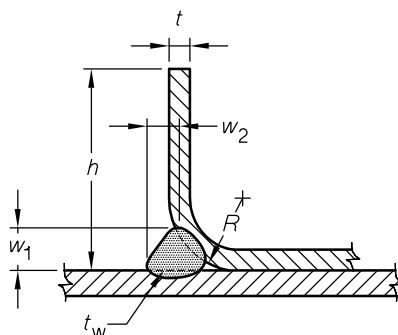
(e) Flare-bevel weld  
(Filled flush to surface,  $w_1 = R$ )



(f) Flare-bevel weld  
(Not filled flush to surface,  $w_1 > R$ )

FIGURE 5.2.6 (in part) SHEAR IN FLARE WELDS

A1



(g) Flare-bevel weld  
(Not filled flush to surface,  $w_1 < R$ )

FIGURE 5.2.6 (in part) SHEAR IN FLARE WELDS

## 5.3 BOLTED CONNECTIONS

### 5.3.1 General

This Clause applies to bolted connections for cold-formed steel structural members in which the thickness of a connected part is less than 3 mm. For bolted connections in which the thickness of a connected part is greater than or equal to 3 mm, AS 4100 or NZS 3404, as appropriate, shall be used.

Bolts shall be installed and tightened to achieve the required performance of the connections involved under service conditions.

Standard holes for bolts shall not exceed the sizes given in Table 5.3.1, except that larger holes may be used in column base details or structural systems connected to concrete provided special plate washers as specified in AS 4100 or NZS 3404 are used.

Oversized and slotted holes given in Table 5.3.1 may be used, provided all bolts are loaded in shear and bolt holes comply with the following:

- (a) *Slotted holes for Australia* The length of slotted holes shall be normal to the direction of the shear force.
- (b) *Purlins and girts for Australia* In situations where lapping or nesting of sections is required, such as purlin and girt applications, it is permissible to have oversized short-slotted holes provided integral washers are used with the bolt head and nut, all bolts are loaded in shear and the length of slotted holes are normal to the direction of the shear force. The dimension of such oversized-slotted holes shall be—

$(d_f + 6.0)\text{mm}$  by  $(d_f + 10.0)\text{mm}$

- (c) *Z-section purlins and girts for New Zealand* In situations where lapping or nesting is required, such as purlin and girt applications, it is permissible to have short-slotted holes provided all bolts are loaded in shear, washers or backup plates are installed and bolts tightened to achieve the required performance of the connection. The dimension of such short-slotted holes shall be—

$(d_f + 2.0)\text{mm}$  by  $(d_f + 10.0)\text{mm}$

Washers and backup plates shall be installed over oversized or short-slotted holes, or long-slotted holes in an outer ply, unless suitable performance is demonstrated by load tests in accordance with Section 8.

TABLE 5.3.1  
MAXIMUM SIZE OF BOLT HOLES

| Nominal bolt diameter<br>$d_f$<br>mm | Standard hole diameter<br>$d_h$<br>mm | Oversized hole diameter<br>$d_h$<br>mm | Short-slotted hole dimensions<br>mm | Long-slotted hole dimensions<br>mm |
|--------------------------------------|---------------------------------------|----------------------------------------|-------------------------------------|------------------------------------|
| <12                                  | $d_f + 1.0$                           | $d_f + 2.0$                            | $(d_f + 1.0)$ by $(d_f + 6.0)$      | $(d_f + 1.0)$ by $2.5d_f$          |
| $\geq 12$                            | $d_f + 2.0$                           | $d_f + 3.0$                            | $(d_f + 2.0)$ by $(d_f + 6.0)$      | $(d_f + 2.0)$ by $2.5d_f$          |

When holes are staggered, the area to be deducted shall be the greater of—

- (i) the deduction for non-staggered holes; or
- (ii) the sum of the areas of all holes in any zig-zag line extending progressively across the member or part of the member, less  $(s_p^2 t / 4s_g)$  for each gauge space in the chain of holes;

where

$s_p$  = staggered pitch, the distance measured parallel to the direction of the design action in the member, centre-to-centre of holes in consecutive lines (see Figure 5.3.1(A))

$t$  = thickness of the holed material

$s_g$  = gauge, the distance measured at right angles to the direction of the design action in the member, centre-to-centre of holes in consecutive lines (see Figure 5.3.1(A))

For sections such as angles with holes in both legs, the gauge shall be taken as the sum of the back marks to each hole, less the leg thickness (see Figure 5.3.1(B))

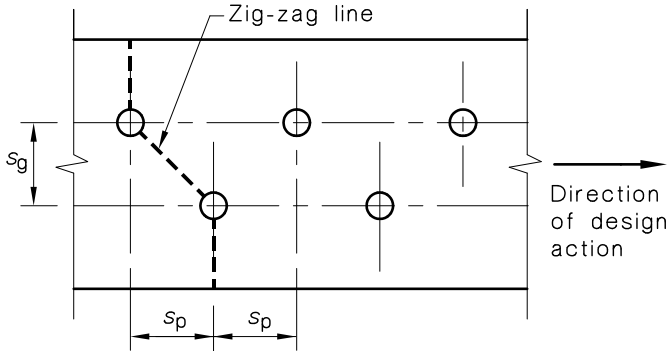


FIGURE 5.3.1(A) STAGGERED HOLES

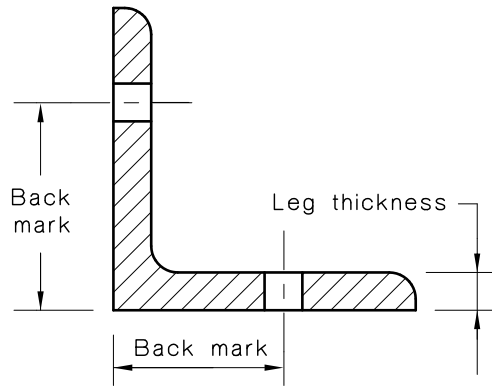


FIGURE 5.3.1(B) ANGLES WITH HOLES IN BOTH LEGS

### 5.3.2 Tearout

A connected part shall have a spacing between bolts and an edge distance from a bolt such that the design shear force ( $V_f^*$ ) of the connected part satisfies—

$$V_f^* \leq \phi V_f \quad \dots 5.3.2(1)$$

where

$\phi$  = capacity reduction factor of bolted connection subject to tearout (see Table 1.6)

$$= 0.70 \text{ for } \frac{f_u}{f_y} \geq 1.08$$

$$= 0.60 \text{ for } \frac{f_u}{f_y} < 1.08$$

$V_f$  = nominal shear capacity of the connected part along two parallel lines in the direction of the applied force

$$= t e f_u \quad \dots 5.3.2(2)$$

$t$  = thickness of the connected part

$e$  = distance measured in the line of force from the centre of a standard hole to the nearest edge of an adjacent hole or to the end of the connected part

In addition, the minimum distance between centres of bolt holes shall provide clearance for bolt heads, nuts, washers and the wrench but shall be not less than 3 times the nominal bolt diameter ( $d_f$ ). Also, the distance from the centre of any standard hole to the end or other boundary of the connecting member shall be not less than  $1.5d_f$ .

For oversized and slotted holes, the distance between edges of two adjacent holes and the distance measured from the edge of the hole to the end or other boundary of the connecting member in the line of force shall be not less than  $[e - (d_h/2)]$ , in which  $e$  is the distance used in Equation 5.3.2(2), and  $d_h$  is the diameter of a hole given in Table 5.3.1. The clear distance between edges of two adjacent holes shall be not less than  $2d_f$  and the distance between the edge of the hole and the end of the member shall be not less than  $d_f$ .

### 5.3.3 Net section tension

The design tensile force ( $N_f^*$ ) on the net section of the connected part shall satisfy Clause 3.2 and—

$$N_f^* \leq \phi N_f \quad \dots 5.3.3(1)$$

where

$\phi$  = capacity reduction factor of bolted connection subject to net section tension (see Table 1.6)

$N_f$  = nominal tensile capacity of the net section of the connected part

The design tensile capacity ( $\phi N_f$ ) of the connected part shall be determined as follows:

(a) Where washers are provided under both the bolt head and the nut—

$\phi = 0.65$  (for double shear connection)

$\phi = 0.55$  (for single shear connection)

A1 |  $N_f = [0.1 + (3d_f/s_f)]A_n f_u \leq A_n f_u$  for a single bolt, or a single row of bolts perpendicular to the force . . . 5.3.3(2)

A1 |  $N_f = A_n f_u$  for multiple bolts in the line parallel to the force . . . 5.3.3(3)

(b) Where either washers are not provided under the bolt head and nut, or only one washer is provided under either the bolt head or nut—

$\phi = 0.65$

A1 |  $N_f = (2.5d_f/s_f)A_n f_u \leq A_n f_u$  for a single bolt, or a single row of bolts perpendicular to the force . . . 5.3.3(4)

$N_f = A_n f_u$  for multiple bolts in the line parallel to the force . . . 5.3.3(5)

where

$s_f$  = spacing of bolts perpendicular to the line of the force; or  
width of sheet, in the case of a single bolt

$A_n$  = net area of the connected part

### 5.3.4 Bearing

#### 5.3.4.1 General

The design bearing capacity ( $\phi V_b$ ) of bolted connections shall be determined in accordance with Clauses 5.3.4.2 and 5.3.4.3. For conditions not specified in this Standard,  $\phi V_b$  of bolted connections shall be determined by tests.

#### 5.3.4.2 Bearing capacity without considering bolt hole deformation

When deformation around the bolt holes is not a design consideration, the nominal bearing capacity ( $V_b$ ) of the connected sheet for each loaded bolt shall be determined as follows:

$$V_b = \alpha C d_f t f_u \quad \dots 5.3.4.2$$

where

$\phi = 0.60$

$\alpha$  = modification factor for type of bearing connection given in Table 5.3.4.2(A)

$C$  = bearing factor given in Table 5.3.4.2(B)

$d_f$  = nominal bolt diameter

$t$  = base metal thickness



$f_u$  = tensile strength of sheet

**TABLE 5.3.4.2(A)**  
**MODIFICATION FACTOR ( $\alpha$ )**  
**FOR TYPE OF BEARING CONNECTION**

| Type of bearing                                                                                                             | $\alpha$ |
|-----------------------------------------------------------------------------------------------------------------------------|----------|
| Single shear and outside sheets of double shear connection with washers under both bolt head and nut                        | 1.00     |
| Single shear and outside sheets of double shear connection without washers under both head and nut, or with only one washer | 0.75     |
| Inside sheet of double shear connection with or without washers                                                             | 1.33     |

**TABLE 5.3.4.2(B)**  
**BEARING FACTOR ( $C$ )**

| Thickness of connected part, $t$<br>mm | Ratio of fastener diameter to member thickness, $d/t$ | $C$              |
|----------------------------------------|-------------------------------------------------------|------------------|
| $0.42 \leq t < 4.76$                   | $d_f/t < 10$                                          | 3.0              |
|                                        | $10 \leq d_f/t \leq 22$                               | $4 - 0.1(d_f/t)$ |
|                                        | $d_f/t > 22$                                          | 1.8              |

### 5.3.4.3 Bearing capacity at a bolt hole deformation of 6 mm

When deformation around a bolt hole is a design consideration, the nominal bearing capacity ( $V_b$ ) shall be determined as follows:

$$V_b = (0.183 t + 1.53) d_f t f_u \quad \dots 5.3.4.3$$

## 5.3.5 Bolts

### 5.3.5.1 Bolt in shear

The design shear force ( $V_{fv}^*$ ) on a bolt shall satisfy—

$$V_{fv}^* \leq \phi V_{fv} \quad \dots 5.3.5.1(1)$$

where

$\phi$  = capacity reduction factor of a bolt subject to shear (see Table 1.6)

$V_{fv}$  = nominal shear capacity of a bolt

$$= 0.62 f_{uf} (n_n A_c + n_x A_o) \quad \dots 5.3.5.1(2)$$

$f_{uf}$  = minimum tensile strength of a bolt

= 400 MPa (for AS 4291.1 (ISO 898-1), Grade 4.6 bolts)

= 830 MPa (for AS 4291.1 (ISO 898-1), Grade 8.8 bolts)

$n_n$  = number of shear planes with threads intercepting the shear plane

$A_c$  = minor diameter area of a bolt, as specified in AS 1275

$n_x$  = number of shear planes without threads intercepting the shear plane

$A_o$  = plain shank area of a bolt

**5.3.5.2 Bolt in tension**

The design tensile force ( $N_{ft}^*$ ) on a bolt shall satisfy—

$$N_{ft}^* \leq \phi N_{ft} \quad \dots 5.3.5.2(1)$$

where

$\phi$  = capacity reduction factor of a bolt subject to tension (see Table 1.6)

$N_{ft}$  = nominal tensile capacity of a bolt

$$= A_s f_{uf} \quad \dots 5.3.5.2(2)$$

$A_s$  = tensile stress area of a bolt, as specified in AS 1275

The pull-over (pull-through) capacity of the connected sheet at the bolt head, nut or washer shall be considered where bolt tension is involved.

**5.3.5.3 Bolt subject to combined shear and tension**

A bolt required to resist simultaneously a design shear force ( $V_{fv}^*$ ) and a design tensile force ( $N_{ft}^*$ ) shall satisfy—

$$\left( \frac{V_{fv}^*}{\phi V_{fv}} \right)^2 + \left( \frac{N_{ft}^*}{\phi N_{ft}} \right)^2 \leq 1.0 \quad \dots 5.3.5.3$$

where  $V_{fv}^*$ ,  $\phi V_{fv}$ ,  $N_{ft}^*$  and  $\phi N_{ft}$  shall be determined in accordance with Clauses 5.3.5.1 and 5.3.5.2, respectively.

**5.4 SCREWED CONNECTIONS****5.4.1 General**

This Clause applies to connections for cold-formed steel structural members using self-tapping screws of nominal diameter ( $d_f$ ) where  $3.0 \text{ mm} \leq d_f \leq 7.0 \text{ mm}$ . The screws shall be thread-forming or thread-cutting, with or without a self-drilling point.

**5.4.2 Screwed connections in shear****5.4.2.1 Minimum spacing and edge distance**

The distance between centres of screws shall provide clearance for screw washers but shall be not less than three times the nominal screw diameter ( $d_f$ ).

*For Australia*, the distance from the centre of a screw to the edge of any part shall be not less than  $1.5d_f$ . If the end distance is parallel to the force on the fastener, the nominal shear capacity ( $V_{fv}$ ) shall be limited to that calculated using Equation 5.4.2.4(2).

*For New Zealand*, the distance from the centre of a screw to the edge of any part shall not be less than  $3d_f$ .

**5.4.2.2 Tension in the connected part**

The design tensile force ( $N_t^*$ ) on the net section of the connected part shall satisfy Clause 3.2 and—

$$N_t^* \leq \phi N_t \quad \dots 5.4.2.2(1)$$

where

$\phi$  = capacity reduction factor of screwed connection subject to tension (see Table 1.6)

$N_t$  = nominal tensile capacity of the net section of the connected part  
 $= (2.5d_f/S_f)A_n f_u \leq A_n f_u$  for a single screw, or a single row of screws perpendicular to the force . . . 5.4.2.2(2)

$= A_n f_u$  for multiple screws in the line parallel to the force . . . 5.4.2.2(3)

$d_f$  = nominal screw diameter

$S_f$  = spacing of screws perpendicular to the line of the force; or width of sheet, in the case of a single screw

$A_n$  = net area of the connected part

#### 5.4.2.3 Tilting and hole bearing

The design bearing force ( $V_b^*$ ) on a screw shall satisfy—

$$V_b^* \leq \phi V_b \quad \dots 5.4.2.3(1)$$

where

$\phi$  = capacity reduction factor of a screw, subject to tilting and hole bearing (see Table 1.6)

$V_b$  = nominal bearing capacity of the connected part

Where the screw is in a single shear connection and where the two connected sheets are in contact at the point of fastening—

(a) for  $t_2/t_1 \leq 1.0$ ,  $V_b$  shall be taken as the smallest of the following:

$$(i) \quad V_b = 4.2\sqrt{t_2^3 d_f} f_{u2} \quad \dots 5.4.2.3(2)$$

$$(ii) \quad V_b = C t_1 d_f f_{u1} \quad \dots 5.4.2.3(3)$$

$$(iii) \quad V_b = C t_2 d_f f_{u2} \quad \dots 5.4.2.3(4)$$

where

$t_2$  = thickness of the sheet not in contact with the screw head

$t_1$  = thickness of the sheet in contact with the screw head

$d_f$  = nominal screw diameter

$f_{u2}$  = tensile strength of the sheet not in contact with the screw head

$f_{u1}$  = tensile strength of the sheet in contact with the screw head

$C$  = bearing factor (see Table 5.4.2.3)

(b) for  $t_2/t_1 \geq 2.5$ ,  $V_b$  shall be taken as the smaller of the following:

$$(i) \quad V_b = C t_1 d_f f_{u1} \quad \dots 5.4.2.3(5)$$

$$(ii) \quad V_b = C t_2 d_f f_{u2} \quad \dots 5.4.2.3(6)$$

(c) for  $1.0 < t_2/t_1 < 2.5$ ,  $V_b$  shall be determined by linear interpolation between the minimum values obtained from Equations 5.4.2.3(2) to 5.4.2.3(4) and the minimum values obtained from Equations 5.4.2.3(5) and 5.4.2.3(6).

For cases where the material is not in contact at the point of fastening, the screw capacity [strength] shall be determined by testing in accordance with Section 8.

**TABLE 5.4.2.3**  
**BEARING FACTOR (C)**

| Ratio of fastener diameter to member thickness, $d_f/t$ | C                  |
|---------------------------------------------------------|--------------------|
| $d_f/t < 6$                                             | 2.7                |
| $6 \leq d_f/t \leq 13$                                  | $3.3 - 0.1(d_f/t)$ |
| $d_f/t > 13$                                            | 2.0                |

**5.4.2.4 Connection shear as limited by end distance (for Australia)**

The design shear force ( $V_{fv}^*$ ) as limited by end distance shall satisfy—

$$V_{fv}^* \leq \phi V_{fv} \quad \dots 5.4.2.4(1)$$

If  $f_u/f_y \geq 1.08$ ,  $\phi = 0.7$

If  $f_u/f_y < 1.08$ ,  $\phi = 0.6$

When the distance to an end of the connected part is parallel to the line of the applied force, the nominal shear force shall be calculated as follows:

$$V_{fv} = t e f_u \quad \dots 5.4.2.4(2)$$

where

$t$  = thickness of the part in which the end distance is measured

$e$  = distance measured in the line of force from the centre of a standard hole to the nearest end of the connected part

**5.4.2.5 Screws in shear**

The nominal shear capacity of the screw shall be determined by testing in accordance with Section 8 and shall be not less than  $1.25V_b$ , where  $V_b$  shall be calculated in accordance with Clause 5.4.2.3.

**5.4.3 Screwed connections in tension**

**5.4.3.1 Minimum edge distance**

The distance from the centre of the screw in tension to the edge of any part shall not be less than  $3d_f$ .

**5.4.3.2 Pull-out and pull-over (pull-through)**

This Clause applies only to screwed connections in tension if the two sheets are in contact at the point of fastening. For other cases, such as crest fixed connections, the design tensile capacity shall be established by prototype testing in accordance with Section 8.

The design tensile force ( $N_t^*$ ) on a screw shall satisfy—

$$N_t^* \leq \phi N_t \quad \dots 5.4.3.2(1)$$

where

$$\phi = 0.5$$

$N_t$  = nominal capacity of the connection in tension

The nominal capacity ( $N_t$ ) shall be the lesser of the following:

(a) The nominal pull-out capacity ( $N_{ou}$ ) calculated as follows:

$$N_{ou} = 0.85t_2d_f f_{u2} \quad \text{for } t_2 > 0.9 \text{ mm} \quad \dots 5.4.3.2(2)$$

(b) The nominal pull-over (pull-through) capacity ( $N_{ov}$ ) calculated as follows:

$$N_{ov} = 1.5t_1d_w f_{u1} \quad \text{for } 0.5 < t_1 > 1.5 \text{ mm} \quad \dots 5.4.3.2(3)$$

where  $d_w$  is the greater of the screw head diameter and the washer diameter, but not greater than 12.5 mm.

For screws subject to tensile forces, the head of the screw or washer shall have a diameter ( $d_w$ ) not less than 8 mm. Washers shall have a minimum thickness of 1.27 mm.

#### 5.4.3.3 Screws in tension

A1 | The nominal tensile capacity of the screw shall be determined by testing in accordance with Section 8 and shall be not less than  $1.25N_t$ , where  $N_t$  shall be calculated in accordance with Clause 5.4.3.1.

For screws not covered by AS 3566, the tensile capacity of the screw shall be determined by testing.

### 5.5 BLIND RIVETED CONNECTIONS

#### 5.5.1 General

This Clause applies to connections for cold-formed steel structural members using blind rivets of nominal diameter ( $d_f$ ), where  $3.0 \text{ mm} \leq d_f \leq 7.0 \text{ mm}$ .

#### 5.5.2 Riveted connections in shear

##### 5.5.2.1 Minimum spacing and edge distance

The distance between centres of rivet holes shall be not less than 3 times the nominal blind rivet diameter ( $d_f$ ).

*For Australia*, the distance from the centre of a rivet to the edge of any part shall be not less than  $1.5d_f$ . If the end distance is parallel to the force on the fastener, the nominal shear capacity ( $V_{fv}$ ) shall be limited to that calculated using Equation 5.5.2.4(2).

*For New Zealand*, the distance from the centre of a rivet to the edge of any part shall be not less than  $3d_f$ .

##### 5.5.2.2 Tension in the connected part

The design tensile force ( $N_t^*$ ) on the net section of the connected part shall satisfy Clause 3.2 and—

$$N_t^* \leq \phi N_t \quad \dots 5.5.2.2(1)$$

where

$\phi$  = capacity reduction factor of blind rivet connection subject to net section tension (see Table 1.6)

$N_t$  = nominal tensile capacity of the net section of the connected part  
 $= (2.5d_f/s_f)A_n f_u \leq A_n f_u$  for a single rivet, or a single row of rivet perpendicular to the force  $\dots 5.5.2.2(2)$

$= A_n f_u$  for multiple rivets in the line parallel to the force  $\dots 5.5.2.2(3)$

$d_f$  = nominal rivet diameter

$s_f$  = spacing of rivets perpendicular to the line of the force; or  
width of sheet, in the case of a single rivet

$A_n$  = net area of the connected part

**5.5.2.3 Tilting and hole bearing**

The design bearing force ( $V_b^*$ ) on a rivet shall satisfy—

$$V_b^* \leq \phi V_b \quad \dots 5.5.2.3(1)$$

where

$\phi$  = capacity reduction factor of a blind rivet subject to tilting and hole bearing (see Table 1.6)

$V_b$  = nominal bearing capacity of the connected part

Where the rivet is in a single shear connection and where the two connected sheets are in contact at the point of fastening—

(a) for  $t_2/t_1 \leq 1.0$ ,  $V_b$  shall be taken as the smallest of the following:

$$(i) \quad V_b = 3.6\sqrt{(t_2^3 d_f)} f_{u2} \quad \dots 5.5.2.3(2)$$

$$(ii) \quad V_b = 2.1 t_1 d_f f_{u1} \quad \dots 5.5.2.3(3)$$

$$(iii) \quad V_b = 2.1 t_2 d_f f_{u2} \quad \dots 5.5.2.3(4)$$

where

$t_1$  = thickness of the sheet in contact with the rivet head

$t_2$  = thickness of the sheet not in contact with the rivet head

$d_f$  = nominal rivet diameter

$f_{u1}$  = tensile strength of the sheet in contact with the rivet head

$f_{u2}$  = tensile strength of the sheet not in contact with the rivet head

(b) for  $t_2/t_1 \geq 2.5$ ,  $V_b$  shall be taken as the smaller of the following:

$$(i) \quad V_b = 2.1 t_1 d_f f_{u1} \quad \dots 5.5.2.3(5)$$

$$(ii) \quad V_b = 2.1 t_2 d_f f_{u2} \quad \dots 5.5.2.3(6)$$

(c) for  $1.0 < t_2/t_1 < 2.5$ ,  $V_b$  shall be determined by linear interpolation between the minimum value obtained from Equations 5.5.2.3(2) to 5.5.2.3(4) and the minimum value obtained from Equations 5.5.2.3(5) and 5.5.2.3(6).

For cases where the material is not in contact at the point of fastening, the rivet capacity shall be determined by testing in accordance with Section 8.

**5.5.2.4 Connection shear as limited by tearout (for Australia)**

The design shear force ( $V_{fv}^*$ ) as limited by end distance shall satisfy—

$$V_{fv}^* \leq \phi V_{fv} \quad \dots 5.5.2.4(1)$$

When the distance to an end of the connected part is parallel to the line of the applied force, the nominal shear force shall be calculated as follows:

$$V_{fv} = t e f_u \quad \dots 5.5.2.4(2)$$

where

$\phi$  = capacity reduction factor of riveted connection subject to tension (see Table 1.6)

$t$  = thickness of the part in which the end distance is measured

$e$  = distance measured in the line of force from the centre of a standard hole to the nearest end of the connected part

### 5.5.2.5 Rivets in shear

The nominal shear capacity of the rivet shall be determined by testing and shall be not less than  $1.25V_b$ , where  $V_b$  shall be calculated in accordance with Clause 5.5.2.3.

NOTE: The design tensile capacity of riveted connections may be established by prototype testing in accordance with Section 8.

## 5.6 RUPTURE

### 5.6.1 Shear rupture

At beam-end connections, where one or more flanges are coped and failure may occur along a plane through the fasteners, the design shear force ( $V_n^*$ ) shall satisfy—

$$V_n^* \leq \phi V_n \quad \dots 5.6.1(1)$$

where

$\phi$  = capacity reduction factor of beam-end connections subject to shear rupture (see Table 1.6)

$$V_n = \text{nominal shear capacity of the beam-end connection} \\ = 0.6f_u A_{wn} \quad \dots 5.6.1(2)$$

$$A_{wn} = \text{net area of the web} \\ = (d_{wc} - n_h d_h)t \quad \dots 5.6.1(3)$$

$d_{wc}$  = coped depth of the web

$n_h$  = number of holes in the critical plane

$d_h$  = diameter of the hole

$t$  = thickness of the coped web

### 5.6.2 Tension rupture

The nominal tension rupture strength along a path in affected elements of connected members shall be determined by Equation 3.2.1(2) with  $k_t$  equal to 1.0.

### 5.6.3 Block shear rupture

At beam-end or tension connections with possible shear and tension rupture planes, the design action effect ( $S^*$ ) shall satisfy—

$$S^* \leq \phi R_n \quad \dots 5.6.3(1)$$

where

$$\phi = \text{capacity reduction factor for block shear rupture of the beam-end or tension member connection} \\ = 0.65 \text{ for bolted connections}$$

$$R_n = \text{nominal capacity for block shear rupture of the beam-end or tension member connection}$$

The nominal capacity for block shear rupture of the beam-end or tension member connection ( $R_n$ ) shall be determined as follows:

$$(a) \quad \text{For } f_u A_{nt} \geq 0.6 f_u A_{nv}: R_n = 0.6 f_y A_{gv} + f_u A_{nt} \quad \dots 5.6.3(2)$$

$$(b) \quad \text{For } 0.6 f_u A_{nv} \geq f_u A_{nt}: R_n = 0.6 f_u A_{nv} + f_y A_{gt} \quad \dots 5.6.3(3)$$

where

$A_{nt}$  = net area subject to tension in block shear rupture

$A_{nv}$  = net area subject to shear in block shear rupture

$A_{gv}$  = gross area subject to shear in block shear rupture

$A_{gt}$  = gross area subject to tension in block shear rupture

## **5.7 OTHER CONNECTIONS USING ANY TYPE OF FASTENERS**

The design capacity for a specific connection using any type of fasteners may be determined by prototype testing in accordance with Section 8.



## SECTION 6 FATIGUE

### 6.1 GENERAL

#### 6.1.1 Requirements

This Section applies to the design of cold-formed steel members and connections subject to cyclic loading within the elastic range stresses of frequency and magnitude sufficient to initiate cracking and progressive failure (fatigue).

The provisions of this Section apply to stresses calculated on the basis of unfactored loads. The maximum permitted tensile stress due to unfactored loads is  $0.6f_y$ .

Stress range is defined as the magnitude of change in stress due to the application or removal of the unfactored live load. In the case of a stress reversal, the stress range shall be computed as the sum of the absolute values of maximum repeated tensile and compressive stresses or the sum of the absolute values of maximum shearing stresses of opposite direction at the point of probable crack initiation.

The occurrence of full design wind or earthquake loads is too infrequent to warrant consideration of fatigue design in buildings. The fatigue design of cladding, fixings and its immediate support shall be in accordance with AS/NZS 1562.1. Wind-induced oscillations can cause fatigue cracks to occur in structures such as masts, lighting poles, traffic sign supports and chimneys, and, therefore, shall be considered.

Evaluation of fatigue resistance to this Section is not required if the number of cycles of application of live load is less than 20 000.

The cyclic load resistance determined by the provisions of this Section is applicable to—

- (a) structures with suitable corrosion protection or subject only to non-aggressive atmospheres; and
- (b) structures subject to temperatures not exceeding 150°C.

The contract document shall provide either—

- (i) complete details including weld sizes; or
- (ii) specify the planned cycle life and the maximum range of moments, shears and reactions for the connections.

#### 6.1.2 Definitions

For the purpose of this Section, the definitions below apply.

##### 6.1.2.1 *Constant stress range fatigue limit*

The highest constant stress range for each detail category at which fatigue cracks are not expected to propagate (see Figure 6.3).

##### 6.1.2.2 *Cut-off limit*

For each detail category, the highest variable stress range that does not require consideration when carrying out cumulative damage calculations (see Figure 6.3).

##### 6.1.2.3 *Design life*

The period over which a structure or structural element is required to perform its function without repair.

##### 6.1.2.4 *Design spectrum*

The sum of the stress spectra from all of the nominal loading events expected during the design life.

#### **6.1.2.5** *Detail category*

A designation given to a particular detail to indicate which of the S-N curves is to be used in the fatigue assessment.

##### NOTES:

- 1 The detail category takes into consideration the local stress concentration at the detail, the size and shape of the maximum acceptable discontinuity, the loading condition, metallurgical effects, residual stresses, the welding process and any post weld improvement.
- 2 The detail category number is defined by the fatigue strength at  $2 \times 10^6$  cycles on the S-N curve (see Figure 6.3).

#### **6.1.2.6** *Discontinuity*

An absence of material, causing a stress concentration.

NOTE: Typical discontinuities include cracks, scratches, corrosion pits, lack of penetration, slag inclusions, cold laps, porosity and undercut.

#### **6.1.2.7** *Fatigue*

Damage caused by repeated fluctuations of stress leading to gradual cracking of a structural element.

#### **6.1.2.8** *Fatigue loading*

A set of nominal loading events described by the distribution of the loads, their magnitudes and the numbers of applications of each nominal loading event.

#### **6.1.2.9** *Fatigue strength*

The stress range defined in Clause 6.3 for each detail category varying with the number of stress cycles (see Figure 6.3).

#### **6.1.2.10** *Miner's summation*

The cumulative damage calculation based on the Palmgren–Miner summation or equivalent.

#### **6.1.2.11** *Nominal loading event*

The loading sequence for the structure or structural element.

NOTE: One nominal loading event may produce one or more stress cycles depending on the type of load and the point in the structure under consideration.

#### **6.1.2.12** *S-N curve*

A curve defining the limiting relationship between the number of stress cycles and stress range for a detail category.

#### **6.1.2.13** *Stress cycle*

One cycle of stress defined by stress cycle counting.

#### **6.1.2.14** *Stress cycle counting method*

Any rational method used to identify individual stress cycles from the stress history.

#### **6.1.2.15** *Stress range*

The algebraic difference between two extremes of stress.

#### **6.1.2.16** *Stress spectrum*

A histogram of the stress cycles produced by a nominal loading event.

### 6.1.3 Notation

For the purpose of this Section, the following applies:

- $f_c$  = fatigue strength corrected for thickness of material
- $f_f$  = uncorrected fatigue strength
- $f_{rn}$  = detail category reference fatigue strength at  $n_r$ -normal stress
- $f_{rnc}$  = corrected detail category reference fatigue strength for normal stress
- $f_{rs}$  = detail category reference fatigue strength at  $n_r$ -shear stress
- $f_{rsc}$  = corrected detail category reference fatigue strength for shear stress
- $f_y$  = yield stress
- $f_3$  = detail category fatigue strength at constant amplitude fatigue limit ( $5 \times 10^6$  cycles)
- $f_{3c}$  = corrected detail category fatigue strength at constant amplitude fatigue limit
- $f_5$  = detail category fatigue strength at cut off limit (108 cycles)
- $f_{5c}$  = corrected detail category reference fatigue strength at cut off limit
- $f^*$  = design stress range
- $f_i^*$  = design stress range for loading event  $i$
- $l$  = length of member
- $n_i$  = number of cycles of nominal loading event  $i$ , producing  $f_i^*$
- $n_r$  = reference number of stress cycles ( $2 \times 10^6$  cycles)
- $n_{sc}$  = number of stress cycles
- $t_p$  = plate thickness
- $\alpha_s$  = inverse of the slope of the S-N curve
- $\beta_{tf}$  = thickness correction factor
- $\phi$  = capacity factor

### 6.1.4 Method

For the reference design condition, the capacity factor ( $\phi$ ) shall be taken as 1.0. The reference design condition implies the following:

- (a) The detail is located on a redundant load path, in a position where failure at that point alone will not lead to overall collapse of the structure.
- (b) The stress history is estimated by conventional methods.
- (c) The load cycles are not highly irregular.
- (d) The detail is accessible for, and subject to, regular inspection.

The capacity factor ( $\phi$ ) shall be reduced when any of the above conditions do not apply.

For non-redundant load paths, the capacity factor ( $\phi$ ) shall be less than or equal to 0.70.

### 6.1.5 Thickness effect

The thickness correction factor ( $\beta_{tf}$ ) shall be taken as—

$$\beta_{tf} = 1.0$$

except for a transverse fillet or butt-welded connection involving a plate thickness ( $t_p$ ) greater than 25 mm, where  $\beta_{tf}$  shall be calculated as follows:

$$\beta_{\text{tf}} = \left( \frac{25}{t_p} \right)^{0.25} \quad \dots 6.1.5(1)$$

The uncorrected fatigue strength ( $f_f$ ) shall be reduced to a corrected fatigue strength ( $f_c$ ) using—

$$f_c = \beta_{\text{tf}} f_f \quad \dots 6.1.5(2)$$

The uncorrected detail category reference fatigue strength for normal stress ( $f_{\text{rn}}$ ) shall be reduced to a corrected detail category reference fatigue strength for normal stress ( $f_{\text{rnc}}$ ) using—

$$f_{\text{rnc}} = \beta_{\text{tf}} f_{\text{rn}} \quad \dots 6.1.5(3)$$

The uncorrected detail category reference fatigue strength for shear stress ( $f_{\text{rs}}$ ) shall be reduced to a corrected detail category reference fatigue strength for shear stress ( $f_{\text{rsc}}$ ) using—

$$f_{\text{rsc}} = \beta_{\text{tf}} f_{\text{rs}} \quad \dots 6.1.5(3)$$

The uncorrected detail category fatigue strength at constant amplitude fatigue limit ( $f_3$ ) shall be reduced to a corrected detail category fatigue strength at constant amplitude fatigue limit ( $f_{3c}$ ) using—

$$f_{3c} = \beta_{\text{tf}} f_3 \quad \dots 6.1.5(3)$$

The uncorrected detail category fatigue strength at cut-off limit ( $f_5$ ) shall be reduced to a corrected detail category reference fatigue strength at cut-off limit ( $f_{5c}$ ) using—

$$f_{5c} = \beta_{\text{tf}} f_5 \quad \dots 6.1.5(3)$$

## 6.2 CALCULATION OF MAXIMUM STRESSES AND STRESS RANGE

Calculated stresses shall be based on elastic analysis. Stresses shall not be amplified by stress concentration factors for geometrical discontinuities.

For bolts and threaded rods subject to axial tension, the calculated stresses shall include the effects of prying action, if applicable.

In the case of axial stress combined with bending, the maximum stresses, of each kind, shall be those determined for concurrent arrangements of applied load.

For members having symmetric cross-sections, the fasteners and welds shall be arranged symmetrically about the axis of the member, or the total stresses including those due to eccentricity shall be included in the calculation of the stress range.

For axially stressed angle members, where the centre of gravity of the connecting welds lies between the line of the centre of gravity of the angle cross-section and the centre of the connected leg, the effects of eccentricity shall be ignored. If the centre of gravity of the connecting welds lies outside this zone, the total stresses, including those due to joint eccentricity, shall be included in the calculation of stress range.

## 6.3 DETAIL CATEGORIES FOR CLASSIFIED DETAILS

For cold-formed sections, the detail categories to be used for the various constructional details are given in Table 6.3(A).

In the classification method, the detail category is a designation given to a particular detail to indicate which of the S-N curves shown in Figure 6.3 shall be used in fatigue assessment. The detail category shall be the nominal stress range corresponding to  $2 \times 10^6$  cycles on a fatigue strength curve of a given construction detail.

The fatigue strength curves for the different detail categories are defined by—

- (a)  $\log(n_{sc}) = \log(a) - 3\log(f_f)$ , for  $n_{sc} \leq 5 \times 10^6$
- (b)  $\log(n_{sc}) = \log(a) - 5\log(f_f)$ , for  $5 \times 10^6 < n_{sc} \leq 10^8$

where

$n_{sc}$  = number of stress cycles

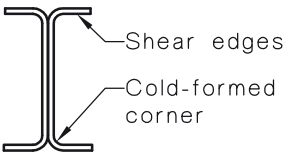
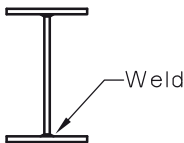
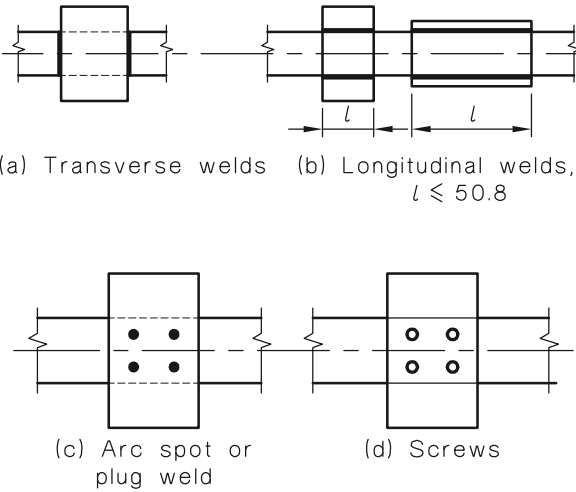
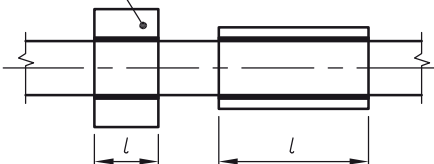
$\log(a)$  = constant which depends on the related part of the slope (see Table 6.3(B))

$f_f$  = uncorrected fatigue strength

The values of the constant ( $\log(a)$ ) for the different parts of the fatigue strength S-N curves, as well as the constant stress range fatigue limit and cut-off limit for each detail category are given in Table 6.3(B).

The welds in the welded details given in Table 6.3(A) for Detail Categories 118 and below shall conform with Category SP as specified in AS/NZS 1554.1.

**TABLE 6.3(A)**  
**DETAIL CATEGORY CLASSIFICATION FOR**  
**COLD-FORMED STEEL MEMBERS AND CONNECTIONS**

| Stress category | Detail category | Construction details                                                                                                                                               |                                                                                                                                                                                                                                                                                  |
|-----------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                 | Illustration                                                                                                                                                       | Description                                                                                                                                                                                                                                                                      |
| I               | 174             |  <p>Cold-formed steel channels</p> <p align="center"><b>Detail Category I</b></p> | <p><i>Non-welded products:</i></p> <p>As-received base metal and components with as-rolled surfaces, including sheared edges and cold-formed corners</p>                                                                                                                         |
| II              | 118             |  <p>Welded I Beam</p> <p align="center"><b>Detail Category II</b></p>             | <p><i>Members connected by continuous longitudinal welds:</i></p> <p>As-received base metal and weld metal in members connected by continuous longitudinal welds loaded in shear</p>                                                                                             |
| III             | 81              |  <p align="center"><b>Detail Category III</b></p>                              | <p><i>Welded attachments and bolt or screw connections:</i></p> <p>(a) and (b) Welded attachments to a plate or a beam, transverse fillet welds and continuous fillet welds loaded in shear less than or equal to 50 mm</p> <p>(c) Bolt and screw connections and spot welds</p> |
| IV              | 55              | <p>Typical plate</p>  <p align="center"><b>Detail Category IV</b></p>          | <p><i>Longitudinal fillet-welded attachments:</i></p> <p>Longitudinal fillet-welded attachments greater than 50 mm parallel to the direction of the applied stress, and intermittent welds parallel to the direction of the applied force</p>                                    |

where

stress category I: non-welded products

stress category II: members connected by continuous longitudinal welds

stress category II: welded attachments and bolt or screw connections

stress category II: longitudinal fillet-welded attachments

**TABLE 6.3(B)**  
**VALUES FOR DEFINING FATIGUE STRENGTH CURVES**  
**FOR DIFFERENT DETAIL CATEGORIES**

| Detail category | Log( <i>a</i> ) for<br>$n_{sc} \leq 5 \times 10^6$<br>( $\alpha_s = 3$ ) | Log( <i>a</i> ) for<br>$5 \times 10^6 \leq n_{sc} \leq 10^8$<br>( $\alpha_s = 5$ ) | Constant stress range<br>fatigue limit, $f_3$<br>MPa | Cut-off limit,<br>$f_5$<br>MPa |
|-----------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------|
| 174             | 13.0197                                                                  | 17.2335                                                                            | 128                                                  | 70                             |
| 118             | 12.5146                                                                  | 16.3916                                                                            | 87                                                   | 48                             |
| 81              | 12.0197                                                                  | 15.5668                                                                            | 59                                                   | 33                             |
| 55              | 11.5146                                                                  | 14.7249                                                                            | 40                                                   | 22                             |

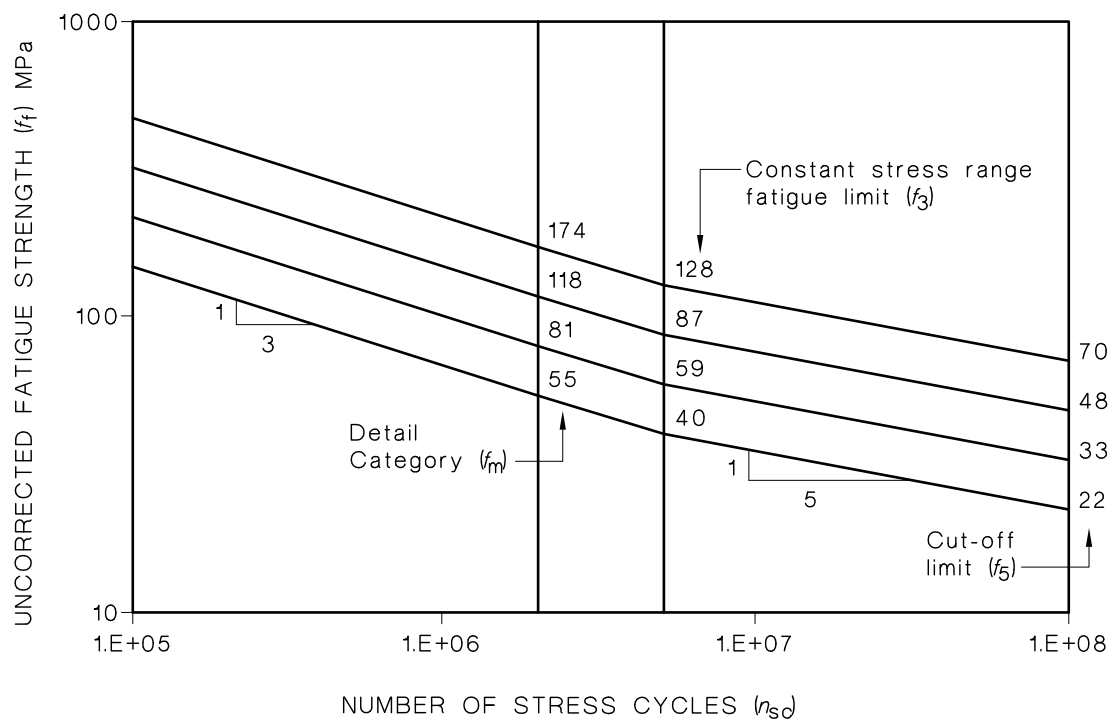


FIGURE 6.3 FATIGUE STRENGTH CURVES FOR DIFFERENT DETAIL CATEGORIES

## 6.4 FATIGUE ASSESSMENT

### 6.4.1 Constant stress range

The design stress range ( $f^*$ ), at any point in the structure subject only to constant stress range cycles, shall satisfy—

$$\frac{f^*}{\phi f_c} \leq 1.0 \quad \dots 6.4.1$$

### 6.4.2 Variable stress range

The design stress range ( $f^*$ ), at any point in the structure at which the stress range varies, shall satisfy the following:

(a) *Normal stresses*

$$\frac{\sum_i n_i (f_i^*)^3}{5 \times 10^6 (\phi f_{3c})^3} + \frac{\sum_j n_j (f_j^*)^5}{5 \times 10^6 (\phi f_{3c})^5} \leq 1.0 \quad \dots 6.4.2(1)$$

(b) *Shear stresses*

$$\frac{\sum_k n_k (f_k^*)^5}{2 \times 10^6 (\phi f_{rsc})^5} \leq 1.0 \quad \dots 6.4.2(2)$$

where—

- (i) the summation  $\sum_i$  is for  $i$  design stress ranges ( $f_i^*$ ) for which  $\phi f_{3c} < f_i^*$ ;
- (ii) the summation  $\sum_j$  is for  $j$  design stress ranges ( $f_j^*$ ) for which  $\phi f_{sc} \leq f_j^* < \phi f_{3c}$ ; and
- (iii) the summation  $\sum_k$  is for  $k$  design stress ranges ( $f_k^*$ ) for shear stresses  $\phi f_{5c} \leq f_k^*$ .



## SECTION 7 DIRECT STRENGTH METHOD

### 7.1 GENERAL

This Section applies to the determination of the nominal axial compression ( $N$ ) and bending ( $M$ ) capacities of cold-formed steel members.

Clauses 7.2.1 and 7.2.2 provide a method applicable to all cold-formed steel compression members and members subject to bending. Those members meeting the geometric and material limitations of Clause 7.1.1 for compression members and Clause 7.1.2 for members subject to bending have been pre-qualified for use, and the calibrated  $\phi$  factors given in Clauses 7.2.1 and 7.2.2 apply. Other compression members and members subject to bending shall use the provisions of Clauses 7.2.1 and 7.2.2 but the  $\phi$  factors for rational analysis given in Clause 1.6.3(c) shall apply.

The direct strength method does not provide explicit provisions for members in shear, combined bending and shear, web crippling, combined bending and web crippling, or combined axial load and bending (beam-column). Further, no provisions are given for structural assemblies or connections and joints. The provisions of Sections 2, 3 and 4, when applicable, shall be used for all cases.

For members or situations that are not applicable to Sections 2, 3 and 4, extensions to the direct strength method may exist. Extensions to the direct strength method are subject to the same provisions as any other rational analysis procedure specified in Clause 1.6.3(c). The applicable provisions of Sections 2, 3 and 4 shall be met when they exist and the reduced  $\phi$  factors shall be used for the design capacity when rational analysis is conducted.

#### 7.1.1 Pre-qualified compression members

Unperforated compression members that fall within the geometric limitations given in Table 7.1.1 have been pre-qualified and shall be permitted to be designed using the  $\phi$  factors given in Clause 7.2.1.1.

#### 7.1.2 Pre-qualified members subject to bending

Unperforated members subject to bending that fall within the geometric limitations given in Table 7.1.2 have been pre-qualified and shall be permitted to be designed using the  $\phi$  factors given in Clause 7.2.2.1.

#### 7.1.3 Elastic buckling

Analysis is required for determining the elastic buckling loads or moments, or both, used in this Section. For compression members, this includes the local and distortional and overall buckling loads specified in Clause 7.2.1. For members subject to bending, this includes the local and distortional and overall buckling moments specified in Clause 7.2.2. For a given compression member or members subject to bending, all three modes may not exist. In this case, the non-existent mode shall be ignored in the calculations of Clauses 7.2.1 and 7.2.2.

#### 7.1.4 Deflection calculation

The bending deflection at any moment ( $M$ ) due to nominal loads, shall be permitted to be determined by reducing the gross second moment of area ( $I_g$ ) to an effective second moment of area ( $I_{\text{eff}}$ ) for deflection, using the following equation:

$$I_{\text{eff}} = I_g \left( \frac{M_n}{M} \right) \leq I_g \quad \dots 7.1.4$$

where

$M_n$  = nominal flexural capacity specified in Clause 7.2.2, but with  $M_y$  replaced by  $M$  in all equations of Clause 7.2.2

$M$  = moment due to nominal loads on member to be considered  
 $\leq M_y$

## 7.2 MEMBERS

### 7.2.1 Design of compression members

#### 7.2.1.1 General

The nominal member capacity of a member in compression ( $N_c$ ) shall be the minimum of the nominal member capacity of a member in compression ( $N_{ce}$ ) for flexural, torsional or flexural-torsional buckling, the nominal member capacity of a member in compression ( $N_{cl}$ ) for local buckling and the nominal member capacity of a member in compression ( $N_{cd}$ ) for distortional buckling as specified in Clauses 7.2.1.2, 7.2.1.3 and 7.2.1.4. For compression members meeting the geometric requirements of Table 7.1.1,  $\phi_c$  shall be taken as 0.85. For all other compression members,  $\phi_c$  specified in Clause 1.6.3(c)(i) applies.

#### 7.2.1.2 Flexural, torsional or flexural-torsional buckling

The nominal member capacity of a member in compression ( $N_{ce}$ ) for flexural, torsional or flexural-torsional buckling shall be calculated as follows:

$$\text{For } \lambda_c \leq 1.5: \quad N_{ce} = (0.658^{\lambda_c^2}) N_y \quad \dots 7.2.1.2(1)$$

$$\text{For } \lambda_c > 1.5: \quad N_{ce} = \left( \frac{0.877}{\lambda_c^2} \right) N_y \quad \dots 7.2.1.2(2)$$

where

$$\begin{aligned} \lambda_c &= \text{non-dimensional slenderness used to determine } N_{ce} \\ &= \sqrt{N_y / N_{oc}} \quad \dots 7.2.1.2(3) \end{aligned}$$

$N_{oc}$  = least of the elastic compression member buckling load in flexural, torsional and flexural-torsional buckling

$$= A f_{oc} \quad \dots 7.2.1.2(4)$$

$N_y$  = nominal yield capacity of the member in compression

$$= A f_y \quad \dots 7.2.1.2(5)$$

#### 7.2.1.3 Local buckling

The nominal member capacity of a member in compression ( $N_{cl}$ ) for local buckling shall be calculated as follows:

$$\text{For } \lambda_l \leq 0.776: \quad N_{cl} = N_{ce} \quad \dots 7.2.1.3(1)$$

$$\text{For } \lambda_l > 0.776: \quad N_{cl} = \left[ 1 - 0.15 \left( \frac{N_{ol}}{N_{ce}} \right)^{0.4} \right] \left( \frac{N_{ol}}{N_{ce}} \right)^{0.4} N_{ce} \quad \dots 7.2.1.3(2)$$

where

$$\begin{aligned} \lambda_l &= \text{non-dimensional slenderness used to determine } N_{cl} \\ &= \sqrt{N_{ce} / N_{ol}} \quad \dots 7.2.1.3(3) \end{aligned}$$

$$\begin{aligned}
 N_{ol} &= \text{elastic local buckling load} \\
 &= Af_{ol}
 \end{aligned}
 \quad \dots 7.2.1.3(4)$$

#### 7.2.1.4 Distortional buckling

The nominal member capacity of a member in compression ( $N_{cd}$ ) for distortional buckling shall be calculated as follows:

$$\text{For } \lambda_d \leq 0.561: \quad N_{cd} = N_y \quad \dots 7.2.1.4(1)$$

$$\text{For } \lambda_d > 0.561: \quad N_{cd} = \left[ 1 - 0.25 \left( \frac{N_{od}}{N_y} \right)^{0.6} \right] \left( \frac{N_{od}}{N_y} \right)^{0.6} N_y \quad \dots 7.2.1.4(2)$$

where

$$\begin{aligned}
 \lambda_d &= \text{non-dimensional slenderness used to determine } N_{cd} \\
 &= \sqrt{N_y / N_{od}}
 \end{aligned}
 \quad \dots 7.2.1.4(3)$$

$$\begin{aligned}
 N_{od} &= \text{elastic distortional compression member buckling load} \\
 &= Af_{od}
 \end{aligned}
 \quad \dots 7.2.1.4(4)$$

### 7.2.2 Design of members subject to bending

#### 7.2.2.1 General

The nominal member moment capacity ( $M_b$ ) shall be the minimum of the nominal member moment capacity ( $M_{be}$ ) for lateral-torsional buckling, the nominal member moment capacity ( $M_{bl}$ ) for local buckling and the nominal member moment capacity ( $M_{bd}$ ) for distortional buckling as specified in Clauses 7.2.2.2, 7.2.2.3 and 7.2.2.4. For members subject to bending, meeting the geometric requirements of Clause 7.1.2,  $\phi_b$  shall be taken as 0.90. For all other members subject to bending,  $\phi_b$  specified in Clause 1.6.3(c)(i) applies.

#### 7.2.2.2 Lateral-torsional buckling

The nominal member moment capacity ( $M_{be}$ ) for lateral-torsional buckling shall be calculated as follows:

$$\text{For } M_o < 0.56M_y: \quad M_{be} = M_o \quad \dots 7.2.2.2(1)$$

$$\text{For } 2.78 M_y \geq M_o \geq 0.56M_y: \quad M_{be} = \frac{10}{9} M_y \left( 1 - \frac{10M_y}{36M_o} \right) \quad \dots 7.2.2.2(2)$$

$$\text{For } M_o > 2.78M_y: \quad M_{be} = M_y \quad \dots 7.2.2.2(3)$$

where

$$\begin{aligned}
 M_o &= \text{elastic lateral-torsional buckling moment as defined in Clause 3.3.3.2} \\
 M_y &= Z_f f_y
 \end{aligned}
 \quad \dots 7.2.2.2(4)$$

where

$$Z_f = \text{full section modulus of the extreme fibre at first yield}$$

#### 7.2.2.3 Local buckling

The nominal member moment capacity ( $M_{bl}$ ) for local buckling shall be calculated as follows:

$$\text{For } \lambda_l \leq 0.776: \quad M_{bl} = M_{be} \quad \dots 7.2.2.3(1)$$

$$\text{For } \lambda_l > 0.776: \quad M_{bl} = \left[ 1 - 0.15 \left( \frac{M_{ol}}{M_{be}} \right)^{0.4} \right] \left( \frac{M_{ol}}{M_{be}} \right)^{0.4} M_{be} \quad \dots 7.2.2.3(2)$$

where

$$\begin{aligned} \lambda_l &= \text{non-dimensional slenderness used to determine } M_{bl} \\ &= \sqrt{M_{be} / M_{ol}} \quad \dots 7.2.2.3(3) \end{aligned}$$

$$\begin{aligned} M_{ol} &= \text{elastic local buckling moment} \\ &= Z_f f_{ol} \quad \dots 7.2.2.3(4) \end{aligned}$$

#### 7.2.2.4 Distortional buckling

The nominal member moment capacity ( $M_{bd}$ ) for distortional buckling shall be calculated as follows:

$$\text{For } \lambda_d \leq 0.673: \quad M_{bd} = M_y \quad \dots 7.2.2.4(1)$$

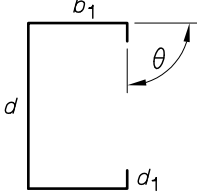
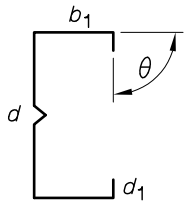
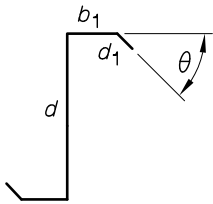
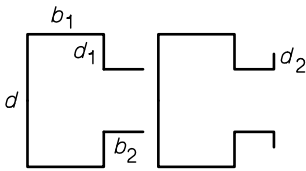
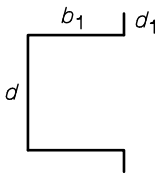
$$\text{For } \lambda_d > 0.673: \quad M_{bd} = \left[ 1 - 0.22 \left( \frac{M_{od}}{M_y} \right)^{0.5} \right] \left( \frac{M_{od}}{M_y} \right)^{0.5} M_y \quad \dots 7.2.2.4(2)$$

where

$$\begin{aligned} \lambda_d &= \text{non-dimensional slenderness used to determine } M_{bd} \\ &= \sqrt{M_y / M_{od}} \quad \dots 7.2.2.4(3) \end{aligned}$$

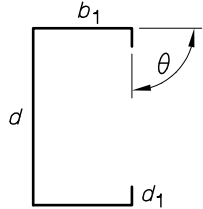
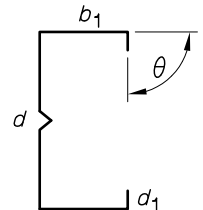
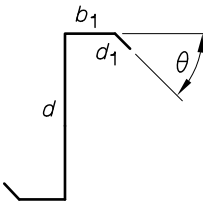
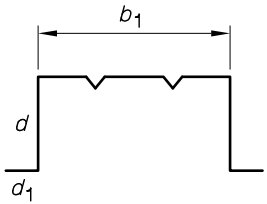
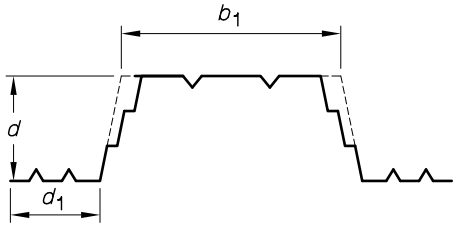
$$\begin{aligned} M_{od} &= \text{elastic distortional buckling moment} \\ &= Z_f f_{od} \quad \dots 7.2.2.4(4) \end{aligned}$$

**TABLE 7.1.1**  
**LIMITS FOR PRE-QUALIFIED COMPRESSION MEMBERS**

| Section                                                                                                                   | Geometric limitation                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lipped channel<br>                       | $d/t < 472$<br>$b_1/t < 159$<br>$4 < d_1/t < 33$<br>$0.7 < d/b_f < 5.0$<br>$0.05 < d_1/b_1 < 0.41$<br>$\theta = 90^\circ$<br>$E/f_y > 340$ ( $f_y < 593$ MPa)                                                                            |
| Lipped channel with web stiffener(s)<br> | $d/t < 489$<br>$b_1/t < 160$<br>$6 < d_1/t < 33$<br>$1.3 < d/b_1 < 2.7$<br>$0.05 < d_1/b_1 < 0.41$<br>One or two intermediate stiffeners<br>$E/f_y > 340$ ( $f_y < 593$ MPa)                                                             |
| Z-section<br>                           | $d/t < 137$<br>$b_1/t < 56$<br>$0 < d_1/t < 36$<br>$1.5 < d/b_1 < 2.7$<br>$0.00 < d_1/b_1 < 0.73$<br>$\theta = 50^\circ$<br>$E/f_y > 590$ ( $f_y < 345$ MPa)                                                                             |
| Rack upright<br>                       | $d/t < 51$<br>$b_1/t < 22$<br>$5 < d_1/t < 8$<br>$2.1 < d/b_1 < 2.9$<br>$1.6 < b_2/d_1 < 2.0$ ( $b_2$ = small outstand parallel to $b_1$ )<br>$d_2/d = 0.3$ ( $d_2$ = second lip parallel to $d_1$ )<br>$E/f_y = 340$ ( $f_y < 593$ MPa) |
| Hat<br>                                | $d/t < 50$<br>$b_1/t < 20$<br>$4 < d_1/t < 6$<br>$1.0 < d/b_1 < 1.2$<br>$d_1/b_1 = 0.13$<br>$E/f_y > 428$ ( $f_y < 476$ MPa)                                                                                                             |

$r/t < 10$ , where  $r$  is the centre-line radius

**TABLE 7.1.2**  
**LIMITS FOR PRE-QUALIFIED MEMBERS SUBJECT TO BENDING**

| Section                                                                                                                                            | Geometric limitation                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Channels</p>                                                   | $d/t < 321$<br>$b_1/t < 75$<br>$0 < d_1/t < 34$<br>$1.5 < d/b_f < 17.0$<br>$0.0 < d_1/b_1 < 0.70$<br>$44^\circ < \theta < 90^\circ$<br>$E/f_y > 421$ ( $f_y < 483$ MPa)                                                                                                                                                                                                  |
| <p>Lipped channels with web stiffener</p>                         | $d/t < 358$<br>$b_1/t < 58$<br>$14 < d_1/t < 17$<br>$5.5 < d/b_1 < 11.7$<br>$0.27 < d_1/b_1 < 0.56$<br>$\theta = 90^\circ$<br>$E/f_y > 578$ ( $f_y < 352$ MPa)                                                                                                                                                                                                           |
| <p>Z-section</p>                                                | $d/t < 183$<br>$b_1/t < 71$<br>$10 < d_1/t < 16$<br>$2.5 < d/b_1 < 4.1$<br>$0.15 < d_1/b_1 < 0.34$<br>$36^\circ < \theta < 90^\circ$<br>$E/f_y > 400$ ( $f_y < 462$ MPa)                                                                                                                                                                                                 |
| <p>Hats (decks) with stiffened flange in compression</p>        | $d/t < 97$<br>$b_1/t < 467$<br>$0 < d_1/t < 26$<br>$0.14 < d/b_1 < 0.87$<br>$0.44 < b_1/2d_1 < 2.0$<br>$0 < n \leq 4$<br>$E/f_y = 492$ ( $f_y < 414$ MPa)                                                                                                                                                                                                                |
| <p>Trapezoids (decks) with stiffened flange in compression</p>  | $d/t < 203$<br>$b_1/t < 231$<br>$42 < (d/\sin\theta)/b_1 < 1.91$<br>$0.55 < d/2d_1 < 1.69$<br>$0 < n_c \leq 2$ ( $n_c$ = number of compression flange stiffeners)<br>$0 < n_w \leq 2$ ( $n_w$ = number of web stiffeners/folds)<br>$0 < n_t \leq 2$ ( $n_t$ = number of tension flange stiffeners)<br>$52^\circ < \theta < 84^\circ$<br>$E/f_y > 310$ ( $f_y < 655$ MPa) |

$r/t < 10$ , where  $r$  is the centre-line radius

## SECTION 8 TESTING

### 8.1 TESTING FOR DETERMINING MATERIAL PROPERTIES

#### 8.1.1 Testing of unformed steel

Where the steels specified in Clause 1.5.1.2 are used or the yield stress of steel is required for the purpose of Clause 6.1.3, unformed steel properties shall be determined by tests in accordance with AS 1391.

Test specimens shall be taken longitudinally (long dimension of specimens in direction of rolling) from positions located one quarter of the coil width from either edge near the outer end of the coil.

#### 8.1.2 Compression testing

Stub-column tests shall be made on flat-end specimens whose length is not less than three times the largest dimension of the section but no more than 20 times the least radius of gyration. If tests of ultimate compressive strength are used to determine yield stress for quality control purposes, the length of the section shall be not less than 15 times the least radius of gyration.

In making compression tests, the specimen in the testing machine shall be centred so that the load is applied concentrically with respect to the centroidal axis of the section.

NOTE: For further information regarding compression testing, reference may be made to ASTM E9, and to Technical Memoranda Nos 2 and 3 of the Column Research Council, 'Notes on Compression Testing of Materials', and 'Stub-Column Test Procedure', reprinted in the Column Research Council Guide to Stability Design Criteria for Metal Structures, Third Edition, 1976.

#### 8.1.3 Testing of full sections

This Clause applies only to the determination of the mechanical properties of a fully formed section for the purposes specified in Clause 1.5.1.3. It shall not be interpreted as forbidding the use of test procedures instead of the usual design calculations.

The procedure shall be as follows:

- (a) Determine the tensile yield stress ( $f_y$ ) in accordance with AS 1391.
- (b) Determine the compressive yield stress ( $f_y$ ) by means of compression tests as specified in Clause 8.1.2.
- (c) Where the principal effect of the loading to which the member will be subjected in service is to produce bending stresses, determine the yield stress for the flanges. In determining the yield stress, carry out tests on specimens cut from the section. Each such specimen shall consist of one complete flange plus a portion of the web of such flat width ratio so that the section is fully effective.
- (d) For acceptance and control purposes, make two full section tests from formed material lots. Material lots shall be considered as parcels or heats as defined in the relevant Standard's material specification in the Clauses on selection and preparation of test samples for mechanical testing.
- (e) Use either tension or compression tests for routine acceptance and control purposes, provided it is demonstrated that such tests reliably indicate the yield stress of the section when subjected to the kind of stress under which the member is to be used.

### 8.1.4 Testing of flat coupons of formed members

#### 8.1.4.1 *Assessment of strength increase*

Tests for determining material properties of flat coupons of formed members and material properties of unformed steel for the purpose of assessing strength increase resulting from cold-forming, as specified in Clause 1.5.1.3, shall be made as follows:

- (a) The yield stress of flats ( $f_{yf}$ ) shall be established by means of a weighted average of the yield stresses of standard tensile coupons taken longitudinally from the major flat portions of a cold-formed member. The weighted average shall be the sum of the products of the average yield stress for each major flat portion times its cross-sectional area, divided by the total area of the major flats in the cross-section.
- (b) If the actual yield stress of the unformed steel exceeds the specified minimum yield stress, the yield stress of the flats ( $f_{yf}$ ) shall be adjusted by multiplying the test values by the ratio of the specified minimum yield stress to the actual yield stress of the unformed steel.

#### 8.1.4.2 *Design properties*

Tests for determining material properties of flat coupons of formed members for the purpose of establishing design properties of the formed members, as specified in Clause 1.5.1.2, shall be made as follows:

- (a) The test specimens shall be taken longitudinally from a major flat portion of the section, midway between corners (excluding the corners) or midway between a corner and a free edge (excluding the corner).
- (b) The test specimen shall be taken from the flat portion with the least strength increase from cold-forming.
- (c) The minimum yield stress ( $f_y$ ) and the minimum tensile strength ( $f_u$ ) used in design shall be determined in accordance with AS 1391.

### 8.1.5 Testing for determining section properties

Flexural section properties, e.g., second moment of area of the cross-section and section modulus, may be determined by tests.

Test specimens prone to lateral displacement shall be suitably braced. The loading apparatus and bracing shall not impose unintentional restraints on the specimen. The loading devices shall be accurately calibrated.

The specimens shall be tested in bending with simply supported spans.

The cross-sectional dimensions of test specimens shall be as close as practicable to nominal dimensions, which are the basis for calculated properties. Where a discrepancy exists, the section properties from tests shall be adjusted by the ratios of nominal to actual dimensions.

The true deflections shall be separated from other causes of specimen movement under load, such as specimen settlement at support, and movement of test frame.

### 8.1.6 Testing of single-point fastener connections

The testing of single-point fastener connections shall be in accordance with Appendix F.

## 8.2 TESTING FOR ASSESSMENT OR VERIFICATION

### 8.2.1 General

This Clause applies to prototype units of complete structures, parts of structures, individual members or connections for design verification.



The methods of test shall be in accordance with AS/NZS1170.0, as an alternative to calculation.

The methods do not apply to the testing of structural models nor to the establishment of general design criteria.

### 8.2.2 Coefficient of variation of structural characteristics

The coefficient of variation of structural characteristics ( $V_{sc}$ ) refers to the variability of the total population of the production units. This includes the total population variation due to fabrication ( $k_f$ ) and material ( $k_m$ ). It can be approximated as follows:

$$V_{sc} = \sqrt{k_f^2 + k_m^2} \quad \dots 8.2.2$$

### 8.2.3 Design capacity of specific products and assemblies

The design capacity ( $R_d$ ) of a specific product or a specific assembly may be established by prototype testing of that specific product or assembly. The design capacity ( $R_d$ ) shall satisfy—

$$R_d \leq \left( \frac{R_{min.}}{k_t} \right) \quad \dots 8.2.3$$

where  $R_{min.}$  is the minimum value of the test results and  $k_t$  is as given in Table 8.2.3.

Testing of sheet roof and wall cladding systems shall be in accordance with AS/NZS 1562.1.

**TABLE 8.2.3**  
**FACTORS ( $k_t$ ) TO ALLOW FOR VARIABILITY OF STRUCTURAL UNITS**

| No. of units<br>to be tested | Coefficient of variation of structural characteristics ( $V_{sc}$ ) |      |      |      |      |      |
|------------------------------|---------------------------------------------------------------------|------|------|------|------|------|
|                              | 5%                                                                  | 10%  | 15%  | 20%  | 25%  | 30%  |
| 1                            | 1.20                                                                | 1.46 | 1.79 | 2.21 | 2.75 | 3.45 |
| 2                            | 1.17                                                                | 1.38 | 1.64 | 1.96 | 2.36 | 2.86 |
| 3                            | 1.15                                                                | 1.33 | 1.56 | 1.83 | 2.16 | 2.56 |
| 4                            | 1.15                                                                | 1.30 | 1.50 | 1.74 | 2.03 | 2.37 |
| 5                            | 1.13                                                                | 1.28 | 1.46 | 1.67 | 1.93 | 2.23 |
| 10                           | 1.10                                                                | 1.21 | 1.34 | 1.49 | 1.66 | 1.85 |
| 100                          | 1.00                                                                | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

## APPENDIX A

### NORMATIVE REFERENCES

(Normative)

The documents listed below are indispensable for the application of this Standard.

#### AS

|             |                                                                                       |
|-------------|---------------------------------------------------------------------------------------|
| 1110        | ISO metric hexagon bolts and screws—Product grades A and B                            |
| 1110.1      | Part 1: Bolts                                                                         |
| 1110.2      | Part 2: Screws                                                                        |
| 1111        | ISO metric hexagon bolts and screws—Product grade C                                   |
| 1111.1      | Part 1: Bolts                                                                         |
| 1111.2      | Part 2: Screws                                                                        |
| 1112        | ISO metric hexagon nuts                                                               |
| 1112.1      | Part 1: Style 1—Product grades A and B                                                |
| 1112.2      | Part 2: Style 2—Product grades A and B                                                |
| 1112.3      | Part 3: Product grade C                                                               |
| 1112.4      | Part 4: Chamfered thin nuts—Product grades A and B                                    |
| 1163        | Structural steel hollow sections                                                      |
| 1170.4      | Part 4: Structural design actions—Minimum design loads on structures—Earthquake loads |
| 1275        | Metric screw threads for fasteners                                                    |
| 1391        | Metallic materials—Tensile testing at ambient temperature                             |
| 1397        | Steel sheet and strip—Hot-dipped zinc-coated or aluminium/zinc-coated                 |
| 3566        | Self-drilling screws for the building and construction industries                     |
| 3566.1      | Part 1: General requirements and mechanical properties                                |
| 3566.2      | Part 2: Corrosion resistance requirements                                             |
| 3623        | Domestic metal framing                                                                |
| 4040        | Methods of testing sheet roof and wall cladding                                       |
| 4040.2      | Part 2: Resistance to wind pressure for non-cyclone regions                           |
| 4100        | Steel structures                                                                      |
| 4291        | Mechanical properties of fasteners made of carbon steel and alloy steel               |
| 4291.1      | Part 1: Bolts, screws and studs                                                       |
| (ISO 898-1) |                                                                                       |

#### AS/NZS

|        |                                                                                       |
|--------|---------------------------------------------------------------------------------------|
| 1170   | Structural design actions                                                             |
| 1170.0 | Part 0: General principles                                                            |
| 1170.1 | Part 1: Permanent, imposed and other actions                                          |
| 1170.2 | Part 2: Wind actions                                                                  |
| 1170.3 | Part 3: Snow and ice actions                                                          |
| 1252   | High strength steel bolts with associated nuts and washers for structural engineering |
| 1554   | Structural steel welding                                                              |
| 1554.1 | Part 1: Welding of steel structures                                                   |

|    |  |        |                                                                                        |
|----|--|--------|----------------------------------------------------------------------------------------|
| A1 |  | AS     |                                                                                        |
|    |  | 1554.2 | Part 2: Stud welding (steel studs to steel)                                            |
|    |  | AS/NZS |                                                                                        |
|    |  | 1554.5 | Part 5: Welding of steel structures subject to high levels of fatigue loading          |
|    |  | 1554.7 | Part 7: Structural steel welding—Welding of sheet steel structures                     |
|    |  | 1559   | Hot-dip galvanized steel bolts with associated nuts and washers for tower construction |
|    |  | 1562   | Design and installation of sheet roof and wall cladding                                |
|    |  | 1562.1 | Part 1: Metal                                                                          |
|    |  | 1594   | Hot-rolled steel flat products                                                         |
|    |  | 1595   | Cold-rolled, unalloyed, steel sheet and strip                                          |
|    |  | 3678   | Structural steel—Hot-rolled plates, floorplates and slabs                              |
|    |  | 4680   | Hot-dip galvanized (zinc) coatings on fabricated ferrous articles                      |
|    |  | NZS    |                                                                                        |
|    |  | 1170.5 | Part 5: Structural design actions—Earthquake actions—New Zealand                       |
| A1 |  | 3404   | Steel Structures Standard                                                              |
|    |  | AWS    |                                                                                        |
|    |  | C1.1   | Recommended Practices for Resistance Welding                                           |
|    |  | C1.3   | Recommended Practices for Resistance Welding Coated Low Carbon Steels                  |
|    |  | F114   | Industrial Fastener Institute                                                          |

NOTE: Appendix G includes a list of informative documents referenced in this Standard.

## APPENDIX B

FLEXURAL MEMBERS SUBJECTED TO POSITIVE AND NEGATIVE  
BENDING

(Informative)

**B1 GENERAL**

If the geometrical properties of flexural members are based on the effective design width accounting for flange curling and such a member is subjected to positive and negative bending moments (e.g., in the case of a continuous beam or a rigid frame), Paragraphs B2 and B3 may apply, subject to the limitations specified in Paragraph B4.

**B2 LOAD-CARRYING CAPACITY**

The bending moments and the support reactions may be determined assuming constant section beams or frames, provided that the ratio of section moduli for positive and negative bending moments does not exceed the values specified in Paragraph B4. The maximum design bending moments ( $M^*$ ) so determined should not exceed the nominal member moment capacity ( $M_b$ ) times  $\phi_b$ .

**B3 DEFLECTIONS**

Deflections may be determined assuming constant section beams or frames, and are based on a mean second moment of area, provided that the ratio of second moments of area for positive and negative bending moment does not exceed the value specified in Paragraph B4.

**B4 LIMITATIONS**

For the purpose of Paragraphs B2 and B3, the ratios of geometrical properties of a member for positive and negative bending moments, determined in accordance with this Standard, should not exceed the following:

- |     |                            |       |
|-----|----------------------------|-------|
| (a) | Section moduli:            |       |
|     | (i) Continuous beams ..... | 1.35. |
|     | (ii) Rigid frames.....     | 1.25. |
| (b) | Second moment of area:     |       |
|     | (i) Continuous beams ..... | 1.20. |
|     | (ii) Rigid frames.....     | 1.16. |

For the purpose of this Paragraph, the section property with the greater value should be taken as the numerator of the ratio.

For members with ratios outside the limits specified in Paragraph B4, a rational analysis approach may be developed based on testing.

## APPENDIX C

### PROTECTION

(Informative)

#### C1 SCOPE

This Appendix applies to the protection of cold-formed structural members, including decking, cladding and structures.

NOTE: See the New Zealand Building Code and the New Zealand Building Code Handbook Verification Method (B2/VM1) and Acceptable Solutions (B2/AS1) for additional requirements, if the New Zealand Building Act is applicable to the project.

#### C2 PROTECTION AGAINST CORROSION

##### C2.1 Members to be protected

A member should be adequately protected against corrosive attacks, with due regard to environmental conditions.

##### C2.2 Protective coating

The protective coating may be applied to steel sheet or strip, either before or after the forming of the members. The type of coating should be specified, after proper account has been taken of the use of the structure, climatic or other local conditions, maintenance provisions, and the effect of the forming process on previously applied coatings.

##### C2.3 Members made from uncoated steel

A member made from uncoated steel should be protected by a rust-inhibitive coating immediately after processing. The coating should possess a permanent adhesion to the steel. Subsequent coatings, before or after assembly, should adequately adhere to, and be compatible with, the first coating. The type and quality of coatings and their application should comply with the recommendations of the appropriate sections of AS/NZS 2312. Coatings destroyed by welding, assembly, or by handling should be restored as specified in Paragraph C4.

##### C2.4 Members made from coated steel

For a member made from coated steel, the coatings applied before forming should have adequate mechanical properties and adhesion to the steel sufficient to withstand the forming process without damage or peeling.

NOTE: Recommendations for corrosion protection may be found in AS/NZS 2311 and AS/NZS 2312.

#### C3 PROTECTION DURING TRANSPORT, HANDLING AND STORAGE

##### C3.1 General

Structural members that have been distorted, and subsequently corrected, may be weakened to the extent that their structural integrity may be impaired or lost. Such members should not be used.

##### C3.2 Transport and handling

Structural members should be adequately protected during handling and transport to prevent damage. Units that are transported in nested bundles should be separable without damage to the units or their coatings. Care should be taken when handling long units or bundles. Consideration should be given to the use of lifting beams with appropriately spaced lifting points and slings or to lifting with properly spaced forklift tines.

### **C3.3 Storage**

Structural members should be protected from the weather. They should be stacked clear of the ground and protected by a waterproof covering. Ventilation adequate to avoid condensation should be ensured. If bundles become wet, the members should be separated, wiped and placed so that air circulation completes the drying.

## **C4 REPAIRS TO COATINGS**

Coatings that have been damaged by welding or other causes should be restored before the structure is put into service. The damaged area should be dry and clean, free from dirt, grease, loose or heavy scale or rust before the protective coating is applied. When preparing welded assemblies for painting, care should be taken that the area at and near welds is thoroughly cleaned down to base metal. The protective coating should be applied as soon as practicable and before noticeable oxidation of the clean surface occurs. Damaged zinc coating should be restored with a suitable zinc paint.

## APPENDIX D

## DISTORTIONAL BUCKLING STRESSES OF GENERAL CHANNELS, LIPPED CHANNELS AND Z-SECTIONS IN COMPRESSION AND BENDING

(Normative)

**D1 GENERAL CHANNELS IN COMPRESSION**

The elastic distortional buckling stress ( $f_{od}$ ) of general channels in compression (see Figure D1(a)) shall be determined as follows:

$$f_{od} = \frac{E}{2A} \left\{ (\alpha_1 + \alpha_2) - \sqrt{[(\alpha_1 + \alpha_2)^2 - 4\alpha_3]} \right\} \quad \dots \text{D1(1)}$$

where

$$\alpha_1 = \frac{\eta}{\beta_1} \left( \beta_2 + 0.039J\lambda^2 \right) + \frac{k_\phi}{\beta_1 \eta E} \quad \dots \text{D1(2)}$$

$$\alpha_2 = \eta \left( I_y + 2y_0 \frac{\beta_3}{\beta_1} \right) \quad \dots \text{D1(3)}$$

$$\alpha_3 = \eta \left( \alpha_1 I_y - \frac{\eta}{\beta_1} \beta_3^2 \right) \quad \dots \text{D1(4)}$$

$$\beta_1 = h_x^2 + \left( \frac{I_x + I_y}{A} \right) \quad \dots \text{D1(5)}$$

$$\beta_2 = I_w + I_x (x_0 - h_x)^2 \quad \dots \text{D1(6)}$$

$$\beta_3 = I_{xy} (x_0 - h_x) \quad \dots \text{D1(7)}$$

$$\beta_4 = \beta_2 + (y_0 - h_y) [I_y (y_0 - h_y) - 2\beta_3] \quad \dots \text{D1(8)}$$

$$\lambda = 4.80 \left( \frac{\beta_4 b_w}{t^3} \right)^{0.25} \quad \dots \text{D1(9)}$$

$$\eta = \left( \frac{\pi}{\lambda} \right)^2 \quad \dots \text{D1(10)}$$

$$k_\phi = \frac{Et^3}{5.46(b_w + 0.06\lambda)} \left[ 1 - \frac{1.11f'_{od}}{Et^2} \left( \frac{b_w^2 \lambda}{b_w^2 + \lambda^2} \right)^2 \right] \quad \dots \text{D1(11)}$$

$f'_{od}$  is obtained from Equation D1(1) with—

$$\alpha_1 = \frac{\eta}{\beta_1} \left( \beta_2 + 0.039J\lambda^2 \right) \quad \dots \text{D1(12)}$$

The values of  $A$ ,  $I_x$ ,  $I_y$ ,  $I_{xy}$ ,  $I_w$  are for the compression flange and lip alone.

## D2 SIMPLE LIPPED CHANNELS IN COMPRESSION

The elastic distortional buckling stress ( $f_{od}$ ) of simple lipped channels in compression (see Figure D1(b)) shall be determined as follows:

$$f_{od} = \frac{E}{2A} \left\{ (\alpha_1 + \alpha_2) - \sqrt{(\alpha_1 + \alpha_2)^2 - 4\alpha_3} \right\} \quad \dots \text{D2(1)}$$

where

$$\alpha_1 = \frac{\eta}{\beta_1} (I_x b_f^2 + 0.039 J \lambda^2) + \frac{k_\phi}{\beta_1 \eta E} \quad \dots \text{D2(2)}$$

$$\alpha_2 = \eta \left( I_y + \frac{2}{\beta_1} \bar{y} b_f I_{xy} \right) \quad \dots \text{D2(3)}$$

$$\alpha_3 = \eta \left( \alpha_1 I_y - \frac{\eta}{\beta_1} I_{xy}^2 b_f^2 \right) \quad \dots \text{D2(4)}$$

$$\beta_1 = \bar{x}^2 + \left( \frac{I_x + I_y}{A} \right) \quad \dots \text{D2(5)}$$

$$\lambda = 4.80 \left( \frac{I_x b_f^2 b_w}{t^3} \right)^{0.25} \quad \dots \text{D2(6)}$$

$$\eta = \left( \frac{\pi}{\lambda} \right)^2 \quad \dots \text{D2(7)}$$

$$k_\phi = \frac{Et^3}{5.46(b_w + 0.06\lambda)} \left[ 1 - \frac{1.11 f'_{od}}{Et^2} \left( \frac{b_w^2 \lambda}{b_w^2 + \lambda^2} \right)^2 \right] \quad \dots \text{D2(8)}$$

$f'_{od}$  is obtained from Equation D2(1) with—

$$\alpha_1 = \frac{\eta}{\beta_1} (I_x b_f^2 + 0.039 J \lambda^2) \quad \dots \text{D2(9)}$$

The values of  $A, \bar{x}, \bar{y}, J, I_x, I_y$  and  $I_{xy}$  for a compression flange with a simple lip are as follows:

$$A = (b_f + d_l) t \quad \dots \text{D2(10)}$$

$$\bar{x} = \frac{(b_f^2 + 2b_f d_l)}{2(b_f + d_l)} \quad \dots \text{D2(11)}$$

$$\bar{y} = \frac{d_l^2}{2(b_f + d_l)} \quad \dots \text{D2(12)}$$

$$J = \frac{t^3 (b_f + d_l)}{3} \quad \dots \text{D2(13)}$$

$$I_x = \frac{b_f t^3}{12} + \frac{t d_l^3}{12} + b_f t \bar{y}^2 + d_l t \left( \frac{d_l}{2} - \bar{y} \right)^2 \quad \dots \text{D2(14)}$$



A1

$$I_y = \frac{tb_f^3}{12} + \frac{d_l t^3}{12} + d_l t (b_f - \bar{x})^2 + b_f t \left( \bar{x} - \frac{b_f}{2} \right)^2 \quad \dots \text{D2(15)}$$

$$I_{xy} = b_f t \left( \frac{b_f}{2} - \bar{x} \right) (-\bar{y}) + d_l t \left( \frac{d_l}{2} - \bar{y} \right) (b_f - \bar{x}) \quad \dots \text{D2(16)}$$

### D3 SIMPLE LIPPED CHANNELS OR Z-SECTIONS IN BENDING ABOUT THE AXIS PERPENDICULAR TO THE WEB

The elastic distortional buckling stress ( $f'_{od}$ ) of simple lipped channels or Z-sections in bending about the axis perpendicular to the web (see Figure D1(c)) shall be determined in accordance with Paragraph D2, except that—

$$\lambda = 4.80 \left( \frac{I_x b_f^2 b_w}{2t^3} \right)^{0.25} \quad \dots \text{D3(1)}$$

A1

$$k_\phi = \frac{2Et^3}{5.46(b_w + 0.06\lambda)} \left[ 1 - \frac{1.11f'_{od}}{Et^2} \left( \frac{b_w^4 \lambda^2}{12.56\lambda^4 + 2.192b_w^4 + 13.39\lambda^2 b_w^2} \right) \right] \quad \dots \text{D3(2)}$$

$f'_{od}$  is obtained from Equation D2(1) with—

$$\alpha_1 = \frac{\eta}{\beta_1} (I_x b_f^2 + 0.039J\lambda^2) \quad \dots \text{D3(3)}$$

If  $k_\phi$  is negative,  $k_\phi$  shall be calculated with  $f'_{od} = 0$ .

If the bracing interval that fully restrains rotation of the flange and lip in the distortional mode is located at an interval less than  $\lambda$  obtained from Equation D3(1), the bracing interval shall be used in place of  $\lambda$ .

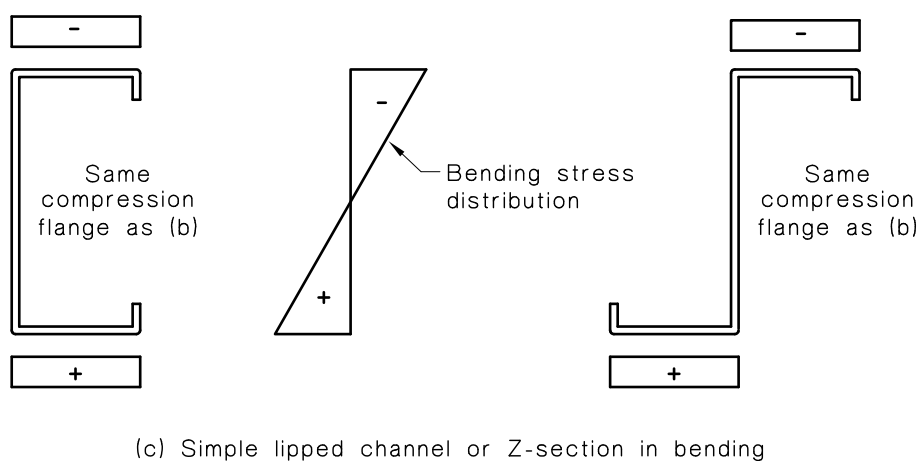
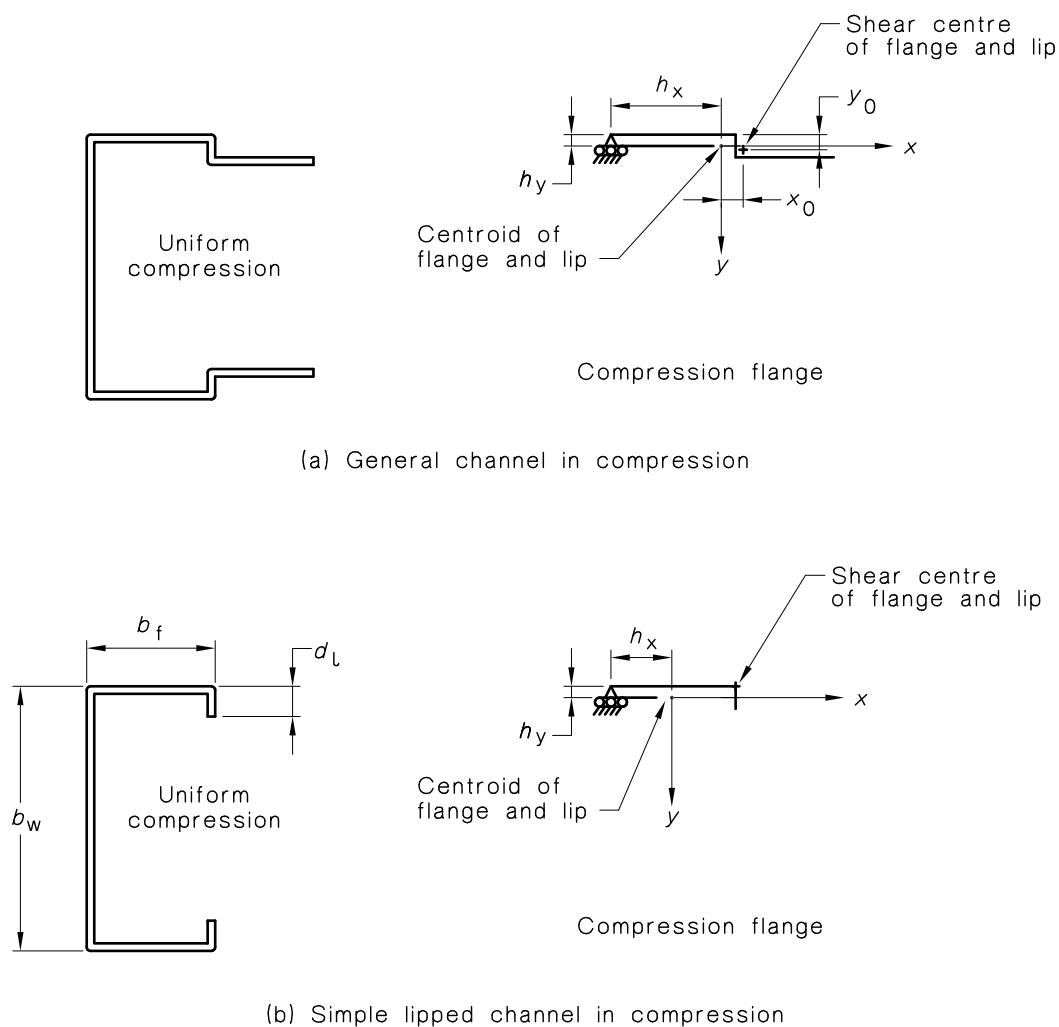


FIGURE D1 GENERAL, LIPPED CHANNEL AND Z-SECTION MODELS FOR DISTORTIONAL BUCKLING

## APPENDIX E

### SECTION PROPERTIES

(Normative)

#### E1 SHEAR CENTRE DISTANCE ( $m$ ), TORSION CONSTANT ( $J$ ) AND WARPING CONSTANT ( $I_w$ )

Values of  $m$ ,  $J$  and  $I_w$  for certain sections are shown in Figure E1.

For  $I_w$  of sections other than those given in Figure E1,  $I_w$  shall be taken as zero for box sections.

#### E2 MONOSYMMETRY SECTION CONSTANTS

Monosymmetry section constants are calculated as follows:

$$\beta_x = \frac{1}{I_x} \left( \int_A x^2 y dA + \int_A y^3 dA \right) - 2y_o \quad \dots \text{E2(1)}$$

$$\beta_y = \frac{1}{I_y} \left( \int_A xy^2 dA + \int_A x^3 dA \right) - 2x_o \quad \dots \text{E2(2)}$$

Where the  $x$ -axis is the axis of symmetry (see Table E1)—

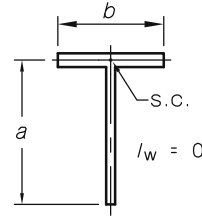
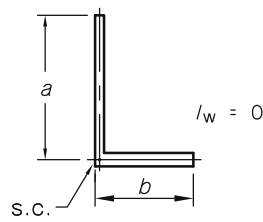
$$\beta_x = 0 \quad \dots \text{E2(3)}$$

$$\beta_y = \frac{\beta_w + \beta_f + \beta_L}{I_y} - 2x_o \quad \dots \text{E2(4)}$$

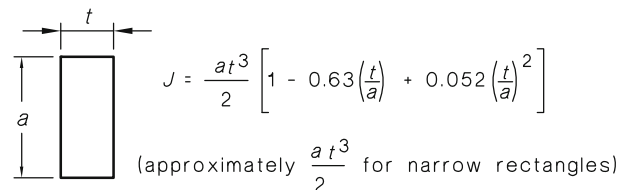
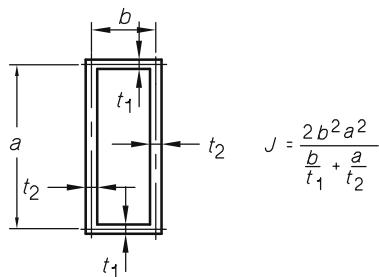
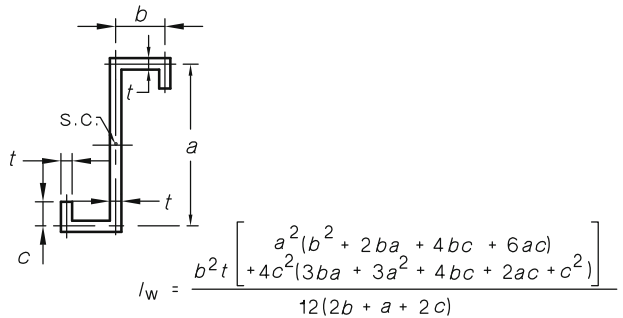
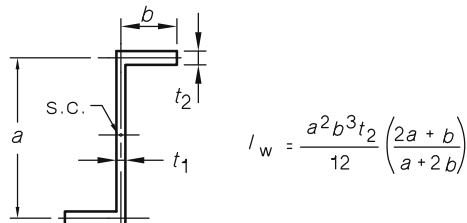
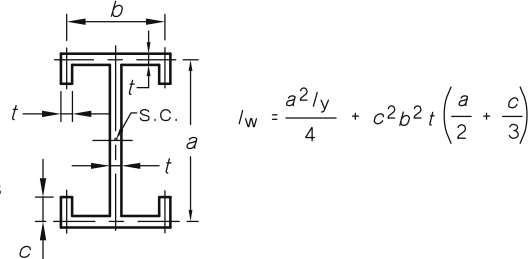
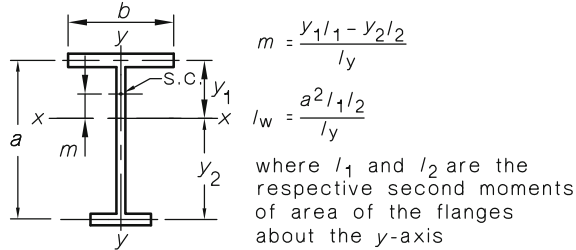
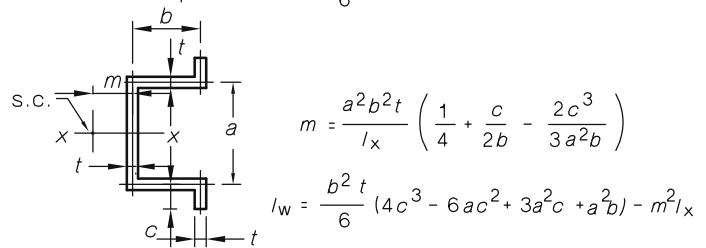
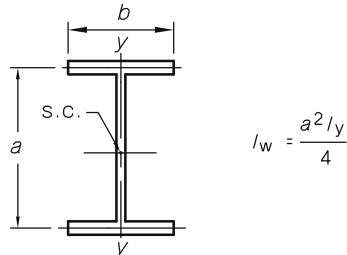
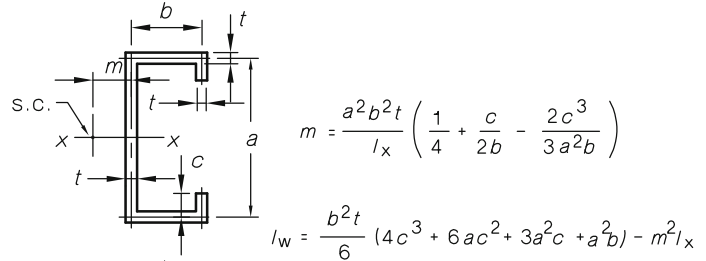
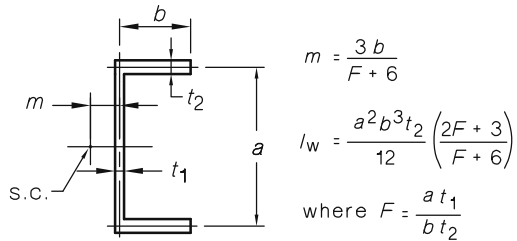
NOTES:

- 1 For doubly symmetric sections,  $\beta_x = 0$  and  $\beta_y = 0$ .
- 2 In the calculation of  $\beta_y$  using the value of  $x_o$ , determined from Table E1,  $x_o$  and  $\bar{x}$  are to be taken as negative.

Where the  $y$ -axis is the axis of symmetry, interchange  $x$  and  $y$  in the equations for the  $x$ -axis of symmetry and Table E1.



A1



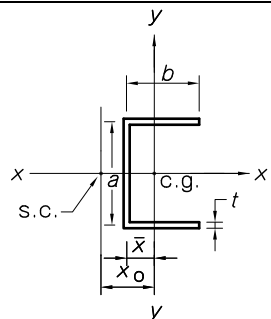
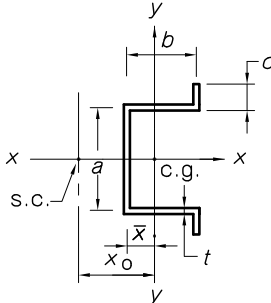
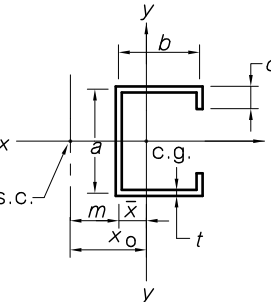
LEGEND: s.c. = shear centre

FIGURE E1 SHEAR CENTRE DISTANCE, TORSION CONSTANT AND WARPING CONSTANT FOR CERTAIN SECTIONS

## NOTES TO FIGURE E1:

- 1 For all open section:  $J = \sum \frac{bt^3}{3}$ .
- 2 For members cold-formed from a single steel sheet of uniform thickness:  $J = \frac{w_f t^3}{3}$ , where  $w_f$  is the feed width of the flat sheet.
- 3 For the box and rectangle sections,  $I_w$  is negligibly small in comparison to  $J$ .

**TABLE E1**  
**CERTAIN MONOSYMMETRIC SECTIONS—CENTROID AND SHEAR-CENTRE**  
**DISTANCES AND MONOSYMMETRY SECTION CONSTANTS**

| Section                                                                             | $\bar{x}$                 | $x_o$                                                           | $\beta_w$                                 | $\beta_r$                                                                                    | $\beta_L$                                                                                                            |
|-------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|    | $\frac{b^2}{a+2b}$        | $\frac{b^2}{a+2b} + \frac{3b^2}{6b+a}$                          | $\frac{1}{12}t\bar{x}a^3 + t(\bar{x})^3a$ | $\frac{1}{2}t[(b+\bar{x})^4 - (\bar{x})^4] +$ $\frac{1}{4}a^2t[(b+\bar{x})^2 - (\bar{x})^2]$ | 0                                                                                                                    |
|   | $\frac{b(b+2c)}{a+2b+2c}$ | $\frac{bt(b+2c)}{A} +$ $\frac{bt}{12I_x}(6ca^2 + 3ba^2 - 8c^3)$ | $\frac{1}{12}t\bar{x}a^3 + t(\bar{x})^3a$ | $\frac{1}{2}t[(b+\bar{x})^4 - (\bar{x})^4] +$ $\frac{1}{4}a^2t[(b+\bar{x})^2 - (\bar{x})^2]$ | $2ct(\bar{x}+b)^3 +$ $\frac{2}{3}t(\bar{x}+b)\left[\left(\frac{a}{2}+c\right)^3 - \left(\frac{a}{2}\right)^3\right]$ |
|  | $\frac{b(b+2c)}{a+2b+2c}$ | $\frac{bt(b+2c)}{A} +$ $\frac{bt}{12I_x}(6ca^2 + 3ba^2 - 8c^3)$ | $\frac{1}{12}t\bar{x}a^3 + t(\bar{x})^3a$ | $\frac{1}{2}t[(b+\bar{x})^4 - (\bar{x})^4] +$ $\frac{1}{4}a^2t[(b+\bar{x})^2 - (\bar{x})^2]$ | $2ct(\bar{x}+b)^3 +$ $\frac{2}{3}t(\bar{x}+b)\left[\left(\frac{a}{2}\right)^3 - \left(\frac{a}{2}-c\right)^3\right]$ |

LEGEND:

s.c. = shear centre

c.g. = centre of gravity

APPENDIX F  
STANDARD TESTS FOR SINGLE-POINT FASTENER CONNECTIONS  
(Normative)

**F1 SCOPE**

This Appendix sets out test methods to evaluate the structural performance of single-point fastener connections and clinching. The following tests shall be made for single-point fastener connections:

- (a) Shear test (see Paragraph F3).
- (b) Cross-tension tests (see Paragraph F4).

**F2 MATERIAL**

A specimen of the steel sheet shall be tested in accordance with AS 1391 to determine its physical properties.

**F3 SHEAR TEST****F3.1 General**

A specimen consisting of two strips of steel sheet, connected by a single fastener through overlapped ends, shall be evaluated for its capacity to resist a tensile force.

**F3.2 Apparatus**

The following apparatus shall be used:

- (a) *Grips* Any device that is capable of holding the ends of the test specimen in such a way as to ensure uniform loading. In addition, for test specimens where the thickness at each end exceeds 2.0 mm, packing shims or adjustable grips shall be used to ensure central loading across the lap joint.
- (b) *Loading device* Any suitable device that is capable of loading the grips uniaxially at a controlled rate.
- (c) *Instrumentation* Capable of measuring a force applied to the test specimen to a minimum accuracy of  $\pm 1\%$ , and displacement across the joint to a minimum accuracy of 0.02 mm.

**F3.3 Test specimen**

The test specimen shall consist of two strips of flat steel joined by lapping the ends and fastening through the centre of the lapped area (see Figure F1). The strips shall be joined together flat and shall be free of any residue. The fastener shall be installed within 3.0 mm of its specified location and in accordance with the manufacturer's recommendations or the actual site practice, as applicable.

**F3.4 Procedure**

The procedure for the shear test shall be as follows:

- (a) Align the test specimen in the grips and clamp.
- (b) Monitor load and displacement.
- (c) Load the specimen at a controlled rate to ensure the test is completed within a 30 s to 240 s time frame.

- (d) Stop the test once the maximum load has been reached and the load has either dropped off, or the joint has undergone a displacement of 6.0 mm or the fastener diameter, whichever is greater.
- (e) Record the maximum load and mode of failure.

F4 CROSS-TENSION TEST

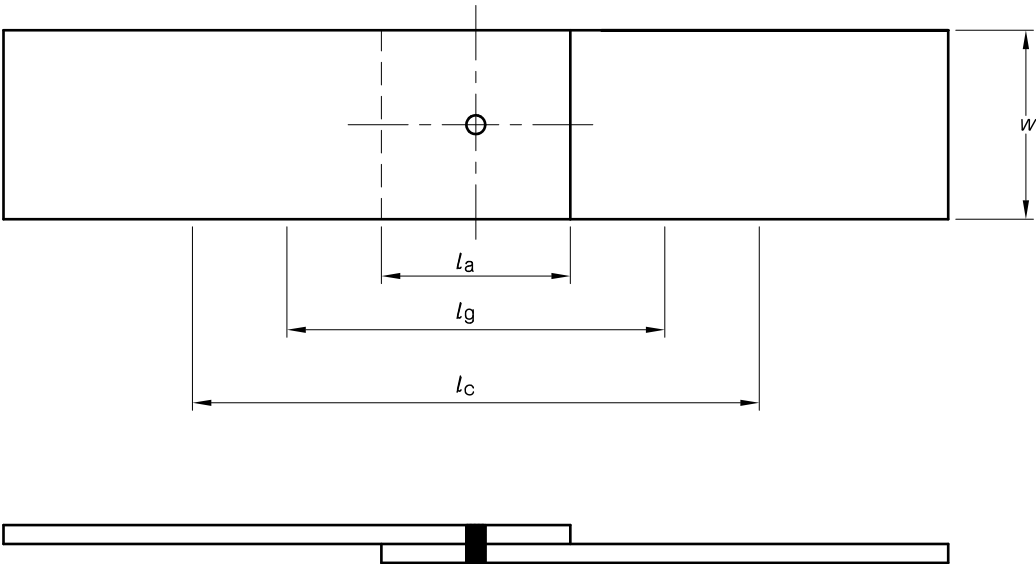
F4.1 General

Specimens consisting of two strips of steel sheet, connected by a single fastener to form a cross, shall be evaluated for their capacity to resist a tensile force applied perpendicular to the plane of the specimen.

F4.2 Apparatus

The following apparatus shall be used:

- (a) *Holding jig* The jig for holding the cross-tension specimen is shown in Figure F2. The attachment of the jig to the loading device shall allow for self-alignment.
- (b) *Loading device* Any suitable device that is capable of loading the halves of the holding jig uniaxially at a controlled rate.



| millimetres                                                         |                               |                      |                                                             |            |                                            |
|---------------------------------------------------------------------|-------------------------------|----------------------|-------------------------------------------------------------|------------|--------------------------------------------|
| Fastener                                                            | Width of the specimen ( $w$ ) | Lap length ( $l_a$ ) | Gauge length for measuring the joint displacement ( $l_g$ ) |            | Unclamped length of the specimen ( $l_c$ ) |
|                                                                     |                               |                      | Min.                                                        | Max.       | Min.                                       |
| Clinches and all other fasteners with shank diameters $\leq 7.0$ mm | 50                            | 50                   | 100                                                         | 150        | 150                                        |
| All fasteners with shank diameters $> 7.0$ mm                       | $8d_{sh}$                     | $8d_{sh}$            | $16d_{sh}$                                                  | $24d_{sh}$ | $24d_{sh}$                                 |

NOTE:  $d_{sh}$  is the nominal shank diameter.

FIGURE F1 SPECIMEN FOR SHEAR TEST



### **F4.3 Test specimen**

The test specimen consists of two strips of flat steel crossed and joined through the centre with a fastener (see Figure G3). The strips shall be joined together flat and shall be free of any residue. The fastener shall be installed within 3 mm of its specified location and in accordance with the manufacturer's recommendations or the actual site practice, as applicable.

### **F4.4 Procedure**

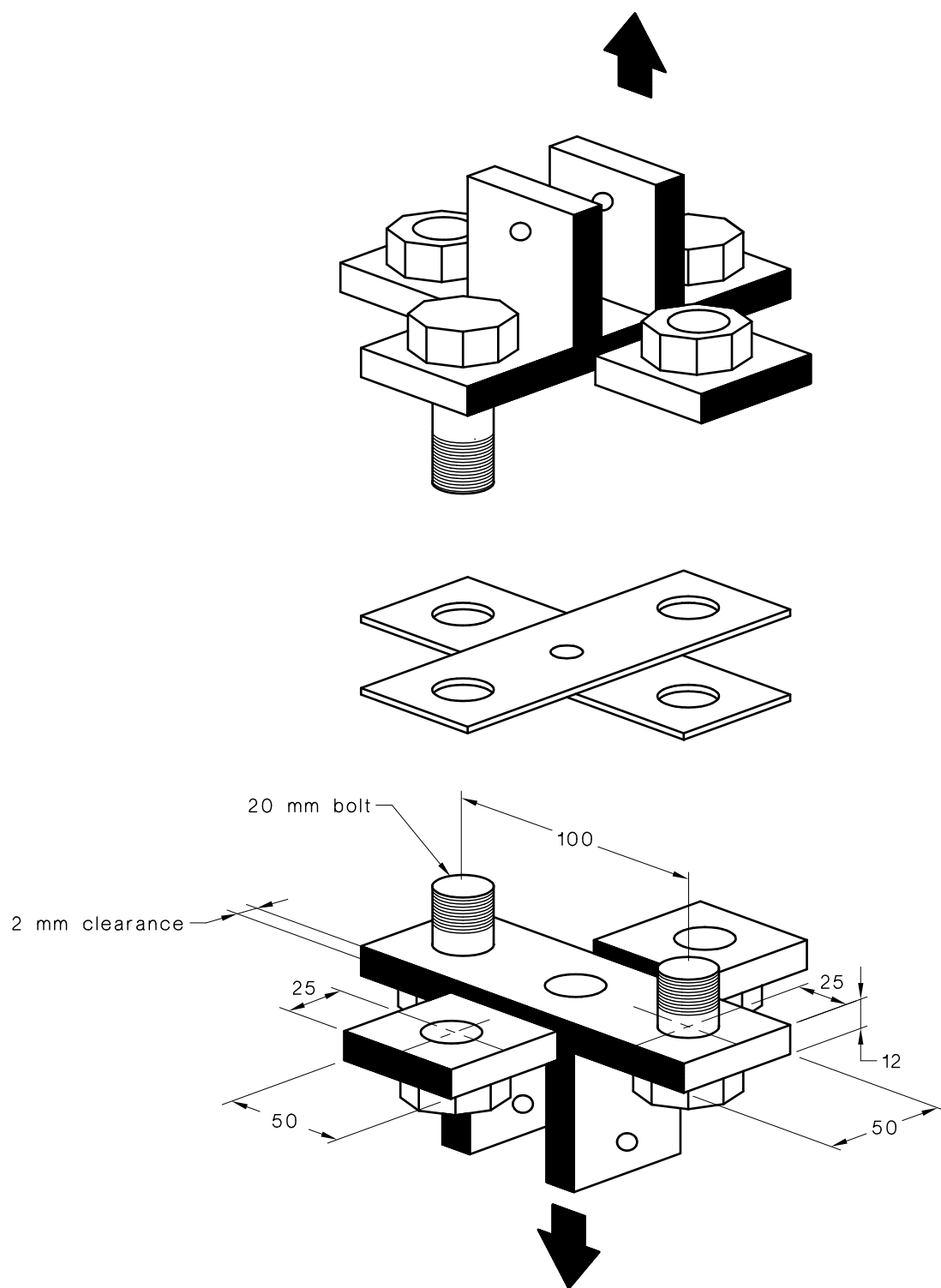
The procedure for the cross-tension test shall be as follows:

- (a) Align the test specimen in the holding jig and clamp.
- (b) Load the specimen at a controlled rate to ensure the test is completed within a 30 s to 240 s time frame.
- (c) Stop the test once the maximum load has been reached and the load has dropped off.
- (d) Record the maximum load and mode of failure.

### **F5 REPORT**

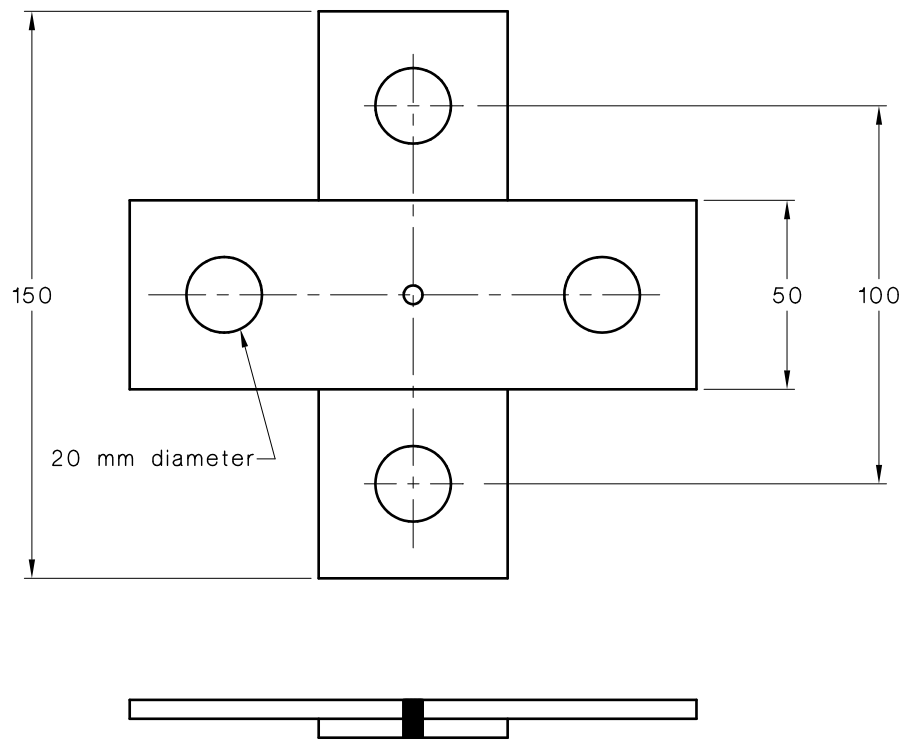
The following shall be reported:

- (a) Number of this Australian/New Zealand Standard, i.e., AS/NZS 4600.
- (b) Testing laboratory.
- (c) Type of test, i.e., shear or cross-tension test.
- (d) Type and properties of the fastener and method of installation.
- (e) Type and properties of the sheet material.
- (f) Duration of test.
- (g) Maximum load.
- (h) Load-displacement curve.
- (i) Mode of failure.



DIMENSIONS IN MILLIMETRES

FIGURE F2 HOLDING JIG FOR CROSS-TENSION TEST SPECIMEN



DIMENSIONS IN MILLIMETRES

FIGURE F3 SPECIMEN FOR CROSS-TENSION TEST

## APPENDIX G

### BIBLIOGRAPHY

(Informative)

The following non-mandatory documents are referred to in this Standard:

- AS  
2311 Guide to the painting of buildings  
2312 Guide to the protection of iron and steel against exterior atmospheric corrosion
- AISI Load and resistance factor design specification for cold-formed steel structural members  
Part IV: Rational-lateral stiffness test method for beam-to-panel assemblies
- ASTM  
E9 Test Methods of Compression Testing of Metallic Materials at Room Temperature
- Technical Memoranda Nos 2 and 3 of the Column Research Council, 'Notes on Compression Testing of Materials', and 'Stub Column Test Procedure', reprinted in the Column Research Council Guide to Stability Design Criteria for Metal Structures, 3<sup>rd</sup> Edition, 1976

**AMENDMENT CONTROL SHEET****AS/NZS 4600:2005**

---

**Amendment No. 1 (2010)**

---

**CORRECTION**

*SUMMARY:* This Amendment applies to the Clauses 1.5.1.4(b)(i), 1.5.3.2, 1.6.3(a), 2.4.2(a), 3.3.2.3(d), 3.3.6.2, 3.3.8.4, 3.4.1, 4.3.3.4, 5.2.1, 5.2.2.2, 5.4.3.3, 7.2.2.2, Equations 2.2.1.2(5), 2.2.1.3(3), 2.2.1.3(5), 2.4.2(9), 3.3.3.2(12), 3.3.3.3(5), 3.3.3.3(7), 3.3.4(3), 3.3.4.2(4), 3.3.7(2), 3.3.8.1(7), 3.3.8.1(8), 3.4.1(3), 3.4.3(1), 3.4.6(2), 3.6.2(1), 3.6.2(2), 3.6.2(3), 5.2.4.2(4), 5.2.6.2(5), 5.3.3(2), 5.3.3(3), 5.3.3(4), 5.3.4.3, 5.4.2.2(2), 5.4.2.3(5), 5.4.2.3(6), D1(11), D2(5), D2(8), D2(10) to D2(16), D3(2), Tables 1.4, 1.6, 2.4.2, 3.3.6.2(A) and 3.3.6.2(B), Figures 2.3.2(A) (a) and (b), 2.3.2(B) (a) and (b), 2.4.2, 5.2.6 (d), (e), (f) and (g), and E1, and Appendices A, B, D and E.

Published on 10 August 2010.

---

NOTES

### **Standards Australia**

Standards Australia is an independent company, limited by guarantee, which prepares and publishes most of the voluntary technical and commercial standards used in Australia. These standards are developed through an open process of consultation and consensus, in which all interested parties are invited to participate. Through a Memorandum of Understanding with the Commonwealth government, Standards Australia is recognized as Australia's peak national standards body.

### **Standards New Zealand**

The first national Standards organization was created in New Zealand in 1932. The Standards Council of New Zealand is the national authority responsible for the production of Standards. Standards New Zealand is the trading arm of the Standards Council established under the Standards Act 1988.

### **Australian/New Zealand Standards**

Under a Memorandum of Understanding between Standards Australia and Standards New Zealand, Australian/New Zealand Standards are prepared by committees of experts from industry, governments, consumers and other sectors. The requirements or recommendations contained in published Standards are a consensus of the views of representative interests and also take account of comments received from other sources. They reflect the latest scientific and industry experience. Australian/New Zealand Standards are kept under continuous review after publication and are updated regularly to take account of changing technology.

### **International Involvement**

Standards Australia and Standards New Zealand are responsible for ensuring that the Australian and New Zealand viewpoints are considered in the formulation of international Standards and that the latest international experience is incorporated in national and Joint Standards. This role is vital in assisting local industry to compete in international markets. Both organizations are the national members of ISO (the International Organization for Standardization) and IEC (the International Electrotechnical Commission).

### **Visit our web sites**

[www.standards.org.au](http://www.standards.org.au)

[www.standards.co.nz](http://www.standards.co.nz)

[www.standards.com.au](http://www.standards.com.au)



GPO Box 476 Sydney NSW 2001  
**Phone** (02) 9237 6000  
**Fax** (02) 9237 6010  
**Email** [mail@standards.org.au](mailto:mail@standards.org.au)  
**Internet** [www.standards.org.au](http://www.standards.org.au)  
**SAI Global Customer Service**  
**Phone** 13 12 42  
**Fax** 1300 65 49 49  
**Email** [sales@saiglobal.com](mailto:sales@saiglobal.com)



Level 10 Radio New Zealand House  
155 The Terrace Wellington 6011  
(Private Bag 2439 Wellington 6140)  
**Phone** (04) 498 5990  
**Fax** (04) 498 5994  
**Customer Services** (04) 498 5991  
**Information Service** (04) 498 5992  
**Email** [snz@standards.co.nz](mailto:snz@standards.co.nz)  
**Internet** [www.standards.co.nz](http://www.standards.co.nz)

ISBN 0 7337 7073 8

Printed in Australia



This page has been left intentionally blank.